

New cosystolic high-dimensional expanders from KMS groups

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Abstract

Cosystolic expansion is a high-dimensional generalization of the Cheeger constant for simplicial complexes. Originally, this notion was motivated by the fact that it implies the topological overlapping property, but more recently it was shown to be connected to problems in theoretical computer science such as list agreement expansion and agreement expansion in the low soundness regime.

There are only a few constructions of high-dimensional cosystolic expanders and, in dimensions larger than 2, the only known constructions prior to our work were (co-dimension 1)-skeletons of quotients of affine buildings. In this paper, we give the first coset complex construction of cosystolic expanders for an arbitrary dimension. Our construction is more symmetric and arguably more elementary than the previous constructions relying on quotients of affine buildings.

The coset complexes we consider arise from finite quotients of Kac–Moody–Steinberg (KMS) groups, called KMS complexes. KMS complexes were introduced in recent work by Grave de Peralta and Valentiner-Branth where it was shown that they are local spectral expanders. Our result is that KMS complexes, satisfying some minor condition, give rise to infinite families of bounded degree cosystolic expanders of arbitrary dimension and for any finitely generated Abelian coefficient group.

This result is achieved by observing that proper links of KMS complexes are joins of opposite complexes in spherical buildings. In order to show that these opposite complexes are coboundary expanders, we develop a new method for constructing cone functions by iteratively adding sets of vertices. Using the prior local-to-global results, we obtain cosystolic expansion for the (co-dimension 1)-skeletons of the KMS complexes.

Keywords: High-dimensional expander, Kac–Moody–Steinberg groups, Cosystolic expansion, Spherical buildings

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1 Introduction

High-dimensional expanders (HDXs) are high-dimensional analogues of expander graphs. The basic idea behind HDXs is to extend notions of expansion from graphs to higher-dimensional objects, namely simplicial complexes. Two main notions of expansion are commonly used in the definition of HDXs: spectral expansion (more precisely, local spectral expansion; see the exact definition in Section 2.2) and coboundary/cosystolic expansion, which are higher-dimensional analogues of the Cheeger constant (see the exact definition in Section 2.3). Unlike the case of expander graphs, however, these two notions are not equivalent.

In this paper, we focus on coboundary/cosystolic expansion rather than spectral expansion. The original motivation for studying cosystolic expansion came from the result of [DKW18], showing that such complexes satisfy the topological overlapping property (see the definition in [DKW18, Introduction]). Since then, coboundary/cosystolic expansion has found numerous applications, including coding theory and property testing [KL14, GK19, BM24], quantum codes [EKZ24, KT21], and Probabilistically Checkable Proofs (PCPs) [DDL24, BMVY25].

In our work below, we construct new examples of bounded degree cosystolic HDXs. Our construction relies on KMS complexes (which are coset complexes introduced in [GdPVB24]) and on a novel idea of constructing a cone function in an iterative process.

1.1 KMS complexes

In [KO18, KO23], Kaufman and Oppenheim gave a construction of spectral HDXs based on the idea of coset complexes (see below). The construction of Kaufman and Oppenheim was based on the group $\mathrm{SL}_{n+1}(\mathbb{F}_p[t])$, and was later generalized by O’Donnell and Pratt [OP22] to other Chevalley groups.

In [GdPVB24], another coset complex construction of spectral HDXs was given using Kac–Moody–Steinberg groups. Namely, in [GdPVB24] it was shown that given a generalized Cartan matrix satisfying some conditions and a finite field $\mathbb{F}_q$, there is an infinite family of finite coset complexes that are spectral HDXs. The construction of KMS complexes involves some technicalities and we refer the reader to Section 4 for the full details and an explicit example. Our results are restricted to a certain class of KMS complexes characterized as follows. We call a generalized Cartan matrix $A = (A_{ij})_{i,j \in I}$ of rank m n -classical (for $n \leq m$) if every submatrix $(A_{ij})_{i,j \in J}$, $J \subset I$, $|J| \leq n$ is of classical (i.e. type A_k, B_k, C_k, D_k) type or if it is reducible, all irreducible parts are of classical type. A KMS complex is called n -classical if its underlying generalized Cartan matrix is n -classical.

While KMS complexes can be seen as a variant of the Chevalley group constructions of [KO23, OP22], they seem to have a significant advantage that is relevant to our result below. Namely, the links of vertices of KMS complexes are associated to opposite complexes which are well-studied simplicial complexes. Opposite complexes are large subcomplexes of spherical buildings and were studied independently to answer questions of finiteness properties of arithmetic groups (see for instance Abramenko [Abr96]). The connection of opposite complexes to KMS complexes is that every link of a KMS complex is either an opposite complex or a join of opposite complexes. Prior work on opposite complexes in [Abr96] suggests that, in some sense, they mimic the properties of spherical buildings. In our work below, this point is explored and we prove that much like spherical buildings, opposite complexes are coboundary expanders.

1.2 Main results

A link of a pure n -dimensional simplicial complex is called *proper* if its dimension is at least 1 and at most $n - 1$ (i.e. it is the link of a face of dimension between 0 and $n - 2$). As noted above, our main result is that, under some restrictions, proper links of the KMS complexes constructed in [GdPVB24] (see Section 4 below) are coboundary expanders:

Theorem 1.1. *Let $n \geq 3$ be an integer. There exists $\varepsilon > 0$ such that for every prime power $q > 2^{2n-1}$ and every n -dimensional, n -classical KMS complex X constructed over $k = \mathbb{F}_q$, all the proper links of X are ε -coboundary expanders¹ (with respect to any finitely generated Abelian coefficient group).*

The theorem relies on the fact that proper links of n -classical KMS complexes are joins of opposite complexes in spherical buildings of classical type. Given an explicit n -classical KMS group with the extra restriction that no subdiagram of type D_k appears in the Dynkin diagram, it is possible to track ε and get an explicit (but very rough) bound. Tracking an explicit bound for ε in the D_n case could probably be done by very carefully analysing the proof, but we did not attempt to do so.

In [GdPVB24], it was shown that n -dimensional KMS complexes over $\mathbb{F}_q$ are $\frac{2}{\sqrt{q}}$ -local spectral expanders for every q that is large enough with respect to n (the KMS complexes are constructed under some assumptions on the corresponding generalized Cartan matrices, see exact formulation in Theorem 4.12 below). Combining this fact with the local-to-global results of [EK24, KM21, DD24] yields the following theorem.

Theorem 1.2. *For an integer $n \geq 3$, let $\{X_i\}_i$ be a family of n -dimensional, n -classical KMS complexes over $\mathbb{F}_q$ (see details in Section 4) and let $\{Y_i\}_i$ be the family of $(n - 1)$ -dimensional skeletons of $\{X_i\}$. There are $\varepsilon > 0, \mu > 0, q_0 \in \mathbb{N}$ such that if $q \geq q_0$ every Y_i is an (ε, μ) -cosystolic expander² (with respect to any finitely generated Abelian group).*

By [DKW18], it follows that the family of complexes $\{Y_i\}_i$ defined in the Theorem above has the topological overlapping property (see exact formulation in [DKW18]).

1.3 Our contributions

There are only a few examples of cosystolic HDXs (with a uniformly bounded degree). Indeed, prior to our work, the known constructions were:

- Constructions stemming from quotients of affine buildings (e.g., Ramanujan complexes [LSV05]). These constructions give cosystolic HDXs of every dimension with respect to any finitely generated Abelian coefficient group via the work of Evra and Kaufman [EK24] and the generalizations of Kaufman and Mass [KM21] and of Dikstein and Dinur [DD24]. However, all known such constructions are less symmetric and less elementary than the construction arising from coset complexes (see below).

¹When we say that a simplicial complex Y is an ε -coboundary expander, we mean that for every $0 \leq j \leq \dim(Y) - 1$, its j -th coboundary expansion is bounded from below by ε - see exact definition in Section 2.3

²When we say that a simplicial complex Y is a (ε, μ) -cosystolic expander, we mean that for every $0 \leq j \leq \dim(Y) - 1$, its j -th cosystolic expansion is bounded from below by ε and that the size j -cosystoles is bounded from below by μ - see exact definition in Section 2.3

- Coset complexes arising from Chevalley groups. There are two constructions of this flavour and both only yield 2-dimensional cosystolic HDXs. First, Kaufman and Oppenheim [KO21] showed that their coset complex construction of spectral HDXs gives rise to 2-dimensional cosystolic HDXs with respect to any group coefficients. Second, in a more recent work, O'Donnell and Singer [OS24] showed that for coset complexes arising from B_3 Chevalley groups, the 2-skeletons of links of vertices are coboundary expanders with respect to $\mathbb{F}_2$ coefficients and that these complexes give rise to 2-dimensional cosystolic HDXs with respect to $\mathbb{F}_2$ coefficients.

Our construction is the first construction of cosystolic HDXs arising from coset complexes that holds beyond dimension 2. Furthermore, our results are more general than previous results on coset complexes arising from Chevalley groups, since our results hold for finitely generated Abelian coefficient groups and for KMS complexes of the types $\tilde{A}_n, \tilde{B}_n, \tilde{C}_n, \tilde{D}_n$.

The method of our proof (as described below) is also novel in the setting of HDX – we introduce a new method of constructing a cone function by iteratively adding sets of vertices to a subcomplex until we added all the vertices of the entire complex.

1.4 Proof idea

As noted above, Theorem 1.2 follows from Theorem 1.1 via the results of [EK24, KM21, DD24]. Thus, we will focus on the idea of the proof of Theorem 1.1, i.e., on the idea of proving that links of KMS complexes are coboundary expanders. Since these links are symmetric, it is enough to show that they have a cone function with bounded radius.

In the constructions of cosystolic expanders stemming from quotients of affine buildings, a cone function was constructed in [LMM16] by utilizing the fact that the links were spherical buildings that have a rich apartment structure. In our construction, the links of KMS complexes do not have such an apartment structure and thus we need a different method for constructing a cone function with bounded radius. The method that we use is based on the idea of constructing a cone function by starting with a subcomplex with a cone function, iteratively adding sets of vertices and extending the cone function to that set.

This idea stems from the works of [Qui73, Vog81, AA93], where this method was used to prove certain simplicial complexes are homotopy equivalent to a wedge of spheres. In [Abr96], Abramenko showed how to apply this idea to prove that certain subcomplexes of spherical classical buildings are homotopy equivalent to a wedge of spheres. The subcomplexes treated in [Abr96] are exactly the links in our n -classical KMS complexes discussed above and we show that the same method can be “quantified” in order to produce a cone function with bounded radius for the links of our n -classical KMS complexes.

The exact formulation of our result is quite technical (see Theorem 3.1 below), and we will explain the idea only on the level of vanishing of homology (as noted above, a cone function can be thought of as a quantification of vanishing of homology with respect to a base point). Let X be an n -dimensional simplicial complex with a full subcomplex X' . Given a vertex $w \in X$, we denote $X' \cup \{w\}$ to be the full subcomplex of X spanned by the vertices of X' and w . The link of w relative to X' is the intersection $X_w \cap X'$, where X_w is the link of w in X . We will show that vanishing of reduced homology for X' and the link of w relative to X' implies vanishing of reduced homology for $X' \cup \{w\}$. Indeed, let $A = X', B = \{w\} * (X_w \cap X')$ (more precisely the geometric

realization of the two subcomplexes, but we will not consider this subtlety at this point). Then

$$X' \cup \{w\} = X' \cup \{\sigma \in X : w \in \sigma, \sigma \setminus \{w\} \in X'\} = A \cup B,$$

and $A \cap B = X_w \cap X'$ (note that the intersection of A and B is in the topological sense, which can be seen as $A \cap B = \{\tau \cap \sigma \mid \tau \in A, \sigma \in B\}$). Furthermore, B can be contracted onto $\{w\}$. Hence we have $\tilde{H}_i(A) = \tilde{H}_i(B) = 0$ for $0 \leq i \leq n-1$ and $\tilde{H}_i(A \cap B) = 0$ for $0 \leq i \leq n-2$. Thus, the Mayer-Vietoris Theorem implies that $\tilde{H}_i(X) = 0$ for $0 \leq i \leq n-1$.

An important observation is that in the procedure above, instead of adding a single vertex w to X' , one can add a set of vertices W to X' as long as no two vertices in W are connected by an edge in X and for every vertex $w \in W$, the link of w relative to X' has vanishing of reduced homology. Theorem 3.1 is a quantification of this idea in the language of cone functions, where the cone radius is bounded as a function of the cone radius of X' and the cone radii of the links of $w \in W$ relative to X' .

As noted above, applying this idea to links of KMS complexes follows the work of Abramenko [Abr96] on opposite complexes of spherical buildings of classical type. Let X be an n -dimensional, n -classical KMS complex constructed over $\mathbb{F}_q$. Under this assumption, the links of vertices in X are subcomplexes of classical spherical buildings known as opposite complexes, or joins of such. Roughly speaking, Abramenko showed that if q is sufficiently large, then the links of X can be constructed by the procedure of adding sets of vertices described above, that the number of steps in this procedure is independent of q and that the (relative) links of the added vertices are joins of opposite complexes of spherical buildings of lower dimension. This gives rise to an inductive bound on the cone radii and thus (by the fact that the links are symmetric) on the coboundary expansion of opposite complexes in classical spherical buildings (and thus of the proper links in n -classical KMS complexes).

1.5 Organization

The paper is organized as follows. In Section 2, we recall the notions of local spectral, coboundary and cosystolic high-dimensional expansion, explain the relevant (co)homological background, define cone functions and show some basic properties. This includes the fact that a cone function with coefficients over $\mathbb{Z}$ gives rise to cone functions with coefficients in arbitrary finitely generated Abelian groups, and how one can construct a cone function for the join of two simplicial complexes, given cone functions for each of them separately. In Section 3, we describe our technique of extending cone functions which is a key tool in our work. In Section 4, we recall the construction of the KMS complexes, and in Section 5 we first describe the connection of KMS complexes to buildings and then explain a geometric way to construct classical buildings. Finally, in Section 6 we combine the previous observations to prove Theorem 1.1.

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2 Preliminaries

2.1 Weighted simplicial complexes

Let X be a finite n -dimensional simplicial complex. A simplicial complex X is called *pure n -dimensional* if every face in X is contained in some face of dimension n . The set of all k -faces of X is denoted $X(k)$, and we will be using the convention in which $X(-1) = \{\emptyset\}$. For every $0 \leq k \leq n$, the k -skeleton of X is the simplicial complex $\bigcup_{i=-1}^k X(i)$. We will say that X is *connected* if its 1-skeleton is a connected graph.

For a complex X , a subcomplex $X' \subseteq X$ is called a *full subcomplex*, if for every $v_0, \dots, v_k \in X'$, if $\{v_0, \dots, v_k\} \in X$, then $\{v_0, \dots, v_k\} \in X'$. Equivalently, X' is a full subcomplex if there is a set S that is a subset of the vertices of X such that $X' = \{\sigma \in X : \sigma \subseteq S\}$.

Given a pure n -dimensional simplicial complex X , the *weight function* $w : \bigcup_{k=-1}^n X(k) \rightarrow \mathbb{R}_+$ is defined to be

$$\forall \tau \in X(k), w(\tau) = \frac{|\{\sigma \in X(n) : \tau \subseteq \sigma\}|}{\binom{n+1}{k+1} |X(n)|}.$$

We note that w is normalized such that for each $-1 \leq k \leq n$, the function w can be thought of as a probability function on $X(k)$.

Given a finite pure n -dimensional complex X and $\tau \in X$, the *link of τ* is the subcomplex X_τ defined as

$$X_\tau = \{\sigma \in X : \tau \cap \sigma = \emptyset, \tau \cup \sigma \in X\}.$$

To clarify the notation in some situations, we will also write $\text{lk}_X(\tau)$ for the link of τ in X . We note that if $\tau \in X(k)$, then X_τ is a pure $(n - k - 1)$ -dimensional complex. We call X_τ a *proper link*, if $1 \leq \dim(X_\tau) \leq n - 1$.

We denote the weight function w_τ to be the weight function on X_τ defined as above. By slight abuse of notation, we will write $v \in X(0)$ to state that v is a vertex of X .

2.2 Local spectral expansion

Let X be a finite pure n -dimensional simplicial complex with $n \geq 1$ and w the weight function on X defined above. We define the stochastic matrix of the random walk on (the 1-skeleton of) X to be the matrix indexed by $X(0) \times X(0)$ and defined as

$$M(\{v\}, \{u\}) = \begin{cases} \frac{w(\{u,v\})}{\sum_{\{u',v\} \in X(1)} w(\{u',v\})} & \{v,u\} \in X(1) \\ 0 & \{v,u\} \notin X(1) \end{cases}.$$

For $\lambda < 1$, we say that X is a *(one-sided) λ -spectral expander* if X is connected and the second largest eigenvalue of M is $\leq \lambda$. We say that X is a *(one-sided) λ -local spectral expander* if for every $-1 \leq k \leq n-2$ and every $\tau \in X(k)$, the simplicial complex X_τ is a (one-sided) λ -spectral expander. We note that every (one-sided) λ -local spectral expander is a (one-sided) λ -spectral expander, since the link of the empty set is X itself.

2.3 Cohomological notations and cosystolic/coboundary expansion

Let X be a pure n -dimensional simplicial complex and $\mathbb{A}$ an Abelian group.

We denote $\vec{X}(k)$ to be the set of oriented k -simplices and for $\sigma \in \vec{X}(k)$, we denote $-\sigma$ to be the simplex with reversed orientation from σ on the same vertex set. For $\{v_0, \dots, v_k\} \in X(k)$, we denote $[v_0, \dots, v_k] \in \vec{X}(k)$ to be the oriented simplex with the orientation induced from the ordering of the vertices. Note that for every permutation $\gamma \in \text{Sym}\{0, \dots, k\}$ it holds that

$$[v_{\gamma(0)}, \dots, v_{\gamma(k)}] = (-1)^{\text{sign}(\gamma)} [v_0, \dots, v_k].$$

We denote $C^k(X; \mathbb{A})$ to be the set of k -cochains defined as follows: $C^k(X; \mathbb{A})$ is the set of functions $\phi : \vec{X}(k) \rightarrow \mathbb{A}$ that are anti-symmetric, i.e., for every $[v_0, \dots, v_k]$, $\phi(-[v_0, \dots, v_k]) = -\phi([v_0, \dots, v_k])$. We recall that $C^k(X; \mathbb{A})$ is an Abelian group with respect to pointwise addition.

Let w be the weight function on X defined above. For $\phi \in C^k(X; \mathbb{A})$, we define

$$\begin{aligned} \text{supp}(\phi) &= \{\{v_0, \dots, v_k\} \in X(k) : \phi([v_0, \dots, v_k]) \neq 0_{\mathbb{A}}\}, \\ \|\phi\| &= \sum_{\sigma \in \text{supp}(\phi)} w(\sigma). \end{aligned}$$

Given a subgroup $K \subseteq C^k(X; \mathbb{A})$ and a cochain $\phi \in C^k(X; \mathbb{A})$, we define

$$\|\phi - K\| = \min_{\psi \in K} \|\phi - \psi\|.$$

The *coboundary map* $d_k : C^k(X; \mathbb{A}) \rightarrow C^{k+1}(X; \mathbb{A})$ is the map defined by mapping $\phi \in C^k(X; \mathbb{A})$ to $d_k\phi$ given by

$$(d_k\phi)([v_0, \dots, v_{k+1}]) = \sum_{i=0}^{k+1} (-1)^i \phi([v_0, \dots, \hat{v}_i, \dots, v_{k+1}]),$$

where $[v_0, \dots, \hat{v}_i, \dots, v_{k+1}]$ is the simplex obtained from $[v_0, \dots, v_{k+1}]$ by removing the vertex v_i . We denote $B^k(X; \mathbb{A}) = \text{Im}(d_{k-1})$ and $Z^k(X; \mathbb{A}) = \text{Ker}(d_k)$ and recall that these are Abelian subgroups of $C^k(X; \mathbb{A})$ and that $B^k(X; \mathbb{A}) \subseteq Z^k(X; \mathbb{A})$. The k -cohomology $H^k(X; \mathbb{A})$ is defined as $H^k(X; \mathbb{A}) = Z^k(X; \mathbb{A})/B^k(X; \mathbb{A})$.

The k -coboundary/cosystolic expansion constant of X (with coefficients in $\mathbb{A}$) are defined to be

$$\begin{aligned} h_{\text{cb}}^k(X; \mathbb{A}) &= \min_{\phi \in C^k(X; \mathbb{A}) \setminus B^k(X; \mathbb{A})} \frac{\|d_k\phi\|}{\|\phi - B^k(X; \mathbb{A})\|}, \\ h_{\text{cs}}^k(X; \mathbb{A}) &= \min_{\phi \in C^k(X; \mathbb{A}) \setminus Z^k(X; \mathbb{A})} \frac{\|d_k\phi\|}{\|\phi - Z^k(X; \mathbb{A})\|}. \end{aligned}$$

We note that by definition $h_{\text{cs}}^k(X; \mathbb{A}) \geq h_{\text{cb}}^k(X; \mathbb{A})$ and that by definition $h_{\text{cs}}^k(X; \mathbb{A}) > 0$. We also note that $h_{\text{cs}}^k(X; \mathbb{A}) = h_{\text{cb}}^k(X; \mathbb{A})$ if and only if $H^k(X; \mathbb{A}) = 0$ and that if $H^k(X; \mathbb{A}) \neq 0$, then $h_{\text{cb}}^k(X; \mathbb{A}) = 0$.

Remark 2.1. We note that in some papers, $h_{\text{cs}}^k(X; \mathbb{A})$ is also called the k -th cocycle expansion (e.g., see [KKL16]).

For a finite pure n -dimensional simplicial complex X and a constant $\varepsilon > 0$, we will say that X is a $(\varepsilon, \mathbb{A})$ -coboundary expander if for every $0 \leq k \leq n-1$, $h_{\text{cb}}^k(X; \mathbb{A}) \geq \varepsilon$. Also, for constants $\varepsilon > 0, \mu > 0$, we will say that X is a $(\varepsilon, \mu, \mathbb{A})$ -cosystolic expander if for every $0 \leq k \leq n-1$,

$h_{\text{cs}}^k(X; \mathbb{A}) \geq \varepsilon$ and for every $\phi \in Z^k(X, \mathbb{A}) \setminus B^k(X, \mathbb{A})$, $\|\phi\| \geq \mu$ (the parameter $\min\{\|\phi\|: \phi \in Z^k(X, \mathbb{A}) \setminus B^k(X, \mathbb{A})\}$ is sometimes called the k -cosystole with respect to $\mathbb{A}$, but we will not use this terminology).

In [KM21, DD24] the following local to global results were proven:

Theorem 2.2. *Let $n \geq 3$ be an integer, $\varepsilon' > 0$ a constant and $\mathbb{A}$ be a finitely generated Abelian group. There are constants $0 < \lambda < 1, \varepsilon > 0, \mu > 0$ such that the following holds: For every finite pure n -dimensional simplicial complex X if*

- *the simplicial complex X is a (one-sided) λ -local spectral expander,*

and

- *for every $0 \leq k \leq n-2$ and every $\tau \in X(k)$, it holds that X_τ is a $(\varepsilon', \mathbb{A})$ -coboundary expander,*

then the $(n-1)$ -skeleton of X is a $(\varepsilon, \mu, \mathbb{A})$ -cosystolic expander.

2.4 Homological notations and cone functions

Let X be a pure n -dimensional simplicial complex and $\mathbb{A}$ an Abelian group.

Keeping the notations regarding oriented simplices above, we denote $C_k(X; \mathbb{A})$ to be the set of k -chains defined as follows: $C_k(X; \mathbb{A})$ is the set of all formal sums of the form

$$\sum_{\sigma \in \vec{X}(k)} t_\sigma \sigma,$$

such that $t_\sigma \in \mathbb{A}$ and $t_{-\sigma} = -t_\sigma$. We note that $C_k(X; \mathbb{A})$ is an Abelian group or equivalently a $\mathbb{Z}$ -module. For $A \in C_k(X; \mathbb{A})$, we define

$$\text{supp}(A) = \{\{v_0, \dots, v_k\} : t_{[v_0, \dots, v_k]} \neq 0_{\mathbb{A}}\}.$$

In the sequel, it will be useful to work with a “basis” for $C_k(X; \mathbb{A})$. For any $\sigma_0 \in \vec{X}(k)$ and any $a \in \mathbb{A}$, we define $a\mathbb{1}_{\sigma_0} \in C_k(X; \mathbb{Z})$ to be

$$a\mathbb{1}_{\sigma_0} = \sum_{\sigma \in \vec{X}(k)} t_\sigma^{\sigma_0} \sigma$$

with

$$t_\sigma^{\sigma_0} = \begin{cases} a & \sigma = \sigma_0 \\ -a & \sigma = -\sigma_0 \\ 0_{\mathbb{A}} & \text{otherwise} \end{cases}.$$

With this notation, every element in $C_k(X; \mathbb{A})$ is a formal sum of elements of the form $a\mathbb{1}_\sigma, \sigma \in \vec{X}(k), a \in \mathbb{A}$.

The *boundary map* $\partial_k : C_k(X; \mathbb{A}) \rightarrow C_{k-1}(X; \mathbb{A})$ is the map given by

$$\partial_k \left(\sum_{[v_0, \dots, v_k] \in \vec{X}(k)} t_{[v_0, \dots, v_k]} \mathbb{1}_{[v_0, \dots, v_k]} \right) = \sum_{[v_0, \dots, v_k] \in \vec{X}(k)} \sum_{i=0}^k (-1)^i t_{[v_0, \dots, v_k]} \mathbb{1}_{[v_0, \dots, \hat{v}_i, \dots, v_k]}.$$

Definition 2.3 (Cone function). *Let X be a finite simplicial complex. Let $k \in \mathbb{N} \cup \{0, -1\}$ be a constant, $\mathbb{A}$ an Abelian group and v be a vertex of X . A $(k, \mathbb{A})$ -cone function with apex v is a function $\text{Cone}_X : \bigoplus_{j=-1}^k C_j(X; \mathbb{A}) \rightarrow \bigoplus_{j=0}^{k+1} C_j(X; \mathbb{A})$ that fulfills the following conditions:*

1. *For every $a \in \mathbb{A}$, $\text{Cone}_X(a\mathbb{1}_\emptyset) = a\mathbb{1}_{[v]}$.*
2. *For every $-1 \leq j \leq k$, $\text{Cone}_X(C_j(X; \mathbb{A})) \subseteq C_{j+1}(X; \mathbb{A})$ and*

$$\text{Cone}_X|_{C_j(X; \mathbb{A})} : C_j(X; \mathbb{A}) \rightarrow C_{j+1}(X; \mathbb{A})$$

is a homomorphism of $\mathbb{Z}$ -modules.

3. *For every $0 \leq j \leq k$ and every $A \in C_j(X; \mathbb{A})$,*

$$\partial_{j+1} \text{Cone}_X(A) + \text{Cone}_X(\partial_j A) = A.$$

Below, we will refer to this equation as the cone equation.

Definition 2.4 (Radius of a cone). *Let X be an n -dimensional simplicial complex and $-1 \leq k \leq n-1$. Given a $(k, \mathbb{A})$ -cone function Cone_X , we define for every $-1 \leq j \leq k$, the j -th radius of Cone_X to be*

$$\text{Rad}_j(\text{Cone}_X) = \max_{\sigma \in \vec{X}(j), a \in \mathbb{A}} |\text{supp}(\text{Cone}_X(a\mathbb{1}_\sigma))|.$$

Define the $(k, \mathbb{A})$ -th cone radius of X to be

$$\text{Rad}_k(X, \mathbb{A}) = \min\{\text{Rad}_k(\text{Cone}_X) : \text{Cone}_X \text{ is a } (k, \mathbb{A})\text{-cone function}\}.$$

If no $(k, \mathbb{A})$ -cone function of X exists, we define $\text{Rad}_k(X, \mathbb{A}) = \infty$.

Remark 2.5. *Note that in the case where $n = 0$ and X is just a non-empty set of vertices, it holds by definition that $\text{Rad}_{-1}(X, \mathbb{A}) = 1$.*

Example 2.6. Let Γ be a simplicial complex of dimension 1, i.e. a graph. Assume that Γ is connected. Then we can construct a $(0, \mathbb{Z})$ -cone function for Γ in the following way. Pick an arbitrary vertex $v \in \Gamma(0)$ and set $\text{Cone}_\Gamma(\mathbb{1}_\emptyset) = \mathbb{1}_{[v]}$. Furthermore, we define $\text{Cone}_\Gamma(\mathbb{1}_{[v]}) = 0$. In this case we have

$$\partial \text{Cone}_\Gamma(\mathbb{1}_{[v]}) = 0 = \mathbb{1}_{[v]} - \text{Cone}_\Gamma(\mathbb{1}_\emptyset) = \mathbb{1}_{[v]} - \text{Cone}_\Gamma(\partial \mathbb{1}_{[v]})$$

and the cone equation is satisfied in this case.

Next, let $w \in \Gamma(0), w \neq v$. Since Γ is connected, there exists a path $w = w_0, \dots, w_n = v$ from w to v . We define

$$\text{Cone}_\Gamma(\mathbb{1}_{[w]}) = \sum_{i=1}^n \mathbb{1}_{[w_{i-1}, w_i]}.$$

Then

$$\partial \text{Cone}_\Gamma(\mathbb{1}_{[w]}) = \sum_{i=1}^n \mathbb{1}_{[w_{i-1}]} - \mathbb{1}_{[w_i]} = \mathbb{1}_{[w_0]} - \mathbb{1}_{[w_n]} = \mathbb{1}_{[w]} - \mathbb{1}_{[v]} = \mathbb{1}_{[w]} - \text{Cone}_\Gamma(\mathbb{1}_\emptyset) = \mathbb{1}_{[w]} - \text{Cone}_\Gamma(\partial \mathbb{1}_{[w]})$$

which shows that the cone equation is satisfied in all cases, since we defined Cone_Γ on the generators of $C_0(\Gamma, \mathbb{Z})$ and extend it to a homomorphism.

Additionally, note that if we choose the path from w to v to be of minimal length we have

$$\text{Rad}_{-1}(\text{Cone}_\Gamma) = 1, \quad \text{Rad}_0(\text{Cone}_\Gamma) \leq \max_{w \in \Gamma(0)} \text{dist}(v, w)$$

where $\text{dist}(v, w)$ is the minimum length of a path between v, w . We note that $\text{Rad}_0(\text{Cone}_\Gamma)$ is upper-bounded by the diameter of Γ in the usual graph theoretic sense. Moreover, when considering $\text{Rad}_0(\Gamma, \mathbb{Z})$, we minimize over all the choices of apex vertices v and thus $\text{Rad}_0(\Gamma, \mathbb{Z})$ is upper-bounded by the radius of Γ in the usual graph theoretic sense.

We will show that a bound on the cone radii of $\mathbb{Z}$ -cones bounds the cone radii for every finitely generated Abelian group. This result is probably well known to experts, but we could not find a reference for it, so we include a sketch of the proof here (leaving some of the details for the reader).

Proposition 2.7. *Let X be an n -dimensional simplicial complex and $-1 \leq k \leq n-1$. For every finitely generated Abelian group $\mathbb{A}$, it holds that $\text{Rad}_k(X, \mathbb{A}) \leq \text{Rad}_k(X, \mathbb{Z})$.*

Proof. If $(k, \mathbb{Z})$ -cone functions of X do not exist, then $\text{Rad}_k(X, \mathbb{Z}) = \infty$ and the inequality holds trivially. Assume that there exists a $(k, \mathbb{Z})$ -cone function.

By the fundamental theorem of finitely generated Abelian groups, there are prime powers $q_1, \dots, q_l$ and an integer $m \in \mathbb{N} \cup \{0\}$ such that $\mathbb{A} = \left(\bigoplus_{j=1}^l \mathbb{Z}/(q_j \mathbb{Z}) \right) \oplus \mathbb{Z}^m$. We will denote the elements of $\mathbb{A}$ as $m+l$ tuples $(a_1, \dots, a_l, a_{l+1}, \dots, a_{l+m})$, where for every $1 \leq j \leq l$, $a_j \in \mathbb{Z}/(q_j \mathbb{Z})$ and $a_{l+1}, \dots, a_{l+m} \in \mathbb{Z}$. We note that for every $-1 \leq i \leq n$,

$$C_i(X, \mathbb{A}) = \left(\bigoplus_{j=1}^l C_i(X, \mathbb{Z}/(q_j \mathbb{Z})) \right) \oplus C_i(X, \mathbb{Z})^m.$$

Explicitly, for every $\sigma \in \vec{X}(i)$ and every $(a_1, \dots, a_{l+m}) \in \mathbb{A}$, we identify $(a_1, \dots, a_{l+m}) \mathbb{1}_\sigma \in C_i(X, \mathbb{A})$ with $\bigoplus_{j=1}^{l+m} a_j \mathbb{1}_\sigma \in \left(\bigoplus_{j=1}^l C_i(X, \mathbb{Z}/(q_j \mathbb{Z})) \right) \oplus C_i(X, \mathbb{Z})^m$.

For $1 \leq j \leq l$, let $\varphi_j : \mathbb{Z} \rightarrow \mathbb{Z}/(q_j \mathbb{Z})$ be the quotient map. For every $-1 \leq i \leq n$, φ_j extends to a surjective homomorphism $\varphi_j : C_i(X, \mathbb{Z}) \rightarrow C_i(X, \mathbb{Z}/(q_j \mathbb{Z}))$ (this is a homomorphism of $\mathbb{Z}$ -modules).

Let Cone_X be a $(k, \mathbb{Z})$ -cone function. For every $1 \leq j \leq l$, we define the function

$$\text{Cone}_X^j : \bigoplus_{i=-1}^k C_i(X; \mathbb{Z}/(q_j \mathbb{Z})) \rightarrow \bigoplus_{i=0}^{k+1} C_i(X; \mathbb{Z}/(q_j \mathbb{Z}))$$

as follows:

$$\text{Cone}_X^j(\varphi_j(A)) = \varphi_j(\text{Cone}_X(A)), \forall -1 \leq i \leq k, \forall A \in C_i(X; \mathbb{Z}).$$

We note that Cone_X^j is well-defined: Let $A, A' \in C_i(X; \mathbb{Z})$ with $\varphi_j(A) = \varphi_j(A')$. Then $A - A' \in \text{Ker}(\varphi_j)$, which implies that there is $A'' \in C_i(X; \mathbb{Z})$ such that $A - A' = q_j A''$. Thus

$$\begin{aligned} \text{Cone}_X^j(\varphi_j(A)) - \text{Cone}_X^j(\varphi_j(A')) &= \varphi_j(\text{Cone}_X(A)) - \varphi_j(\text{Cone}_X(A')) = \\ \varphi_j(\text{Cone}_X(A - A')) &= \varphi_j(\text{Cone}_X^{\mathbb{Z}}(q_j A'')) = \varphi_j(q_j \text{Cone}_X^{\mathbb{Z}}(A'')) = 0_{\mathbb{Z}/(q_j \mathbb{Z})} \end{aligned}$$

and it follows that $\text{Cone}_X^j(\varphi_j(A)) = \text{Cone}_X^j(\varphi_j(A'))$. Also, by the fact that φ_j is surjective on every $C_i(X, \mathbb{Z}/(q_j\mathbb{Z}))$ it follows that Cone_X^j is defined on every $A \in C_i(X; \mathbb{Z}/(q_j\mathbb{Z}))$.

We leave it to the reader to verify that Cone_X^j is a $(k, \mathbb{Z}/(q_j\mathbb{Z}))$ -cone function and note that for every i , every $\sigma \in \overrightarrow{X}(i)$ and every $a \in \mathbb{Z}/(q_j\mathbb{Z})$ it holds that

$$\text{supp}(\text{Cone}_X^j(a\mathbb{1}_\sigma)) \subseteq \text{supp}(\text{Cone}_X(\mathbb{1}_\sigma)).$$

For every $l+1 \leq j \leq m+l$, we denote $\text{Cone}_X^j = \text{Cone}_X$ (this is a $(k, \mathbb{Z})$ -cone function for every $l+1 \leq j \leq l+m$). For every $-1 \leq i \leq k$ and every

$$\bigoplus_{j=1}^{l+m} A_j \in \left(\bigoplus_{j=1}^l C_i(X, \mathbb{Z}/(q_j\mathbb{Z})) \right) \oplus C_i(X, \mathbb{Z})^m,$$

we define

$$\text{Cone}_X^{\mathbb{A}}\left(\bigoplus_{j=1}^{l+m} A_j\right) = \bigoplus_{j=1}^{l+m} \text{Cone}_X^j(A_j).$$

Extending this map $\mathbb{Z}$ -linearly yields a map

$$\text{Cone}_X^{\mathbb{A}} : \bigoplus_{i=-1}^k \left(\left(\bigoplus_{j=1}^l C_i(X, \mathbb{Z}/(q_j\mathbb{Z})) \right) \oplus C_i(X, \mathbb{Z})^m \right) \rightarrow \bigoplus_{i=0}^{k+1} \left(\left(\bigoplus_{j=1}^l C_i(X, \mathbb{Z}/(q_j\mathbb{Z})) \right) \oplus C_i(X, \mathbb{Z})^m \right),$$

i.e., a map

$$\text{Cone}_X^{\mathbb{A}} : \bigoplus_{i=-1}^k C_i(X, \mathbb{A}) \rightarrow \bigoplus_{i=0}^{k+1} C_i(X, \mathbb{A}).$$

We conclude the proof by observing that $\text{Cone}_X^{\mathbb{A}}$ is a $(k, \mathbb{A})$ -cone function that for every $(a_1, \dots, a_{l+m}) \in \mathbb{A}$, every $-1 \leq i \leq k$ and every $\sigma \in \overrightarrow{X}(i)$, it holds that

$$\text{supp}(\text{Cone}_X^{\mathbb{A}}((a_1, \dots, a_{l+m})\mathbb{1}_\sigma)) \subseteq \text{supp}(\text{Cone}_X(\mathbb{1}_\sigma)).$$

Since Cone_X was an arbitrary $(k, \mathbb{Z})$ -cone function, it follows that $\text{Rad}_k(X, \mathbb{A}) \leq \text{Rad}_k(X, \mathbb{Z})$. $\square$

In light of this Proposition, in the sequel, we will be interested in bounding $\mathbb{Z}$ -radii for cone functions. Below, we will omit the $\mathbb{Z}$ notation when discussing cone functions: we will write k -cone functions in lieu of $(k, \mathbb{Z})$ -cone functions, $\text{Rad}_k(X)$ in lieu of $\text{Rad}_k(X, \mathbb{Z})$ etc.

Observation 2.8. *We observe that for every j , $C_j(X; \mathbb{Z})$ is a free $\mathbb{Z}$ -module and thus the condition in the definition of the cone function that*

$$\text{Cone}_X|_{C_j(X; \mathbb{Z})} : C_j(X; \mathbb{Z}) \rightarrow C_{j+1}(X; \mathbb{Z})$$

is a homomorphism is equivalent to the condition that this map is a $\mathbb{Z}$ -linear map of $\mathbb{Z}$ -modules.

By this linearity, the cone equation is equivalent to the condition:

$$\partial_{j+1} \text{Cone}_X(\mathbb{1}_\sigma) + \text{Cone}_X(\partial_j \mathbb{1}_\sigma) = \mathbb{1}_\sigma, \forall \sigma \in \overrightarrow{X}(j).$$

In other words, in order to verify that a linear function is a cone function, it is enough to check the cone equation on $\mathbb{1}_\sigma$ for every oriented simplex σ .

Moreover, when one defines a cone function, it is enough to define it as a linear function that fulfills the cone equation on $\mathbb{1}_\sigma$ on the subset of oriented simplices that contain at least one orientation for every simplex.

The work of Gromov [Gro10] and subsequent works [KO21, KM19] showed that for symmetric simplicial complexes, a bound on the cone radius gives rise to a bound on the coboundary expansion. For a simplicial complex X , we denote $\text{Aut}(X)$ to be the simplicial automorphisms from X to itself. In [KO21] the following was proven:

Theorem 2.9. *Let $n \geq 1$, $\mathcal{R} > 0$ be constants and $\mathbb{A}$ an Abelian group. For every finite pure n -dimensional simplicial complex, if*

- *the group $\text{Aut}(X)$ acts transitively on $X(n)$,*

and

- *for every $0 \leq k \leq n-1$ it holds that $\text{Rad}_k(X, \mathbb{A}) \leq \mathcal{R}$,*

then for every $0 \leq k \leq n-1$, $h_{\text{cb}}^k(X, \mathbb{A}) \geq \frac{1}{\mathcal{R} \binom{n+1}{k+1}}$.

Combining this Theorem with Proposition 2.7 yields the following:

Theorem 2.10. *Let $n \geq 1$, $\mathcal{R} > 0$ be constants. For every finite pure n -dimensional simplicial complex, if*

- *The group $\text{Aut}(X)$ acts transitively on $X(n)$*

and

- *For every $0 \leq k \leq n-1$ it holds that $\text{Rad}_k(X, \mathbb{Z}) \leq \mathcal{R}$*

Then for every $0 \leq k \leq n-1$ and every finitely generated Abelian group $\mathbb{A}$ it holds that

$$h_{\text{cb}}^k(X, \mathbb{A}) \geq \frac{1}{\mathcal{R} \binom{n+1}{k+1}}.$$

Remark 2.11. *The statement in [KO21] is for $\mathbb{A} = \mathbb{F}_2$, but the proof given there generalizes almost verbatim for a general finitely generated Abelian group.*

2.5 Cone radius of joins

In this section, we denote $\sqcup$ to be a disjoint union. Given two non-empty finite simplicial complexes Y_1 and Y_2 with vertex sets $V(Y_1)$ and $V(Y_2)$, the *join* of Y_1 and Y_2 , denoted $Y_1 * Y_2$, is the simplicial complex with the vertex set $V(Y_1) \sqcup V(Y_2)$ and simplices $\sigma \sqcup \tau$ for every $\sigma \in Y_1$ and $\tau \in Y_2$. We note that $Y_1 * Y_2$ includes all the simplices of the form $\tau = \emptyset \sqcup \tau$ and $\sigma = \sigma \sqcup \emptyset$ for every $\sigma \in Y_1$ and $\tau \in Y_2$.

Below, we will need the following definition (which will also be used in the next section): Given a simplicial complex X and a vertex u in X , we define the following operation; For every $0 \leq j \leq k-1$ and every $[v_0, \dots, v_j] \in \overrightarrow{X}_u(j)$, we define

$$[u, \mathbb{1}_{[v_0, \dots, v_j]}] = \mathbb{1}_{[u, v_0, \dots, v_j]}.$$

Extending this definition linearly, for every $A \in C_j(X_u; \mathbb{Z})$, we define $[u, A] \in C_{j+1}(X; \mathbb{Z})$. We observe that for every $[v_0, \dots, v_j] \in \overrightarrow{X_u}(j)$,

$$\partial_{j+1} \mathbb{1}_{[u, v_0, \dots, v_j]} = \mathbb{1}_{[v_0, \dots, v_j]} - [u, \partial_j \mathbb{1}_{[v_0, \dots, v_j]}]$$

and thus by linearity, for every $A \in C_j(X_u; \mathbb{Z})$,

$$\partial_{j+1} [u, A] = A - [u, \partial_j A]. \quad (1)$$

Proposition 2.12. *Let Y be a non-empty finite simplicial complex and v a vertex that is not in Y . Then for every $k \in \mathbb{N} \cup \{0\}$, there is a k -cone function $\text{Cone}_{\{v\} * Y}$ such that for every $-1 \leq j \leq k$, $\text{Rad}_j(\text{Cone}_{\{v\} * Y}) \leq 1$.*

Proof. Fix $k \in \mathbb{N} \cup \{0\}$ and define a function $\text{Cone}_{\{v\} * Y}$ as follows:

- $\text{Cone}_{\{v\} * Y}(\emptyset) = \mathbb{1}_{[v]}$.
- For every $0 \leq j \leq k$ and every $[v_0, \dots, v_j] \in \overrightarrow{Y}(j)$,

$$\text{Cone}_{\{v\} * Y}(\mathbb{1}_{[v_0, \dots, v_j]}) = [v, \mathbb{1}_{[v_0, \dots, v_j]}] = \mathbb{1}_{[v, v_0, \dots, v_j]}.$$

- For every $0 \leq j \leq k$ and every $\{v_1, \dots, v_j\} \in Y$,

$$\text{Cone}_{\{v\} * Y}(\mathbb{1}_{[v, v_1, \dots, v_j]}) = 0.$$

Extend the function $\text{Cone}_{\{v\} * Y}$ linearly.

We check that the cone equation holds for the above function. In the case of $[v_0, \dots, v_j] \in \overrightarrow{Y}(j)$ this is due to (1). For $\{v_1, \dots, v_j\} \in Y(j-1)$, we note that

$$\begin{aligned} & \partial_{j+1} \text{Cone}_{\{v\} * Y}(\mathbb{1}_{[v, v_1, \dots, v_j]}) + \text{Cone}_{\{v\} * Y}(\partial_j \mathbb{1}_{[v, v_1, \dots, v_j]}) = \\ & 0 + \text{Cone}_{\{v\} * Y}(\mathbb{1}_{[v_1, \dots, v_j]}) - \text{Cone}_{\{v\} * Y}([v, \partial_{j-1}[v_1, \dots, v_j]]) = \\ & 0 + \mathbb{1}_{[v, v_1, \dots, v_j]} + 0 = \mathbb{1}_{[v, v_1, \dots, v_j]} \end{aligned}$$

as needed. Thus, by Observation 2.8, the above function is a k -cone function and it is easy to see that $\text{Rad}_j(\text{Cone}_{\{v\} * Y}) \leq 1$ for every $-1 \leq j \leq k$. $\square$

Remark 2.13. *The reader should note that geometrically $\{v\} * Y$ is a cone (in the usual geometric definition) with apex v and base Y .*

Remark 2.14. *We note that in the above construction, the cone function is well-defined for every k , i.e., it is well-defined even in the cases where $k \geq \dim(\{v\} * Y)$.*

Given a simplicial complex X with $\{v_0, \dots, v_{j_1}\}, \{u_0, \dots, u_{j_2}\} \in X$ such that $\{v_0, \dots, v_{j_1}\} \cap \{u_0, \dots, u_{j_2}\} = \emptyset$ and $\{v_0, \dots, v_{j_1}\} \cup \{u_0, \dots, u_{j_2}\} \in X$, we define for $\sigma = [v_0, \dots, v_{j_1}]$ and $\tau = [u_0, \dots, u_{j_2}]$,

$$[\sigma, \tau] = [v_0, \dots, v_{j_1}, u_0, \dots, u_{j_2}] \in \overrightarrow{X}(j_1 + j_2 + 1)$$

and

$$[\mathbb{1}_\sigma, \mathbb{1}_\tau] = \mathbb{1}_{[\sigma, \tau]}.$$

For $X = Y_1 * Y_2$, we note that for every $\sigma \in \vec{Y}_1(j_1)$ and $\tau \in \vec{Y}_2(j_2)$, $[\mathbb{1}_\sigma, \mathbb{1}_\tau]$ is well-defined. Also note that for σ and τ as above,

$$\partial_{j_1+j_2+1}[\mathbb{1}_\sigma, \mathbb{1}_\tau] = [\partial_{j_1} \mathbb{1}_\sigma, \mathbb{1}_\tau] + (-1)^{j_1+1}[\mathbb{1}_\sigma, \partial_{j_2} \mathbb{1}_\tau].$$

Extending this equation linearly on $C_{j_1}(Y_1; \mathbb{Z})$ and $C_{j_2}(Y_2; \mathbb{Z})$ yields that for every $A_1 \in C_{j_1}(Y_1; \mathbb{Z})$ and every $A_2 \in C_{j_2}(Y_2; \mathbb{Z})$,

$$\partial_{j_1+j_2+1}[A_1, A_2] = [\partial_{j_1} A_1, A_2] + (-1)^{j_1+1}[A_1, \partial_{j_2} A_2]. \quad (2)$$

We note that this equation is actually a generalization of (1) above.

Furthermore, observe that $\text{supp}([A_1, A_2]) = \text{supp}(A_1) \times \text{supp}(A_2)$.

The main idea of the following Proposition is taken from P. Wild's PhD Thesis [Wil22] and we claim no originality here.

Proposition 2.15. *Let Y_1, Y_2 be finite simplicial complexes of dimensions n_1, n_2 respectively. Assume that for $i = 1, 2$ there is an n_i -cone function Cone_{Y_i} and constants $R_j^{(i)} \in \mathbb{N}$, $-1 \leq j \leq n_i - 1$ such that $\text{Rad}_j(\text{Cone}_{Y_i}) \leq R_j^{(i)}$. Then there is an $(n_1 + n_2)$ -cone function $\text{Cone}_{Y_1 * Y_2}$ such that for every $-1 \leq j \leq n_1 + n_2$,*

$$\text{Rad}_j(\text{Cone}_{Y_1 * Y_2}) \leq \begin{cases} \max_{-1 \leq j_1 \leq j} R_{j_1}^{(1)} & j < n_1 \\ \max\{\max_{-1 \leq j_1 \leq n_1-1} R_{j_1}^{(1)}, ((n_1 + 1)R_{n_1-1}^{(1)} + 1)R_{j-n_1-1}^{(2)}\} & n_1 \leq j \leq n_1 + n_2 \end{cases}.$$

Proof. Let v be the apex of Cone_{Y_1} . Define

$$\text{Cone}_{Y_1 * Y_2}(\mathbb{1}_\emptyset) = \mathbb{1}_{[v]}.$$

Let $-1 \leq j_1 \leq n_1$ and $-1 \leq j_2 \leq n_2$ such that $0 \leq j_1 + j_2 + 1 \leq n_1 + n_2$. For $\sigma_i \in \vec{X}(j_i), i = 1, 2$, we define

$$\text{Cone}_{Y_1 * Y_2}(\mathbb{1}_{[\sigma_1, \sigma_2]}) = \begin{cases} [\text{Cone}_{Y_1}(\mathbb{1}_{\sigma_1}), \mathbb{1}_{\sigma_2}] & \dim(\sigma_1) < n_1 \\ (-1)^{n_1+1}[\mathbb{1}_{\sigma_1} - \text{Cone}_{Y_1}(\partial_{n_1} \mathbb{1}_{\sigma_1}), \text{Cone}_{Y_2}(\mathbb{1}_{\sigma_2})] & \dim(\sigma_1) = n_1 \end{cases} \quad (3)$$

and extend it linearly.

We note that by linearity,

$$\text{Cone}_{Y_1 * Y_2}(\partial_{j_1+j_2+1} \mathbb{1}_{[\sigma_1, \sigma_2]}) \stackrel{(2)}{=} \text{Cone}_{Y_1 * Y_2}([\partial_{j_1} \mathbb{1}_{\sigma_1}, \mathbb{1}_{\sigma_2}]) + (-1)^{j_1+1} \text{Cone}_{Y_1 * Y_2}([\mathbb{1}_{\sigma_1}, \partial_{j_2} \mathbb{1}_{\sigma_2}]). \quad (4)$$

We will check that the cone equation is fulfilled by dealing with two separate cases.

Assume first that $\dim(\sigma_1) < n_1$. Then

$$\begin{aligned} & \partial_{j_1+j_2+2} \text{Cone}_{Y_1 * Y_2}(\mathbb{1}_{[\sigma_1, \sigma_2]}) \stackrel{(3)}{=} \partial_{j_1+j_2+2}[\text{Cone}_{Y_1}(\mathbb{1}_{\sigma_1}), \mathbb{1}_{\sigma_2}] \stackrel{(2)}{=} \\ & [\partial_{j_1+1} \text{Cone}_{Y_1}(\mathbb{1}_{\sigma_1}), \mathbb{1}_{\sigma_2}] + (-1)^{j_1+2}[\text{Cone}_{Y_1}(\mathbb{1}_{\sigma_1}), \partial_{j_2} \mathbb{1}_{\sigma_2}] = \\ & [\mathbb{1}_{\sigma_1} - \text{Cone}_{Y_1}(\partial_{j_1} \mathbb{1}_{\sigma_1}), \mathbb{1}_{\sigma_2}] + (-1)^{j_1+2}[\text{Cone}_{Y_1}(\mathbb{1}_{\sigma_1}), \partial_{j_2} \mathbb{1}_{\sigma_2}] = \\ & [\mathbb{1}_{\sigma_1}, \mathbb{1}_{\sigma_2}] - ([\text{Cone}_{Y_1}(\partial_{j_1} \mathbb{1}_{\sigma_1}), \mathbb{1}_{\sigma_2}] + (-1)^{j_1+1}[\text{Cone}_{Y_1}(\mathbb{1}_{\sigma_1}), \partial_{j_2} \mathbb{1}_{\sigma_2}]) \stackrel{(4)}{=} \\ & \mathbb{1}_{[\sigma_1, \sigma_2]} - \text{Cone}_{Y_1 * Y_2}(\partial_{j_1+j_2+1} \mathbb{1}_{[\sigma_1, \sigma_2]}), \end{aligned}$$

thus the cone equation holds as needed.

Next, assume that $\dim(\sigma_1) = n_1$. Then by (4),

$$\begin{aligned} \text{Cone}_{Y_1 * Y_2}(\partial_{n_1+j_2+1} \mathbb{1}_{[\sigma_1, \sigma_2]}) &=^{(4)} \\ \text{Cone}_{Y_1 * Y_2}([\partial_{n_1} \mathbb{1}_{\sigma_1}, \mathbb{1}_{\sigma_2}]) + (-1)^{n_1+1} \text{Cone}_{Y_1 * Y_2}([\mathbb{1}_{\sigma_1}, \partial_{j_2} \mathbb{1}_{\sigma_2}]) &=^{(3)} \\ [\text{Cone}_{Y_1}(\partial_{n_1} \mathbb{1}_{\sigma_1}), \mathbb{1}_{\sigma_2}] + (-1)^{n_1+1} (-1)^{n_1+1} [\mathbb{1}_{\sigma_1} - \text{Cone}_{Y_1}(\partial_{n_1} \mathbb{1}_{\sigma_1}), \text{Cone}_{Y_2}(\partial_{j_2} \mathbb{1}_{\sigma_2})] &= \\ [\text{Cone}_{Y_1}(\partial_{n_1} \mathbb{1}_{\sigma_1}), \mathbb{1}_{\sigma_2}] + [\mathbb{1}_{\sigma_1} - \text{Cone}_{Y_1}(\partial_{n_1} \mathbb{1}_{\sigma_1}), \text{Cone}_{Y_2}(\partial_{j_2} \mathbb{1}_{\sigma_2})], \end{aligned}$$

i.e.,

$$\text{Cone}_{Y_1 * Y_2}(\partial_{n_1+j_2+1} \mathbb{1}_{[\sigma_1, \sigma_2]}) = [\text{Cone}_{Y_1}(\partial_{n_1} \mathbb{1}_{\sigma_1}), \mathbb{1}_{\sigma_2}] + [\mathbb{1}_{\sigma_1} - \text{Cone}_{Y_1}(\partial_{n_1} \mathbb{1}_{\sigma_1}), \text{Cone}_{Y_2}(\partial_{j_2} \mathbb{1}_{\sigma_2})]. \quad (5)$$

We also note that

$$\partial_{n_1} (\mathbb{1}_{\sigma_1} - \text{Cone}_{Y_1}(\partial_{n_1} \mathbb{1}_{\sigma_1})) = \partial_{n_1} \mathbb{1}_{\sigma_1} - (\partial_{n_1} \mathbb{1}_{\sigma_1} - \text{Cone}_{Y_1}(\partial_{n_1-1} \mathbb{1}_{\sigma_1})) = 0. \quad (6)$$

Then

$$\begin{aligned} \partial_{n_1+j_2+2} \text{Cone}_{Y_1 * Y_2}(\mathbb{1}_{[\sigma_1, \sigma_2]}) &=^{(3)} \\ \partial_{n_1+j_2+2} ((-1)^{n_1+1} [\mathbb{1}_{\sigma_1} - \text{Cone}_{Y_1}(\partial_{n_1} \mathbb{1}_{\sigma_1}), \text{Cone}_{Y_2}(\mathbb{1}_{\sigma_2})]) &=^{(2)} \\ (-1)^{n_1+1} [\partial_{n_1} (\mathbb{1}_{\sigma_1} - \text{Cone}_{Y_1}(\partial_{n_1} \mathbb{1}_{\sigma_1})), \text{Cone}_{Y_2}(\mathbb{1}_{\sigma_2})] + \\ (-1)^{n_1+1} (-1)^{n_1+1} [\mathbb{1}_{\sigma_1} - \text{Cone}_{Y_1}(\partial_{n_1} \mathbb{1}_{\sigma_1}), \partial_{j_2+1} \text{Cone}_{Y_2}(\mathbb{1}_{\sigma_2})] &=^{(6)} \\ [\mathbb{1}_{\sigma_1} - \text{Cone}_{Y_1}(\partial_{n_1} \mathbb{1}_{\sigma_1}), \partial_{j_2+1} \text{Cone}_{Y_2}(\mathbb{1}_{\sigma_2})] &= \text{the cone equation for } \text{Cone}_{Y_2} \\ [\mathbb{1}_{\sigma_1} - \text{Cone}_{Y_1}(\partial_{n_1} \mathbb{1}_{\sigma_1}), \mathbb{1}_{\sigma_2} - \text{Cone}_{Y_2}(\partial_{j_2} \mathbb{1}_{\sigma_2})] &= \\ [\mathbb{1}_{\sigma_1} - \text{Cone}_{Y_1}(\partial_{n_1} \mathbb{1}_{\sigma_1}), \mathbb{1}_{\sigma_2}] - [\mathbb{1}_{\sigma_1} - \text{Cone}_{Y_1}(\partial_{n_1} \mathbb{1}_{\sigma_1}), \text{Cone}_{Y_2}(\partial_{j_2} \mathbb{1}_{\sigma_2})] &= \\ [\mathbb{1}_{\sigma_1}, \mathbb{1}_{\sigma_2}] - ([\text{Cone}_{Y_1}(\partial_{n_1} \mathbb{1}_{\sigma_1}), \mathbb{1}_{\sigma_2}] + [\mathbb{1}_{\sigma_1} - \text{Cone}_{Y_1}(\partial_{n_1} \mathbb{1}_{\sigma_1}), \text{Cone}_{Y_2}(\partial_{j_2} \mathbb{1}_{\sigma_2})]) &=^{(5)} \\ \mathbb{1}_{[\sigma_1, \sigma_2]} - \text{Cone}_{Y_1 * Y_2}(\partial_{j_1+j_2+1} \mathbb{1}_{[\sigma_1, \sigma_2]}), \end{aligned}$$

thus the cone equation holds as needed.

Let $0 \leq j \leq n_1 + n_2$. Then every $\sigma \in \vec{X}(j)$ is of the form $\sigma = [\sigma_1, \sigma_2]$ where $\sigma_i \in \vec{X}(j_i), i = 1, 2$ and $j_1 + j_2 + 1 = j$. Note that $j_2 \geq -1$ and thus $j_1 \leq j$. In particular, if $j < n_1$, then $j_1 < n_1$.

For such j_1, j_2 , it holds that if $j_1 < n_1$, then

$$|\text{supp}(\text{Cone}_{Y_1 * Y_2}(\mathbb{1}_{[\sigma_1, \sigma_2]}))| \leq R_{j_1}^{(1)}.$$

If $j_1 = n_1$, then $j_2 = j - n_1 - 1$ and

$$|\text{supp}(\text{Cone}_{Y_1 * Y_2}(\mathbb{1}_{[\sigma_1, \sigma_2]}))| \leq |\text{supp}(\mathbb{1}_{\sigma_1} - \text{Cone}_{Y_1}(\partial_{n_1} \mathbb{1}_{\sigma_1}))| R_{j_2}^{(2)} \leq ((n_1 + 1) R_{n_1-1}^{(1)} + 1) R_{j-n_1-1}^{(2)}.$$

Combining these bounds yields the bound stated above for $\text{Rad}_j(\text{Cone}_{Y_1 * Y_2})$. $\square$

3 Constructing a cone function by adding vertices

The idea is to construct a cone function for a given complex X , by starting with a given subcomplex for which we have a cone function and adding a set of vertices which is “well-behaved”. The exact formalism of this idea is given in Theorem 3.1 that can be thought of as a special version of the Mayer-Vietoris theorem for cone functions.

Theorem 3.1. *Let X be an n -dimensional simplicial complex and X' be a full subcomplex. Assume there is an $(n-1)$ -cone function $\text{Cone}_{X'}$ and constants $R'_k \in \mathbb{N}$, $-1 \leq k \leq n-1$ such that for every $-1 \leq j \leq n-1$, $\text{Rad}_j(\text{Cone}_{X'}) \leq R'_j$. Also let $W \subseteq X(0)$ be a set of vertices such that the following holds:*

1. $W \cap X' = \emptyset$.
2. For every $w_1, w_2 \in W$, $\{w_1, w_2\} \notin X(1)$.
3. For every $w \in W$, $X_w \cap X'$ is a non-empty simplicial complex, i.e., for every $w \in W$ there exists $v \in X'(0)$ such that $\{w, v\} \in X(1)$.
4. There are $(n-2)$ -cone functions $\text{Cone}_{X_w \cap X'}$, $w \in W$ and constants $R''_j \in \mathbb{N}$, $-1 \leq k \leq n-2$ such that for every $w \in W$ and every $-1 \leq j \leq n-2$, $\text{Rad}_j(\text{Cone}_{X_w \cap X'}) \leq R''_j$.

Let $X' \cup W$ be the full subcomplex of X spanned by $X'(0) \cup W$. Then there is an $(n-1)$ -cone function $\text{Cone}_{X' \cup W}$ such that for every $-1 \leq j \leq n-1$, $\text{Rad}_j(\text{Cone}_{X' \cup W}) \leq R''_{j-1}(R'_j + 1)$.

Proof. We will define $\text{Cone}_{X' \cup W}$ by “adding” the cone functions of $X_w \cap X'$, $w \in W$ to $\text{Cone}_{X'}$.

We note that by linearity and Observation 2.8, it is enough to define the cone function on every $\mathbb{1}_{[v_0, \dots, v_j]}$ for every $0 \leq j \leq n-1$ and every $[v_0, \dots, v_j] \in \overrightarrow{X' \cup W}(j)$. We also note that by our assumptions, every $[v_0, \dots, v_j] \in \overrightarrow{X' \cup W}(j)$ is either in $\overrightarrow{X'}(j)$ or that it has a single vertex in W . Thus, it is enough to define the cone function on $\mathbb{1}_{[v_0, \dots, v_j]}$ in the following two cases: either $[v_0, \dots, v_j] \in \overrightarrow{X'}(j)$ or $v_0 = w \in W$ and $[v_1, \dots, v_j] \in \overrightarrow{X_w \cap X'}(j-1)$.

After the preceding discussion, we define an $(n-1)$ -cone function as a linear continuation of the following: First, we define $\text{Cone}_{X' \cup W}(\emptyset) = \mathbb{1}_{[v]}$ where v is the apex of the cone function $\text{Cone}_{X'}$. Second, for every $0 \leq j \leq n-1$ and every $[v_0, \dots, v_j] \in \overrightarrow{X'}(j)$, we define

$$\text{Cone}_{X' \cup W}(\mathbb{1}_{[v_0, \dots, v_j]}) = \text{Cone}_{X'}(\mathbb{1}_{[v_0, \dots, v_j]}).$$

Last, for every $0 \leq j \leq n-1$, every $w \in W$ and every $[v_1, \dots, v_j] \in \overrightarrow{X_w \cap X'}(j-1)$, we define

$$\text{Cone}_{X' \cup W}(\mathbb{1}_{[w, v_1, \dots, v_j]}) = \text{Cone}_{X'}(\text{Cone}_{X_w \cap X'}(\mathbb{1}_{[v_1, \dots, v_j]})) - [w, \text{Cone}_{X_w \cap X'}(\mathbb{1}_{[v_1, \dots, v_j]})].$$

We need to verify that the cone equation holds for every $\mathbb{1}_{[w, v_1, \dots, v_j]}$.

We will start by computing $\text{Cone}_{X' \cup W}(\partial_j \mathbb{1}_{[w, v_1, \dots, v_j]})$:

$$\begin{aligned} & \text{Cone}_{X' \cup W}(\partial_j \mathbb{1}_{[w, v_1, \dots, v_j]}) \stackrel{(1)}{=} \\ & \text{Cone}_{X' \cup W}(\mathbb{1}_{[v_1, \dots, v_j]}) - \text{Cone}_{X' \cup W}([w, \partial_{j-1} \mathbb{1}_{[v_1, \dots, v_j]}]) = \\ & \text{Cone}_{X'}(\mathbb{1}_{[v_1, \dots, v_j]}) - \text{Cone}_{X'}(\text{Cone}_{X_w \cap X'}(\partial_{j-1} \mathbb{1}_{[v_1, \dots, v_j]})) + \\ & [w, \text{Cone}_{X_w \cap X'}(\partial_{j-1} \mathbb{1}_{[v_1, \dots, v_j]})] \stackrel{\text{the cone equation for } \text{Cone}_{X_w \cap X'}}{=} \\ & \text{Cone}_{X'}(\mathbb{1}_{[v_1, \dots, v_j]}) - \text{Cone}_{X'}(\mathbb{1}_{[v_1, \dots, v_j]}) + \text{Cone}_{X'}(\partial_j \text{Cone}_{X_w \cap X'}(\mathbb{1}_{[v_1, \dots, v_j]})) + \\ & [w, \text{Cone}_{X_w \cap X'}(\partial_{j-1} \mathbb{1}_{[v_1, \dots, v_j]})] = \\ & \text{Cone}_{X'}(\partial_j \text{Cone}_{X_w \cap X'}(\mathbb{1}_{[v_1, \dots, v_j]})) + [w, \text{Cone}_{X_w \cap X'}(\partial_{j-1} \mathbb{1}_{[v_1, \dots, v_j]})]. \end{aligned}$$

Using this computation, we will verify the cone equation for $\text{Cone}_{X' \cup W}$ (below we use the cone equation for $\text{Cone}_{X'}$ and $\text{Cone}_{X_w \cap X'}$ without noting it):

$$\begin{aligned}
& \partial_{j+1} \text{Cone}_{X' \cup W}(\mathbb{1}_{[w, v_1, \dots, v_j]}) = \\
& \partial_{j+1} \text{Cone}_{X'}(\text{Cone}_{X_w \cap X'}(\mathbb{1}_{[v_1, \dots, v_j]})) - \partial_{j+1}[w, \text{Cone}_{X_w \cap X'}(\mathbb{1}_{[v_1, \dots, v_j]})] = \\
& - \text{Cone}_{X'}(\partial_j \text{Cone}_{X_w \cap X'}(\mathbb{1}_{[v_1, \dots, v_j]})) + \text{Cone}_{X_w \cap X'}(\mathbb{1}_{[v_1, \dots, v_j]}) - \\
& \left(\text{Cone}_{X_w \cap X'}(\mathbb{1}_{[v_1, \dots, v_j]}) - [w, \partial_j \text{Cone}_{X_w \cap X'}(\mathbb{1}_{[v_1, \dots, v_j]})] \right) = \\
& - \text{Cone}_{X'}(\partial_j \text{Cone}_{X_w \cap X'}(\mathbb{1}_{[v_1, \dots, v_j]})) - [w, \text{Cone}_{X_w \cap X'}(\partial_{j-1} \mathbb{1}_{[v_1, \dots, v_j]})] + [w, \mathbb{1}_{[v_1, \dots, v_j]}] = \\
& - \text{Cone}_{X' \cup W}(\partial_j \mathbb{1}_{[w, v_1, \dots, v_j]}) + \mathbb{1}_{[w, v_1, \dots, v_j]}
\end{aligned}$$

as needed.

In order to conclude the proof, we will show that $\text{Rad}_j^{n-1}(\text{Cone}_{X' \cup W}) \leq R''_{j-1}(R'_j + 1)$. For $[v_0, \dots, v_j] \in \overrightarrow{X'}(j)$,

$$|\text{supp}(\text{Cone}_{X' \cup W}(\mathbb{1}_{[v_0, \dots, v_j]}))| = |\text{supp}(\text{Cone}_{X'}(\mathbb{1}_{[v_0, \dots, v_j]}))| \leq R'_j \leq R''_{j-1}(R'_j + 1).$$

For $[w, v_1, \dots, v_j] \in \overrightarrow{X' \cup W}(j)$ where $w \in W$,

$$\begin{aligned}
& |\text{supp}(\text{Cone}_{X' \cup W}(\mathbb{1}_{[w, v_0, \dots, v_j]}))| \leq \\
& |\text{supp}(\text{Cone}_{X'}(\text{Cone}_{X_w \cap X'}(\mathbb{1}_{[v_1, \dots, v_j]})))| + |\text{supp}([w, \text{Cone}_{X_w \cap X'}(\mathbb{1}_{[v_1, \dots, v_j]})])| \leq \\
& R'_j R''_{j-1} + R''_{j-1} = R''_{j-1}(R'_j + 1)
\end{aligned}$$

as needed. □

4 The Kac–Moody–Steinberg complexes

In this section, we describe the families of simplicial complexes, for which we will prove that they give rise to cosystolic expanders. The main source, what we call KMS complexes, are infinite families of coset complexes over finite quotients of so-called Kac–Moody–Steinberg groups. They were introduced in [GdPVB24], where it was shown that they are local spectral high-dimensional expanders. Our methods also apply to the complexes introduced in [DLZ23], since the links of these complexes are isomorphic to links of certain KMS complexes.

4.1 Root systems and Chevalley groups

We start by recalling some facts about generalized Cartan matrices, the associated root systems and Chevalley groups. These are the building blocks of KMS groups.

Most of the structure of a KMS group is encoded in its generalized Cartan matrix, which also gives rise to a Dynkin diagram and a root system.

Definition 4.1. *A generalized Cartan matrix (GCM) is a matrix $A = (A_{ij})_{i,j \in I} \in \text{Mat}_n(\mathbb{Z})$ such that $A_{ii} = 2$ for all $i \in I$, $A_{ij} \leq 0$ for all $i \neq j \in I$ and $A_{ij} = 0 \iff A_{ji} = 0$. We call $\text{rank}(A) := |I|$ the rank of A .*

Every GCM $A = (A_{i,j})_{i,j \in I}$ corresponds to a Dynkin diagram in the following way. As vertex set, we take the index set I and two vertices i, j are connected by $|A_{i,j}|$ edges if $A_{i,j}A_{j,i} \leq 4$ and $|A_{i,j}| \geq |A_{j,i}|$, and these edges are equipped with an arrow pointing towards i if $|A_{i,j}| > 1$ (a bi-directed edge is possible). If $A_{i,j}A_{j,i} > 4$, the vertices i and j are connected by a bold-faced labelled edge labelled with the ordered pair of integers $|A_{i,j}|, |A_{j,i}|$.

A GCM is irreducible if there exists no non-trivial partition $I = I_1 \cup I_2$ such that $A_{i_1, i_2} = 0$ for all $i_1 \in I_1, i_2 \in I_2$.

Definition 4.2. Let A be a $(d+1) \times (d+1)$ generalized Cartan matrix with index set $I = \{0, \dots, d\}$, and let $J \subseteq I$.

- (i) A gives rise to a Coxeter matrix M of the same size (different GCMs can give rise to the same Coxeter matrix) which in turn defines a (finite or infinite) Coxeter group (see Section 5.1).
- (ii) The subset J is called spherical if $A_J = (A_{i,j})_{i,j \in J}$ is of spherical type, meaning that the associated Coxeter group is finite (see e.g. [Bou08, Chapter 6.4.1]). Given $n \geq 2$, A is n -spherical if every subset $J \subseteq I$ of size n is spherical.
- (iii) We denote by Q_A the set of spherical subsets of I associated to the generalized Cartan matrix A .
- (iv) A generalized Cartan matrix A is purely n -spherical if every spherical subset $J \subseteq I$ is contained in a spherical subset of size n . (In particular, no set of size $n+1$ is spherical for a purely n -spherical generalized Cartan matrix.) Note that any such GCM is irreducible (if it could be decomposed, each irreducible component needs to be spherical and thus the whole GCM would be spherical).
- (v) Analogously, we introduce the following terminology and call a GCM n -classical if each irreducible factor of $A_J = (A_{i,j})_{i,j \in J}$ is of classical type A_k, B_k, C_k, D_k (see e.g. [Bou08, Chapter 6.4.1]) for each $J \subseteq I$ with $|J| \leq n$.
- (vi) To a generalized Cartan matrix A we can associate the following sets:
 - a set of simple roots $\Pi = \{\alpha_i \mid i \in I\}$,
 - a set of real roots $\Phi \subseteq \bigoplus_{i \in I} \mathbb{Z}\alpha_i$,
 - two sets, one of positive and one of negative real roots $\Phi^+ = \bigoplus_{i \in I} \mathbb{N}\alpha_i \cap \Phi, \Phi^- = -\Phi^+$.

Note that $\Phi = \Phi^+ \sqcup \Phi^-$. More details can be found e.g. in [Mar18, Chapter 3.5].

Remark 4.3. (i) Irreducible spherical diagrams have been classified. All possible diagrams are given in Figure 1.

- (ii) All rank $n+1$, n -spherical diagrams have been classified and are precisely the affine diagrams and the so-called compact hyperbolic diagrams. The latter have rank at most 5, where the diagram of rank 5 is exactly the one that we will have to exclude to get n -classicality below. See e.g. [Hum90, Chapter 6.9] for the classification result.

The non-spherical, rank $n+1$, n -classical diagrams, with $n \geq 3$ are precisely the non-spherical rank $n+1$ diagrams which are n -spherical but not one of the following

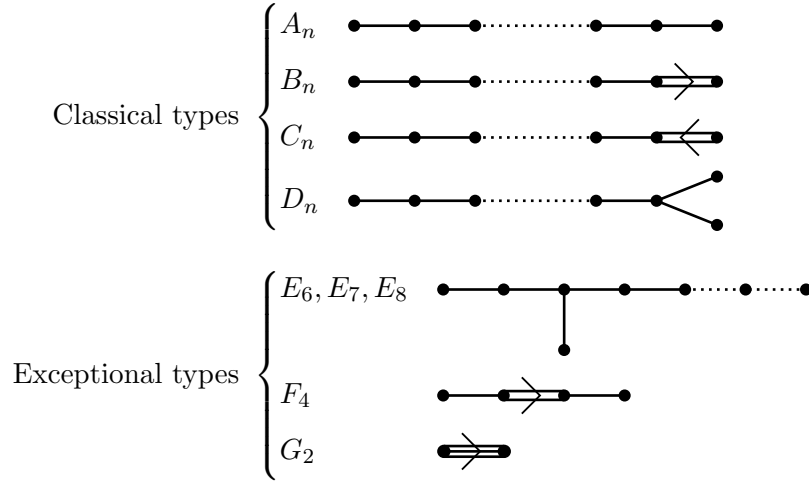
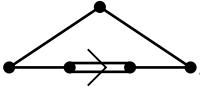


Figure 1: The Dynkin diagrams of irreducible spherical root systems

- affine diagrams $\tilde{E}_6, \tilde{E}_7, \tilde{E}_8, \tilde{F}_4, \tilde{G}_2$,
- twisted affine diagrams $E_6^{(2)}, D_4^{(3)}$ (in the notation of [Mar18, Table 5.1])

- the compact hyperbolic diagram .

Definition 4.4. Corresponding to any irreducible, spherical GCM $\check{A}$ with root system $\check{\Phi}$ of rank at least 2, and any finite field K , there is an associated universal (or simply connected) Chevalley group, denoted $\text{Chev}_{\check{A}}(K)$. Abstractly, it is generated by symbols $x_{\alpha}(s)$ for $\alpha \in \check{\Phi}$ and $s \in K$, subject to the relations

$$\begin{aligned}
 x_{\alpha}(s)x_{\alpha}(u) &= x_{\alpha}(s+u) \\
 [x_{\alpha}(s), x_{\beta}(u)] &= \prod_{i,j>0} x_{i\alpha+j\beta} \left(C_{ij}^{\alpha,\beta} s^i u^j \right) \quad (\text{for } \alpha + \beta \neq 0) \\
 h_{\alpha}(s)h_{\alpha}(u) &= h_{\alpha}(su) \quad (\text{for } s, u \neq 0), \\
 \text{where } h_{\alpha}(s) &= n_{\alpha}(s)n_{\alpha}(-1) \\
 \text{and } n_{\alpha}(s) &= x_{\alpha}(s)x_{-\alpha}(-s^{-1})x_{\alpha}(s).
 \end{aligned}$$

Note that the $C_{ij}^{\alpha,\beta}$ are integers called structure constants that can be found in [Car89].

Remark 4.5. Let K be a field and let $K[t]$ denote the polynomial ring in one variable over K . The simply-connected Chevalley group $\text{Chev}_{\check{A}}(K[t])$ is, similar to the case of a Chevalley group over a field, generated by elements $x_{\alpha}(s)$ for $\alpha \in \check{\Phi}, s \in K[t]$ that satisfy the relations above, where for the third relation we have to add the extra assumption that u, s are invertible in $K[t]$. This can be found e.g. in [Reh75].

4.2 Kac–Moody–Steinberg groups

We can now put the ingredients together to define what a Kac–Moody–Steinberg group is and present some of its properties. For more details, see [GdPVB24, Chapter 3] and references therein.

Definition 4.6. *Let K be a field. Let $A = (A_{i,j})_{i,j \in I}$ be a GCM over the index set I which is 2-spherical. Let $Q_A = \{J \subseteq I \mid A_J := (A_{i,j})_{i,j \in J} \text{ is spherical}\}$. Let Φ be the root system associated to A with simple roots $\Pi = \{\alpha_i \mid i \in I\}$. For $J \in Q_A$ set*

$$U_J := \langle x_{\alpha_i}(s) \mid s \in K, i \in J \rangle \leq \text{Chev}_{A_J}(K)$$

the group generated by all simple roots in the Chevalley group of type A_J over K . Note that if $L \subset J$ then we have a natural inclusion $U_L \hookrightarrow U_J$ by sending $x_{\alpha_i}(s) \in U_L$ to the same generator in U_J . The KMS group of type A over K is defined as the free product of the $U_J, J \in Q_A$ modulo the natural inclusions:

$$\mathcal{U}_A(K) = \bigstar_{J \in Q_A} U_J / (U_L \hookrightarrow U_J, L \subseteq J).$$

Note that $U_J \hookrightarrow \mathcal{U}_A(K)$ and we will denote the image of U_J in $\mathcal{U}_A(K)$ again by U_J . These subgroups are called the local groups of $\mathcal{U}_A(K)$. For $i, j \in I$ we will write $U_i := U_{\{i\}}$ and $U_{i,j} = U_{\{i,j\}}$.

Remark 4.7. *For two roots $\alpha, \beta \in \Phi$ we write $]\alpha, \beta[_{\mathbb{N}} = \{n_1\alpha + n_2\beta \in \Phi \mid n_1, n_2 \in \mathbb{N}^*\}$ and $[\alpha, \beta]_{\mathbb{N}} =]\alpha, \beta[_{\mathbb{N}} \cup \{\alpha, \beta\}$.*

We have the following abstract presentation for the KMS groups:

$$\mathcal{U}_A(K) = \left\langle u_{\beta}(t) \text{ for } t \in K, i, j \in I, \beta \in [\alpha_i, \alpha_j]_{\mathbb{N}} \mid \mathcal{R} \right\rangle$$

where the set of relations $\mathcal{R}$ is defined as:

for all $i, j \in I$, $\{\alpha, \beta\} \subseteq [\alpha_i, \alpha_j]_{\mathbb{N}}$, $t, s \in K$:

$$\begin{aligned} u_{\alpha}(t)u_{\alpha}(s) &= u_{\alpha}(t+s) \\ [u_{\alpha}(t), u_{\beta}(s)] &= \prod_{\gamma=k\alpha+l\beta \in]\alpha, \beta[_{\mathbb{N}}} u_{\gamma} \left(C_{k,l}^{\alpha,\beta} t^k s^l \right). \end{aligned}$$

The constants $C_{k,l}^{\alpha,\beta}$ are the same structure constants as in the presentation of the Chevalley group.

Since A is 2-spherical, the subgroups U_{β} , for $\beta \in [\alpha_i, \alpha_j]$, are contained in the group generated by the root groups $U_{\alpha_i}, U_{\alpha_j}$ (see [Abr96, Proposition 7]).

4.3 KMS complexes

In this section, we recall some facts about coset complexes and put everything together to define KMS complexes.

KMS complexes are coset complexes. More details on coset complexes can be found for example in [KO23], and we will only give a brief overview.

Definition 4.8. *Let G be a group and $\mathcal{H} = (H_0, \dots, H_n)$ be a family of subgroups of G . Then the coset complex $\text{CC}(G; (H_0, \dots, H_n))$ is defined to be the simplicial complex*

- with vertex set $\bigsqcup_{i=0}^n G/H_i$
- such that a set of vertices $\{g_1 H_{i_1}, \dots, g_k H_{i_k}\}$ forms a $(k-1)$ -simplex in $\mathcal{CC}(G; \mathcal{H})$ if and only if $\bigcap_{j=1}^k g_j H_{i_j} \neq \emptyset$.

Note that the set of vertices is partitioned into $n+1$ subsets G/H_i for $i = 0, \dots, n$ such that for any simplex $\sigma \in \mathcal{CC}(G; \mathcal{H})$ we have $|\sigma \cap G/H_i| \leq 1$ for all i . Hence, we say that $\mathcal{CC}(G; \mathcal{H})$ is a $(n+1)$ -partite simplicial complex.

Definition 4.9. The type of a simplex $\{g_1 H_{i_1}, \dots, g_k H_{i_k}\}$ in $\mathcal{CC}(G; \mathcal{H})$ is the set of indices $\{i_1, \dots, i_k\} \subseteq \{0, \dots, n\}$. Given a type $\emptyset \neq T \subseteq \{0, \dots, n\}$ we write

$$H_T := \bigcap_{i \in T} H_i \quad \text{and set} \quad H_\emptyset := \langle H_0, \dots, H_n \rangle \leq G.$$

The following well-known fact will be very useful.

Proposition 4.10. Let σ be a face in $\mathcal{CC}(G; \mathcal{H})$ of type $T \neq \emptyset$. Then the link of σ is isomorphic to the coset complex $\mathcal{CC}(H_T, (H_{T \cup \{i\}} : i \notin T))$.

KMS complexes are coset complexes over certain finite quotients of KMS groups. The precise definition is as follows.

Definition 4.11. Let A be a GCM over the index set I which is $(|I|-1)$ -spherical but non-spherical. Let K be a finite field of size $q \geq 4$. Let $\phi : \mathcal{U}_A(K) \rightarrow G$ be a finite quotient of $\mathcal{U}_A(K)$ such that

1. $\phi|_{U_J}$ is injective for all $J \in Q_A$ (i.e. ϕ is injective on the local groups),
2. $\phi(U_J \cap U_L) = \phi(U_J) \cap \phi(U_L)$ for all $J, L \in Q_A$.

Then we set

$$X = \mathcal{CC}(G; (\phi(U_{I \setminus \{j\}}))_{j \in I})$$

and call X a KMS complex of type A over K .

Theorem 4.12. [GdPVB24, Theorem 4.3] Let A be a GCM over the index set I which is non-spherical but $|I|-1$ -spherical. Let K be a finite field of size $q \geq 4$. Let $\phi_i : \mathcal{U}_A(K) \rightarrow G_i, i \in I$ be a family of finite quotients of $\mathcal{U}_A(K)$ such that $X_i = \mathcal{CC}(G_i, (\phi_i(U_{I \setminus \{j\}}))_{j \in I}), i \in \mathbb{N}$ is a family of KMS complexes and $|G_i| \xrightarrow{i \rightarrow \infty} \infty$. If q is such that $\sqrt{\frac{3}{q}} \leq \frac{1}{|I|-1}$ then the family $(X_i)_{i \in \mathbb{N}}$ is a family of bounded degree local spectral expanders.

Moreover, if $q \gg (|I|-1)^2$, then the family $(X_i)_{i \in \mathbb{N}}$ is a family of bounded degree $\frac{2}{\sqrt{q}}$ -local spectral expanders.

Example 4.13. The following is an example of an n -dimensional, n -classical KMS complex. Consider $\text{SL}_{n+1}(\mathbb{F}_q[t])$ where $n \geq 2$ and q is an odd prime power such that $q \geq 5$. For $1 \leq i, j \leq n+1, i \neq j$ and $f \in \mathbb{F}_q[t]$, we denote $e_{i,j}(f) \in \text{SL}_{n+1}(\mathbb{F}_q[t])$ to be the matrix with 1's along the main diagonal, f in the (i, j) -th entry and 0's in all other entries. Let

$$B = \{e_{i,i+1}(1); i = 1, \dots, n\} \cup \{e_{n+1,1}(t)\}$$

and let $\mathcal{U}(\mathbb{F}_q) < \mathrm{SL}_{n+1}(\mathbb{F}_q[t])$ be the subgroup generated by B , i.e.

$$\mathcal{U}(\mathbb{F}_q) = \langle e_{1,2}(1), e_{2,3}(1), \dots, e_{n,n+1}(1), e_{n+1,1}(t) \rangle.$$

We define subgroups $H_0, \dots, H_n < \mathcal{U}(\mathbb{F}_q)$ as

$$\begin{aligned} H_0 &= \langle e_{1,2}(1), e_{2,3}(1), \dots, e_{n,n+1}(1) \rangle = \langle B \setminus \{e_{n+1,1}(t)\} \rangle, \\ H_1 &= \langle e_{2,3}(1), \dots, e_{n,n+1}(1), e_{n+1,1}(t) \rangle = \langle B \setminus \{e_{1,2}(1)\} \rangle, \\ &\vdots \\ H_i &= \langle B \setminus \{e_{i,i+1}\} \rangle, \\ &\vdots \\ H_n &= \langle e_{1,2}(1), e_{2,3}(1), \dots, e_{n-1,n}(1), e_{n+1,1}(t) \rangle, \end{aligned}$$

and denote $\mathcal{H} = (H_0, \dots, H_n)$.

Let $f \in \mathbb{F}_q[t]$ be an irreducible polynomial of degree $s > 1$. Denote $\phi_f : \mathrm{SL}_{n+1}(\mathbb{F}_q[t]) \rightarrow \mathrm{SL}_{n+1}(\mathbb{F}_q[t]/(f)) \cong \mathrm{SL}_{n+1}(\mathbb{F}_{q^s})$ defined as follows: For every matrix $A \in \mathrm{SL}_{n+1}(\mathbb{F}_q[t])$, $A = (A_{i,j})$ where $A_{i,j} \in \mathbb{F}_q[t]$ define

$$\phi_f(A) = (A_{i,j} + (f)).$$

Restricting ϕ_f to $\mathcal{U}(\mathbb{F}_q)$, we get $G_f = \phi_f(\mathcal{U}(\mathbb{F}_q))$ which is a finite quotient of $\mathcal{U}(\mathbb{F}_q)$. It is proven in [GdPVB24], that ϕ_f is injective on $H_0, \dots, H_n$. The KMS complex of $G_f = \phi_f(\mathcal{U}(\mathbb{F}_q))$ is defined to be the coset complex $X_f = \mathcal{CC}(G_f, \mathcal{H}_f)$, where $\mathcal{H}_f = (\phi_f(H_0), \dots, \phi_f(H_n))$.

Remark 4.14. *Similar examples of n -classical KMS complexes can be constructed for all Chevalley groups of classical type, see [GdPVB24, Chapter 5] for more details.*

5 Describing classical buildings and their opposite complexes

In the first part of this section, we briefly discuss Coxeter complexes. We then describe buildings and their opposite complexes from a group theoretic view point. In this setting, we show that the links of KMS complexes are opposite complexes of spherical buildings. In the third part, we describe a more geometric view point using flag complexes. This context will be used to show that opposite complexes are coboundary expanders.

5.1 Coxeter complex

To each Dynkin diagram with set of vertices I as in Theorem 4.1 we can associate a group, called a Coxeter group, in the following way:

$$\langle s_i; i \in I \mid s_i^2 = 1, (s_i s_j)^{m_{ij}} = 1; i, j \in I, i \neq j \rangle$$

where

$$\begin{array}{ll} \begin{array}{c} i \quad j \\ \bullet \quad \bullet \end{array} \Rightarrow m_{ij} = 2, & \begin{array}{c} i \quad j \\ \bullet \text{---} \bullet \end{array} \Rightarrow m_{ij} = 3 \\ \begin{array}{c} i \quad j \\ \bullet \text{---} \bullet \\ \diagup \quad \diagdown \end{array} \Rightarrow m_{ij} = 4, & \begin{array}{c} i \quad j \\ \bullet \text{---} \bullet \\ \diagup \quad \diagdown \end{array} \Rightarrow m_{ij} = 6 \end{array}$$

and $m_{ij} = +\infty$ if i, j are connected by a bold-faced labelled edge, meaning that in this case there is no relation between s_i and s_j . Note that m_{ij} is defined to be symmetric in i, j . This implies that the root systems of type B_n and C_n give rise to the same Coxeter group. The Coxeter complex associated to the Coxeter group W with set of generators S can be viewed as the coset complex

$$\mathcal{CC}(W, (\langle S \setminus \{t\} \rangle)_{t \in S}).$$

More details can be found for example in [AB08].

5.2 Buildings and BN-pairs

One way to define a building is as a simplicial complex with a family of subcomplexes called apartments such that each apartment is isomorphic to the same Coxeter complex. The type of the Coxeter complex, which can be described by its Coxeter matrix, is also what we call the type of the building. Any two simplices of the building have to be contained in a common apartment and any two apartments are isomorphic with an isomorphism that acts as the identity on the intersection.

One way to construct buildings is via groups with BN-pair (and for thick, irreducible buildings of dimension 2 and larger, all buildings can be constructed that way, as was proven by Jacques Tits in [Tit74]). Note that if we construct a building from a BN-pair of a Chevalley group with underlying root system of some type X , then the building will have the same type. In particular, since type B_n and C_n give rise to the same Coxeter complex, they also determine the same type of building.

Given a group G , a BN-pair of G is a pair of subgroups $B, N \leq G$ satisfying certain properties, see [AB08, Definition 6.55]. In particular, $G = BN$ and setting $T = B \cap N$ we have that $W = N/T$ admits a set of generators S , such that (W, S) is a Coxeter system. W is also called the Weyl group of the BN-pair. To describe the building, we need to define the parabolic subgroups $P_J := B \langle J \rangle B$ for $J \subset S$ (here $\langle J \rangle$ is the subgroup generated by the elements of J inside W). The building can then be described as the following coset complex:

$$\Delta(G, B) := \mathcal{CC}(G; (P_{S \setminus \{j\}})_{j \in S}).$$

Since $P_J \cap P_L = P_{J \cap L}$ for any $J, L \subseteq S$ we can see $\Delta(G, B)$ also as the poset $\{gP_J \mid g \in G, J \subseteq S\}$ ordered by reversed inclusion. Then the simplex $\{gP_{S \setminus \{j\}} \mid j \in J\} \in \mathcal{CC}(G; (P_{S \setminus \{j\}})_{j \in S})$ corresponds to $gP_{S \setminus J}$ for $J \subseteq S, g \in G$. In particular, the set of maximal simplices, called chambers, corresponds to G/B .

If the Weyl group is finite, we call the building spherical. This coincides with our definition of calling a GCM/root system spherical. A finite Weyl group has a unique longest element denoted by w_0 , where longest means that is generated by the largest number of generators, counted with multiplicities.

Two chambers $gB, hB, g, h \in G$ are called opposite if and only if $g^{-1}h \in Bw_0B$, denoted by $gB \text{ op } hB$.

Given a simplex $\sigma \in \Delta(G, B)$, the complex opposite σ is defined as follows:

$$\Delta^0(\sigma) := \{\tau \in \Delta(G, B) \mid \text{there exist chambers } c, d \in \Delta(G, B) \text{ such that } \tau \subseteq c, \sigma \subseteq d, c \text{ op } d\}.$$

To see the connection between opposite complexes and the links of KMS complexes, note that every Chevalley group has a BN-pair in the following way. Let $A = (A_{ij})_{i,j \in I}$ be a spherical GCM

of rank at least 2, K a field and $G = \text{Chev}_A(K) = \langle x_\alpha(\lambda) \mid \alpha \in \Phi, \lambda \in K \rangle$, using the notation of Theorem 4.4. We set

$$\begin{aligned} U_\alpha &:= \langle x_\alpha(\lambda) \mid \lambda \in K \rangle, \alpha \in \Phi, & U^+ &:= \langle U_\alpha \mid \alpha \in \Phi^+ \rangle, & T &:= \langle h_\alpha(\lambda) \mid \alpha \in \Phi, \lambda \in K \rangle, \\ B &:= U^+T, & N &:= \langle T, n_{-\alpha_i}(-\lambda^{-1}) \mid i \in I, \lambda \in K^* \rangle. \end{aligned}$$

For example [AB08, Theorem 7.115] shows that this is indeed a BN-pair.

Let $C_0 = 1B$ denote the fundamental chamber of $\Delta(G, B)$. Recall that w_0 denotes the longest element on the Weyl group W . It satisfies $Bw_0B = U^+w_0B$. We can describe the opposite complex in this set-up as

$$\begin{aligned} \Delta^0(C_0) &= \{gP_J \mid g \in G, J \subseteq S : \exists h \in U^+w_0B : gP_J \supseteq hB\} \\ &= \{gP_J \mid \exists h \in U^+w_0B : h \in gP_J\} \\ &= \{hP_J \mid h \in U^+w_0B\} \\ &= \{uw_0P_J \mid u \in U^+\}. \end{aligned}$$

This leads to the following proposition.

Proposition 5.1. [GdPVB24, Chapter 3.2] *Let G be a Chevalley group with BN-pair as described above. Then*

$$\mathcal{CC}(U^+, (U_{I \setminus \{i\}})_{i \in I}) \cong \Delta^0(C_0).$$

Since we assume that A is 2-spherical and $|K| \geq 4$ we have $U^+ = \langle U_\alpha \mid \alpha \in \Phi^+ \rangle = \langle U_{\alpha_i} \mid i \in I \rangle$ (see e.g. the comment before [GdPVB24, Lemma 3.7]). Combining this with the above proposition and Theorem 4.10 gives the following corollary.

Corollary 5.2. *Let X be a KMS complex, and $\sigma \in X$ a face of dimension less than or equal to $\dim X - 2$ of type $\emptyset \neq T \subset I$. Then*

$$\text{lk}_X(\sigma) \cong \Delta^0(C_0)$$

where Δ is the building obtained from the BN-pair structure of the Chevalley group of type $A_{I \setminus T}$.

5.3 Geometric constructions of buildings

In the following section, we describe how buildings of type A_n , C_n and D_n can be constructed as flag complexes of certain sets equipped with an incidence relation. We furthermore describe the opposite complexes in this set-up. The general framework is as follows.

Definition 5.3. *Let X be a set and let $I \subseteq X \times X$ be a reflexive and symmetric relation. For $a, b \in X$ we write aIb if and only if $(a, b) \in I$ and call a and b incident. The structure gives rise to the following simplicial complex, called the flag complex of X :*

$$\text{Flag}(X) = \{\sigma \subseteq X \mid \forall a, b \in \sigma : aIb\}.$$

Remark 5.4. *We note that by definition, $\text{Flag}(X)$ is always a clique complex.*

5.3.1 Buildings of type A_n

Let $n \in \mathbb{N}, n \geq 1$. Any building of type A_n can be described in the following way, see e.g. [Tit74]. Let K be a field or skew field, V an $(n+2)$ -dimensional vector space over K . Set $X := X(V) := \{0 < U < V \mid U \text{ is a } K\text{-subspace of } V\}$. Two subspaces $U, W \in X$ are called incident if $U \subseteq W$ or $W \subseteq U$. Then $\Delta = \text{Flag } X$ is a building of type A_{n+1} .

To describe the opposite complex in this context, we need the following notation, see [Abr96, Definition 9].

Definition 5.5. 1. Two subspaces U, W of V are called transversal (in V) if $U \cap W = 0$ or $U + W = V$. If this is the case, we write $U \pitchfork_V W$ or simply $U \pitchfork W$.

2. Let $\mathcal{E}$ be a set of subspaces of V . Then we write $U \pitchfork \mathcal{E}$ if $U \pitchfork E$ for all $E \in \mathcal{E}$. We set

$$X_{\mathcal{E}}(V) := \{U \in X \mid U \pitchfork \mathcal{E}\} \quad T_{\mathcal{E}}(V) = \text{Flag } X_{\mathcal{E}}(V).$$

Note that if $0 < U \leq W < V$ then U and W are not transversal in V . We get the following result.

Proposition 5.6. [Abr96, Corollary 12] For any simplex $\sigma = \{E_1 < \dots < E_r\} \in \Delta$ set $\mathcal{E}(\sigma) = \{E_i \mid 1 \leq i \leq r\}$. Then

$$\Delta^0(\sigma) = T_{\mathcal{E}(\sigma)}(V).$$

Definition 5.7. We denote by $\mathbf{C}_A$ the class of all simplicial complexes $T_{\mathcal{E}}(V)$ where $\dim(T_{\mathcal{E}}(V)) = n \geq 0$, K is a (skew) field, V an $(n+2)$ -dimensional vector space over K , $\mathcal{E}$ is a finite set of subspaces of V such that if we set $e_j = |\{E \in \mathcal{E} \mid \dim E = j\}|$ we have $\sum_{j=1}^{n+1} \binom{n}{j-1} e_j \leq |K|$. Furthermore, $T_{\mathcal{E}}(V)$ is defined as described in Theorem 5.5.

5.3.2 Hermitian and pseudo-quadratic forms

In this part, we recall facts about hermitian and pseudo-quadratic forms that are needed to describe the buildings of type C_n and D_n . We follow [Abr96, Chapter 5], who in turn mostly follows [Tit74].

Let K be a skew field, $\sigma : K \rightarrow K, a \mapsto a^\sigma$ be an involution, i.e. an anti-automorphism of K such that $\sigma^2 = \text{id}_K$. Let $\varepsilon \in \{1, -1\} \subset K$. If $\sigma \neq \text{id}_K$ then we require $\varepsilon = -1$. Let V be a right K -vector space of dimension $m \in \mathbb{N} \cup \{\infty\}$. Let $K_{\sigma, \varepsilon} := \{\alpha - \alpha^\sigma \varepsilon \mid \alpha \in K\}$ and $K^{\sigma, \varepsilon} := \{\alpha \in K \mid \alpha + \alpha^\sigma \varepsilon = 0\}$. Let Λ be a form parameter relative to (σ, ε) , i.e. Λ is a subgroup of $(K, +)$ satisfying $K_{\sigma, \varepsilon} \subseteq \Lambda \subseteq K^{\sigma, \varepsilon}$ and $\alpha^\sigma \Lambda \alpha \subseteq \Lambda$ for all $\alpha \in K$.

Let $f : V \times V \rightarrow K$ be a (σ, ε) -hermitian form, i.e. f is biadditive and for all $x, y \in V, a, b \in K$ we have $f(xa, yb) = a^\sigma f(x, y)b$ and $f(y, x) = f(x, y)^\sigma \varepsilon$. Let $Q : V \rightarrow K/\Lambda$ be a (σ, ε) -quadratic form with associated (σ, ε) -hermitian form f , i.e. $Q(xa) = a^\sigma Q(x)a + \Lambda$ and $Q(x+y) - Q(x) - Q(y) = f(x, y) + \Lambda$ for all $x, y \in V, a \in K$. If $\Lambda = K$ we require that f is alternating. If $\Lambda \neq K$ then f is uniquely defined by Q .

For $M \subseteq V$ we write $M^\perp := \{x \in V \mid f(x, M) = 0\}$. A subspace $U < V$ is called

- non-degenerate if $U \cap U^\perp = 0$,
- totally isotropic if $U \subseteq U^\perp$ and $Q(U) = 0$.

We require that there is at least one finite-dimensional maximal totally isotropic subspace. In that case, all maximal totally isotropic subspaces have the same dimension. We denote this dimension by n and call it the Witt index of (V, Q, f) . We require $0 < n < \infty$. Furthermore, we require that $V^\perp = 0$. We say that the triple (V, Q, f) is a pseudo-quadratic space if it is of the form described above.

If it additionally satisfies $(m, \Lambda) \neq (2n, 0)$ then we say (V, Q, f) is a thick pseudo-quadratic space.

Note that if (V, Q, f) is a thick pseudo-quadratic space and K is a finite field, then $\dim V \leq 2n + 1$. In general, if one only considers finite fields, especially of characteristic different from 2, the set-up can be substantially simplified, see [AB08, Remark 9.3, 9.4].

5.3.3 Buildings of type C_n

Let $n \in \mathbb{N}, n \geq 1$ and (V, Q, f) be a thick pseudo-quadratic space with Witt index n . Set $X := X(V) = \{0 < U < V \mid U \text{ is totally isotropic}\}$ with incidence relation given by containment as in the A_n case and $\Delta = \text{Flag } X$. Then Δ is a thick building of type C_n and every classical C_n building can be obtained in this way (here classical means that the links of type A_2 correspond to Desarguesian planes, which is always the case for $n \geq 4$), see [Tit74, Theorem 8.22].

Definition 5.8. Set $\mathcal{U} := \{0 \leq U \leq V \mid \dim U < \infty\}$, $\mathcal{U}^\perp := \{U^\perp \mid u \in \mathcal{U}\}$ and $\mathcal{W} := \mathcal{U} \cup \mathcal{U}^\perp$.

Let $\mathcal{E}$ be a finite subset of $\mathcal{W}$ such that $\mathcal{E}^\perp = \mathcal{E}$. Assume $K = \mathbb{F}_q$ is a finite field. Set $\mathcal{E}_j := \{E \in \mathcal{E} \mid \dim E = j\}$, $e_j := |\mathcal{E}_j|$ and $e_h^{(s)} := \sum_{j=0}^{2s} \binom{2s}{j} e_{h+j}$ for $h \in \mathbb{N}, s \in \mathbb{N}_0$ and $h + 2s < m$. We define

- $N(\mathcal{E}) := e_1^{(n-1)}$ if f is alternating ($\Rightarrow m = 2n$) and $Q = 0$,
- $N(\mathcal{E}) := \left(e_1^{(n-1)}\right)^2$ if $m = 2n$ and $\sigma \neq \text{id}$,
- $N(\mathcal{E}) := 2e_2^{(n-1)}$ if $m = 2n + 1$,
- $N(\mathcal{E}) := \max \left\{ e_2^{(n-1)} + e_3^{(n-1)} + 1, 2e_3^{(n-1)} \right\}$ if $m = 2n + 2$.

Definition 5.9. For a set $\mathcal{E}$ of subspaces of V we define, similar to Theorem 5.5,

$$X_{\mathcal{E}}(V) := \{U \in X \mid U \triangleleft \mathcal{E}\}, \quad T_{\mathcal{E}}(V) = \text{Flag } X_{\mathcal{E}}(V).$$

Proposition 5.10. [Abr96, Corollary 15] For any simplex $\tau = \{E_1 < \dots < E_r\} \in \Delta$ set $\mathcal{E}(\tau) = \{E_i, E_i^\perp \mid 1 \leq i \leq r\}$. Then $\Delta^0(\tau) = T_{\mathcal{E}(\tau)}(V)$.

Definition 5.11. With $\mathbf{C}_C$ we denote the class of simplicial complexes $T_{\mathcal{E}}(V)$ where $n = \dim(T_{\mathcal{E}}(V)) + 1 \geq 1$, K a (skew) field, (V, Q, f) is a thick pseudo-quadratic space with Witt index n , $\mathcal{E}$ a finite subset of $\mathcal{W}$ (see Theorem 5.8) such that $\mathcal{E}^\perp = \mathcal{E}$. If K is a finite field, we further require that $|K| \geq N(\mathcal{E})$. We set $T_{\mathcal{E}}(V)$ as in Theorem 5.9.

5.3.4 Buildings of type D_n

For this section let (V, Q, f) be a pseudo-quadratic space but in the specific case where K is a field, $\sigma = \text{id}_K$, $\varepsilon = 1$, $\Lambda = 0$, V a K -vector space of dimension $m = 2n \geq 4$, Q an ordinary quadratic form and f the corresponding symmetric bilinear form. We further assume that V is non-degenerate and of Witt index n .

We set as before $X = X(V) = \{0 < U < V \mid U \text{ is totally isotropic}\}$. Then $\Delta = \text{Flag } X$ is a weak C_n building, in particular it is not thick, since each totally isotropic subspace of dimension $n-1$ is contained in exactly two totally isotropic subspaces of dimension n . Furthermore, the space $Y = \{U \in X \mid \dim U = n\}$ is partitioned into two sets $Y = Y_1 \sqcup Y_2$ such that for $U_1, U_2 \in Y_i$ we have $n - \dim(U_1 \cap U_2) \in 2\mathbb{Z}$ for $i = 1, 2$ and for $U_1 \in Y_1, U_2 \in Y_2$ we have $n - \dim(U_1 \cap U_2) \in 2\mathbb{Z} + 1$. In particular, if for $U_1, U_2 \in Y$ we have $\dim(U_1 \cap U_2) = n - 1$ then U_1 and U_2 are in different sets of the partition.

Example 5.12. We illustrate the situation with the following example for $n = 2, m = 4$ (i.e. V is a 4-dimensional vector space over the field K) and bilinear form f given by $f(x, y) = x_1y_2 + x_2y_1 + x_3y_4 + x_4y_3$. Then the line generated by $e_1 = (1, 0, 0, 0)$ is totally isotropic (since $f(e_1, e_1) = 0$). Let $y = (y_1, y_2, y_3, y_4)$ be another vector and assume that the plane spanned by e_1 and y is totally isotropic. This gives the conditions $f(e_1, y) = f(y, y) = 0$. The first one implies $y_2 = 0$ and the latter $y_3y_4 = 0$. Hence there are exactly two isotropic planes $\langle e_1, e_3 \rangle$ and $\langle e_1, e_4 \rangle$ containing $\langle e_1 \rangle$.

To obtain a thick building of type D_n we set $\tilde{X} = \tilde{X}(V) = \{U \in X \mid \dim U \neq n - 1\}$ and we say that $U, W \in \tilde{X}$ are incident (write UIW) if and only if

$$U \subseteq W \text{ or } W \subseteq U \text{ or } \dim(W \cap U) = n - 1.$$

Then $\tilde{\Delta} = \text{Flag } \tilde{X}$ is desired building. We will write $\text{Orifl}(\tilde{X})$ for $\text{Flag } \tilde{X}$ to stress that we consider $\tilde{X}$ not with the usual incidence relation given by inclusion, but with this new one. Note that $\text{Orifl } \tilde{X}$ is called the oriflamme complex of $\tilde{X}$.

For $n \geq 4$ every building of type D_n is of the form $\tilde{\Delta}$ for some field K [Tit74, Proposition 8.4.3]. We also consider the construction for $n = 2, 3$ in which case we get certain (but not all) buildings of type $D_2 = A_1 \times A_1$ and $D_3 = A_3$, which were already covered earlier.

To describe the opposite complexes, we need to introduce some further notation.

Definition 5.13. For arbitrary subspaces U, W of V , we define

$$U \tilde{\cap}_V W \iff U \cap_V W \text{ or } (U, W \in \tilde{X}, U = U^\perp, W = W^\perp \text{ and } \dim(U \cap W) = 1).$$

For a set $\mathcal{E}$ of subsets of V we define (different from Theorem 5.5)

$$\begin{aligned} X_{\mathcal{E}}(V) &= \{U \in X \mid U \tilde{\cap} \mathcal{E}\}, & T_{\mathcal{E}}(V) &= \text{Flag } X_{\mathcal{E}}(V) \\ \tilde{X}_{\mathcal{E}}(V) &= \{U \in \tilde{X} \mid U \tilde{\cap} \mathcal{E}\}, & \tilde{T}_{\mathcal{E}}(V) &= \text{Orifl } \tilde{X}_{\mathcal{E}}(V) \end{aligned}$$

For $T_{\mathcal{E}}(V)$ we consider the usual inclusion of subspaces as incidence relation, while for $\tilde{T}_{\mathcal{E}}(V)$ we consider the incidence relation specified above.

Proposition 5.14. [Abr96, Corollary 17] Let $\tau = \{E_1 < \dots < E_r\} \in \tilde{\Delta}$ and set $\mathcal{E}(\tau) = \{E_i, E_i^\perp \mid 1 \leq i \leq r\}$ then $\tilde{\Delta}^0(\tau) = \tilde{T}_{\mathcal{E}(\tau)}(V)$.

6 Coboundary expansion for sub-complexes of spherical buildings

The goal of this section is to show that the opposite complex of an irreducible spherical building is a coboundary expander by constructing a cone function. We achieve this for the opposite complexes of type A_n, C_n and D_n using their geometric description from Section 5.3 (recall that for Coxeter complexes and hence buildings there is no difference between type B_n and type C_n). Using that we can construct cone functions for joins given we have cone functions for the factors, we cover all opposite complexes of classical type, i.e. where each connected component of the Dynkin diagram is of type A_n, B_n, C_n or D_n . We start by generalizing an induction argument from [Abr96, Lemma 22] to our setting of cone functions. It gives a list of conditions on a class of simplicial complexes $\mathbf{C}$ that can be checked by induction on the dimension of the complexes which results in the existence of cone functions with bounded radius for all complexes in the class. We then show that this theorem can be applied to the classes $\mathbf{C}_A, \mathbf{C}_A \cup \mathbf{C}_C$ from Theorems 5.7 and 5.11 which shows the existence of cone functions for opposite complexes of type A_n and C_n . The argument for type D_n is more involved and we will discuss it in the corresponding section.

6.1 General induction argument

The general procedure will be similar for all three cases of buildings. We use a quantitative version of [Abr96, Lemma 22]. We basically only change assumption 1. and the conclusion.

Theorem 6.1. *Let $\mathbf{C}$ be a class of non-empty simplicial complexes. Assume that for any $n \in \mathbb{N}, n \geq 1$ there exists $\ell_n \in \mathbb{N}$ such that for all $\kappa \in \mathbf{C}, \dim \kappa = n$, there exists a filtration $\kappa_0 \subset \kappa_1 \subset \dots \subset \kappa_\ell = \kappa$ of length $\ell \leq \ell_n$ such that the following is satisfied (set $V_i := \kappa_i(0) \setminus \kappa_{i-1}(0)$):*

1. *There is an $(n-1)$ -cone function Cone_{κ_0} such that for every $0 \leq j \leq n-1$,*

$$\text{Rad}_j(\text{Cone}_{\kappa_0}) \leq f(n),$$

where $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function depending only on $\mathbf{C}$.

2. *κ_i is a full subcomplex of κ and for all vertices $w_1, w_2 \in V_i$ we have $\{w_1, w_2\} \notin \kappa$, for all $0 \leq i \leq \ell$.*
3. *$\text{lk}_{\kappa_i}(w) \cap \kappa_{i-1}$ is either in $\mathbf{C}$ or can be written as the join of two elements from $\mathbf{C}$ for any $w \in V_i, i \geq 1$.*

Then for every $n \in \mathbb{N}_0$ there exists a constant $\mathcal{R}(n)$ such that for every $\kappa \in \mathbf{C}$ with $\dim \kappa = n$ we have: κ has an $(n-1)$ -cone function Cone_κ such that for all $-1 \leq j \leq n-1$ we have

$$\text{Rad}_j(\text{Cone}_\kappa) \leq \mathcal{R}(n).$$

We can describe $\mathcal{R}(n)$ recursively. We have $\mathcal{R}(0) = 1$. For $n \geq 1$, assume $\mathcal{R}(k)$ is known for $k < n$. Set

$$S(n) = \max_{a,b \in \mathbb{N}: a+b+1=n} ((a+1)\mathcal{R}(a) + 1)\mathcal{R}(b).$$

Then

$$\mathcal{R}(n) = S(n)^{\ell_n} f(n) + \sum_{j=1}^{\ell_n} S(n)^j.$$

Remark 6.2. Note that the theorem does not assume that the κ_i are in $\mathbf{C}$. Furthermore, we have that

$$\text{lk}_{\kappa_i}(w) \cap \kappa_{i-1} = \text{lk}_{\kappa}(w) \cap \kappa_{i-1}.$$

The $\subseteq$ inclusion is clear since $\kappa_i \subseteq \kappa$. The other inclusion $\supseteq$ holds since κ_i is a full subcomplex of κ thus for $\sigma \in \kappa_{i-1} \subseteq \kappa_i$ and $w \in \kappa_i$ if $\sigma \cup w \in \kappa$ we can deduce $\sigma \cup w \in \kappa_i$.

Proof. We will prove the result by induction on the dimension n .

For $n = 0$, since $\kappa \neq \emptyset$ there exists $v \in \kappa(0)$ and thus we can define a (-1) -cone function via $\emptyset \mapsto \mathbb{1}_{[v]}$. Every cone function has (-1) -cone radius $= 1$, thus we take $\mathcal{R}(0) = 1$.

Fix $n \geq 1$ and $\kappa \in \mathbf{C}$, $\dim(\kappa) = n$ with filtration $\kappa_0 \subset \dots \subset \kappa_\ell = \kappa$ as in the requirements. We will prove by induction on $0 \leq i \leq \ell$, that there is an $(n-1)$ -cone function Cone_{κ_i} such that for every $-1 \leq k \leq n-1$

$$\text{Rad}_k(\text{Cone}_{\kappa_i}) \leq \mathcal{R}^{(i)}(n)$$

where $\mathcal{R}^{(i)}(n)$ is given inductively by

$$\mathcal{R}^{(0)}(n) = f(n), \quad \mathcal{R}^{(i)}(n) = S(n)(\mathcal{R}^{(i-1)}(n) + 1) = S(n)^i f(n) + \sum_{j=1}^i S(n)^j.$$

For $i = 0$ this follows from Assumption 1. We proceed by induction on i . Fix $1 \leq i \leq \ell$ and assume there exists a constant $\mathcal{R}^{(i-1)}(n)$ and an $(n-1)$ -cone function $\text{Cone}_{\kappa_{i-1}}$ such that $\text{Rad}_j(\text{Cone}_{\kappa_{i-1}}) \leq \mathcal{R}^{(i-1)}(n)$ for all $-1 \leq j \leq n-1$. We want to apply Theorem 3.1 to the following setting:

$$X = \kappa, X' = \kappa_{i-1}, \text{Cone}_{X'} = \text{Cone}_{\kappa_{i-1}}, R'_j = \mathcal{R}^{(i-1)}(n), W = V_i = \kappa_i(0) \setminus \kappa_{i-1}(0).$$

We check that the conditions of the theorem are satisfied.

1. $W \cap X' = \emptyset$ by definition.
2. For every $w_1, w_2 \in W : \{w_1, w_2\} \not\subseteq \kappa_i \subset \kappa$ by assumption.
3. For every $w \in W$, $\text{lk}_X(w) \cap X'$ is a non-empty simplicial complex, since by Assumption 3 we have $\text{lk}_X(w) \cap X' \in \mathbf{C}$ or a join of two complexes in $\mathbf{C}$.
4. By Assumption 3, $\text{lk}_X(w) \cap X'$ is either in $\mathbf{C}$ or is a join of two elements of $\mathbf{C}$. In the first case we use the induction hypothesis and get an $(n-2)$ -cone function $\text{Cone}_{\text{lk}_X(w) \cap X'}$ such that

$$\text{Rad}_j(\text{Cone}_{\text{lk}_X(w) \cap X'}) \leq \mathcal{R}(n-1).$$

In the second case, we can write $\text{lk}_X(w) \cap X' = A * B$ for some $A, B \in \mathbf{C}$. Set $n_A = \dim A, n_B = \dim B$ then $n-1 \geq \dim(\text{lk}_X(w) \cap X') = n_A + n_B + 1$, hence $n_A, n_B < n$. Thus by induction there exists an (n_A-1) -cone function Cone_A with $\text{Rad}_j(\text{Cone}_A) \leq \mathcal{R}(n_A)$ for all $-1 \leq j \leq n_A-1$ and similarly for B . Using Theorem 2.15 we get an $(n-1)$ -cone function $\text{Cone}_{\text{lk}_X(w) \cap X'}$ such that

$$\text{Rad}_j(\text{Cone}_{\text{lk}_X(w) \cap X'}) \leq ((n_A+1)\mathcal{R}(n_A) + 1)\mathcal{R}(n_B) \leq S(n) \text{ for all } -1 \leq j \leq n-2.$$

Note that we simplified the bound coming from Theorem 2.15 by only considering the largest possible of the cases. We chose $S(n)$ to be the maximum over all the bounds that can appear at this step. Note that in particular $\mathcal{R}(n-1) \leq S(n)$. We set $R''_j = S(n)$.

Applying Theorem 3.1 we get an $(n - 1)$ -cone function $\text{Cone}_{X' \cup W}$ such that

$$\text{Rad}_j(\text{Cone}_{X' \cup W}) \leq R''_{j-1}(R'_j + 1)$$

which in the original formulation means

$$\text{Rad}_j(\text{Cone}_{\kappa_i}) \leq S(n)(\mathcal{R}^{(i-1)}(n) + 1).$$

Thus we have $\mathcal{R}^{(i)}(n) = S(n)(\mathcal{R}^{(i-1)}(n) + 1)$ and solving the recursion gives the second expression of $\mathcal{R}^{(i)}(n)$.

Now, since $\kappa_\ell = \kappa$, we get an $(n - 1)$ -cone function Cone_κ with $\text{Rad}(\text{Cone}_\kappa) \leq \mathcal{R}^{(\ell)}(n)$. We would like to set $\mathcal{R}(n) = \mathcal{R}^{(\ell)}(n)$. But the bound has to hold for all possible $\kappa \in \mathbf{C}$ of dimension n . Since $\mathcal{R}^{(i)}(n)$ is increasing in i , we take $\mathcal{R}(n) = \mathcal{R}^{(\ell_n)}(n) = S(n)^{\ell_n} f(n) + \sum_{j=1}^{\ell_n} S(n)^j$. $\square$

Remark 6.3. Compared to [Abr96, Lemma 22] we changed condition 1. from asking for κ_0 to be contractible to requiring an explicit cone function, and we removed the dependency on the underlying building, since the theorem also works in this wider generality. In particular, if we want to show that a certain class, for which Abramenko already showed that it satisfies the assumptions of [Abr96, Lemma 22], also satisfies the assumptions of this theorem, we need to check condition 1., while conditions 2. and 3. are usually already covered.

We will apply this theorem to classes of simplicial complexes which are flag complexes over sets of vector subspaces. In this case, condition 2 can be achieved by only adding vector subspaces with the same dimension in one step of the filtration. By not altering the incidence relation considered for the flag complex, we can make sure that the κ_i are full subcomplexes.

To show that condition 3 is satisfied in our class of complexes, we use that for flag complexes over vector subspaces we can describe the links more explicitly using the following definition.

Definition 6.4. Let K be a field, V a K -vector space, X a collection of subspaces of V . Let $Y = \text{Flag } X$ and $U \in X$. Then we define

$$\begin{aligned} X^{<U} &:= \{W \in X \mid W < U\}, \quad Y^{<U} := \text{Flag } X^{<U} \\ X^{>U} &:= \{W \in X \mid W > U\}, \quad Y^{>U} := \text{Flag } X^{>U}. \end{aligned}$$

Lemma 6.5. Note that

$$\text{lk}_Y(U) = Y^{<U} * Y^{>U}.$$

Moreover, if $Y_i = \text{Flag } X_i$ for $i = 1, 2$ such that $Y_1 \subseteq Y_2$ and $U \in X_2$, then

$$\text{lk}_{Y_2}(U) \cap Y_1 = Y_1^{<U} * Y_1^{>U}$$

Proof. Straightforward application of the definitions. $\square$

Note that a similar statement holds for flag complexes over more general posets.

6.2 Buildings of type A_n

To show that the complex opposite the fundamental chamber in a spherical building of type A_n has a cone function with bounded cone radius, we show that the class $\mathbf{C}_A$ from Theorem 5.7 satisfies the assumptions of Theorem 6.1.

We follow the machinery presented in [Abr96] closely, which itself builds on previous work [Qui73, Vog81, AA93].

We proceed as follows

1. Each complex κ in the classes $\mathbf{C}_A$ is the flag complex of a certain set of vector subspaces. We recall the fact from [Abr96] that at least one of these subspaces has dimension 1.
2. We use this subspace of dimension 1 to define a filtration $\kappa_0 \subset \dots \subset \kappa_\ell = \kappa$ as in [Abr96].
3. We show that κ_0 admits a cone function and we bound its radius.
4. We recall from the work in [Abr96] that $\text{lk}_{\kappa_i}(U) \cap \kappa_{i-1}$ for all $U \in \kappa_i(0) \setminus \kappa_{i-1}(0)$ has the desired structure, i.e. is in the class we consider or can be written as a join of two elements in there.
5. We conclude that the class satisfies the required assumptions and hence get a cone function with bounded radius for each element, in particular for the opposite complexes.

We recall some facts that appear in the proof of [Abr96, Step 1, Proposition 12], see also [AA93].

Proposition 6.6. *If $T_{\mathcal{E}}(V) \in \mathbf{C}_A$, then there exists $\ell \in X_{\mathcal{E}}(V)$ such that ℓ is a line, i.e., $\dim(\ell) = 1$.*

Proof. Since similar statements will appear for each case, we will give a proof of this statement here.

A 1-dimensional subspace $\ell < V$ will be transversal to an arbitrary proper subspace $E < V$ if and only if $\ell \not\subseteq E$. Let $\mathcal{E}$ be a collection of subspaces such that $T_{\mathcal{E}}(V) \in \mathbf{C}_A$. In particular, $\mathcal{E}$ contains finitely many subspaces. Thus, if the field K is infinite, there will always exist a 1-dimensional subspace $\ell \not\subseteq \bigcup_{E \in \mathcal{E}} E$. Now, if K is a finite field of size q , then each E contains $\frac{q^{\dim E} - 1}{q - 1}$ lines. Thus in total $\bigcup_{E \in \mathcal{E}} E$ contains at most

$$\sum_{E \in \mathcal{E}} \frac{q^{\dim E} - 1}{q - 1} \leq \frac{q^n - 1}{q - 1} \sum_{i=1}^n e_i \leq q \frac{q^n - 1}{q - 1} < \frac{q^{n+1} - 1}{q - 1}$$

where $e_i = |\{E \in \mathcal{E} \mid \dim E = i\}|$ as above. Thus there exists at least one line transversal to all of $\mathcal{E}$. $\square$

The important property of ℓ being a line, that we will use, is that for any other subspace W we have either $\ell \leq W$ or $\ell \cap W = 0$ in which case $W < W + \ell$.

Let $\ell \in X_{\mathcal{E}}(V)$ be a line. Define

$$Y_0 = \{U \in X_{\mathcal{E}}(V) : \ell + U \in X_{\mathcal{E}}(V)\},$$

and further define κ_0 to be the subcomplex of $T_{\mathcal{E}}(V)$ spanned by Y_0 .

Assume now that $\dim V = n + 2$ and hence $\dim T_{\mathcal{E}}(V) = n$ (and $\text{Flag } X$ is a building of type A_{n+1}). For $1 \leq i \leq n + 1$, define $Y_i \subseteq X_{\mathcal{E}}(V)$ as follows:

$$Y_i = \{U \in X_{\mathcal{E}}(V) \mid \dim(U) \geq n + 2 - i \text{ or } U \in Y_0\}.$$

For every such i , let κ_i be the subcomplex of $T_{\mathcal{E}}(V)$ spanned by Y_i . We have $\kappa_{n+1} = T_{\mathcal{E}}(V)$.

Lemma 6.7. *There is an $(n - 1)$ -cone function Cone_{κ_0} such that for every $0 \leq k \leq n - 1$,*

$$\text{Rad}_k(\text{Cone}_{\kappa_0}) \leq n + 2.$$

Proof. Let Γ_0 be the subcomplex of κ_0 spanned by $Z_0 = \{U \in X_{\mathcal{E}}(V) : \ell \leq U\}$. For $1 \leq i \leq n + 1$, we denote

$$Z_i = \{U \in Y_0 \mid \dim(U) \leq i \text{ or } U \in Z_0\}.$$

Further denote $\Gamma_i = \text{Flag } Z_i$. Note that $\Gamma_{n+1} = \kappa_0$. We will show by induction that for every $0 \leq i \leq n + 1$, there is an $(n - 1)$ -cone function Cone_{Γ_i} such that for every $-1 \leq k \leq n - 1$, $\text{Rad}_k(\text{Cone}_{\Gamma_i}) \leq i + 1$.

For $i = 0$, we note that Γ_0 is the ball of radius 1 around ℓ in κ and can be written as $\Gamma_0 = \{\ell\} * \text{lk}_{\kappa}(\ell)$. By Proposition 2.12, there is an $(n - 1)$ -cone function Cone_{Γ_0} such that for every $-1 \leq k \leq n - 1$, $\text{Rad}_k(\text{Cone}_{\Gamma_0}) \leq 1$.

Let $1 \leq i \leq n + 1$ and assume there is an $(n - 1)$ -cone function $\text{Cone}_{\Gamma_{i-1}}$ such that for every $-1 \leq k \leq n - 1$, $\text{Rad}_k(\text{Cone}_{\Gamma_{i-1}}) \leq i$. We will apply Theorem 3.1 in order to bound the cone radii of Γ_i . We note that any two $U_1, U_2 \in Z_i \setminus Z_{i-1}, U_1 \neq U_2$ are of the same dimension and thus are not connected by an edge. Let $U \in Z_i \setminus Z_{i-1}$. We will show that $\text{lk}_{\kappa_0}(U) \cap \Gamma_{i-1}$ is a join of the vertex $\{U + \ell\}$ with the complex spanned by all the other vertices in $\text{lk}_{\kappa_0}(U) \cap \Gamma_{i-1}$. Note that $U + \ell \neq U$ since $U \notin Z_0$, and $U + \ell \in Z_0 \subseteq Z_{i-1}$.

Since $\text{lk}_{\kappa_0}(U) \cap \Gamma_{i-1}$ is a clique complex, it is enough to show that for every vertex U' in $\text{lk}_{\kappa_0}(U) \cap \Gamma_{i-1}$ that is not $\ell + U$, there is an edge connecting U' and $\ell + U$. Let U' be such a vertex.

First, we will deal with the case where $\dim(U') < i$. In that case (using the fact that $\dim(U) = i$), U' being in $\text{lk}_{\kappa_0}(U)$ implies that $U' \leq U$ and thus $U' \leq U + \ell$, i.e., U' is connected by an edge to $U + \ell$ as needed.

Second, assume that $\dim(U') > i$. From the definition of Z_{i-1} , it follows that $U' \in Z_0$, i.e., that $\ell \leq U'$. Also, U' is in $\text{lk}_{\kappa_0}(U)$ and thus $U \leq U'$. It follows that $U + \ell \leq U'$, i.e., U' is connected by an edge to $U + \ell$ as needed.

By Proposition 2.12, for every $U \in Z_i \setminus Z_{i-1}$, there is an $(n - 2)$ -cone function $\text{Cone}_{\text{lk}_{\kappa_0}(U) \cap \Gamma_{i-1}}$ such that for every $-1 \leq k \leq n - 2$, $\text{Rad}_k(\text{Cone}_{\text{lk}_{\kappa_0}(U) \cap \Gamma_{i-1}}) \leq 1$.

By Theorem 3.1 (with $R_k'' = 1$ for every k), it follows that there is an $(n - 1)$ -cone function Cone_{Γ_i} such that for every $-1 \leq k \leq n - 1$, $\text{Rad}_k(\text{Cone}_{\Gamma_i}) \leq i + 1$ as needed. $\square$

Fact 6.8 ([Abr96, Step 3, Proposition 12]). *Let $Y = X_{\mathcal{E}}(V), \kappa = T_{\mathcal{E}}(V) \in \mathbf{C}_A$ and $\kappa_0 \subseteq \dots \subseteq \kappa_{n+1} = \kappa$ with $\kappa_i = \text{Flag } Y_i$ be the filtration as above. For every $1 \leq i \leq n + 1$ and every $U \in Y_i \setminus Y_{i-1}$, we have*

1. $\text{lk}_{\kappa_i}(U) \cap \kappa_{i-1} = Y_0^{<U} * Y^{>U}$ and
2. $\text{Flag } Y_0^{<U}, \text{Flag } Y^{>U} \in \mathbf{C}_A$.

Proof. We sketch the idea of the proof. For part 1, we know from Theorem 6.5 that $\text{lk}_{\kappa_i}(U) \cap \kappa_{i-1} = Y_{i-1}^{<U} * Y_{i-1}^{>U}$. We have $Y_{i-1}^{<U} = Y_0^{<U}$ since $\dim U = n + 2 - i < n + 2 - i + 1$. On the other hand, any $W > U$ has $\dim W > n + 2 - i$ and thus $Y_{i-1}^{>U} = Y^{>U}$.

For part 2, we observe

- if $W < U$ then $W \dashv_V \mathcal{E}$ and $W + \ell \dashv_V \mathcal{E}$ if and only if $W \dashv_U \mathcal{E}'$ with $\mathcal{E}' = (\mathcal{E} \cap U) \cup ((\mathcal{E} + \ell) \cap U)$.
- If $U < W < V$ then $W \dashv_V \mathcal{E}$ if and only if $W + U \dashv_{V/U} \bar{\mathcal{E}}$ with $\bar{\mathcal{E}} = \{E + U \mid E \in \mathcal{E}\}$.

Thus $Y_0^{<U} \cong X_{\mathcal{E}'}(U)$ and $Y^{>U} \cong X_{\bar{\mathcal{E}}}(V/U)$. In Step 3 [Abr96, Proposition 12] it is shown that $X_{\mathcal{E}'}(U), X_{\bar{\mathcal{E}}}(V/U) \in \mathbf{C}_A$. $\square$

Corollary 6.9. *The class $\mathbf{C}_A$ satisfies the assumptions of Theorem 6.1.*

Proof. The result follows from Theorem 6.7 and Fact 6.8. In the notation of Theorem 6.1, we have $f(n) = n + 2$ and $\ell_n = n + 1$. $\square$

Using Theorem 5.6 we can then get the desired conclusion for opposite complexes of type A_n .

Corollary 6.10. *Let Δ be a building of type A_n over a field K with $|K| \geq 2^{n-1}$ and $a \in \Delta$ a simplex, then there exists an $(n - 2)$ -cone function $\text{Cone}_{\Delta^0(a)}$ of $\Delta^0(a)$ such that*

$$\text{Rad}_j(\text{Cone}_{\Delta^0(a)}) \leq \mathcal{R}(n - 1), \forall -1 \leq j \leq n - 2.$$

6.3 Buildings of type C_n

The class $\mathbf{C}$ that will cover the case of opposite complexes in buildings of type C_n will be the union of class $\mathbf{C}_A$ (see Theorem 5.7) and $\mathbf{C}_C$ (see Theorem 5.11).

We start by considering $\kappa = T_{\mathcal{E}}(V) = \text{Flag } X_{\mathcal{E}}(V) \in \mathbf{C}_C$ of dimension $n \geq 0$ with $V, \mathcal{E}$ as in Theorem 5.11. The corresponding building is then of type C_{n+1} .

We recall from [Abr96, Proposition 13, Step 1] the following.

Fact 6.11. *There exists a 1-dimensional subspace $\ell \in X_{\mathcal{E}}(V)$.*

We will also need the following fact, which can be found in [Abr96, Proposition 13, Step 2].

Fact 6.12. *Given $\mathcal{E}$ and ℓ as above, for $0 < U < V$, if $\ell \not\leq U$ then we have $U \dashv \mathcal{E} + \ell$ if and only if $U + \ell \dashv \mathcal{E}$. Furthermore, if $W, E \not\leq \ell^\perp$ then $W \cap \ell^\perp \dashv E \iff W \dashv E \cap \ell^\perp$.*

We define the filtration $\kappa_0 \subset \dots \subset \kappa_{2n+2}$ of κ as follows. We start by setting $X_0 = X^1 \cup X^2 \cup X^3$ where

1. $X^1 = \{U \in X \mid \ell \leq U \text{ and } U \dashv \mathcal{E}\};$
2. $X^2 = \{U \in X \mid \ell \not\leq U \leq \ell^\perp \text{ and } U \dashv \mathcal{E} \cup (\mathcal{E} + \ell)\};$
3. $X^3 = \{U \in X \mid U \not\leq \ell^\perp, \dim U > 1 \text{ and } U \dashv \mathcal{E} \cup (\mathcal{E} \cap \ell^\perp) \cup (\mathcal{E} + \ell) \cup ((\mathcal{E} \cap \ell^\perp) + \ell)\}.$

Set $\kappa_0 = \text{Flag } X_0$.

Next, we define the rest of the desired filtration in two steps. Set $Z = \{U \in X_{\mathcal{E}}(V) \mid \ell \leq U \text{ or } U \cap \mathcal{E} + \ell\}$ and $Y_0 = X_0$. Note that $Y_0 \subseteq Z$. For the first half of the filtration we define

$$Y_i = \{U \in Z \mid U \in Y_0 \text{ or } \dim U \leq i\}, \kappa_i = \text{Flag } Y_i, \quad 1 \leq i \leq n+1.$$

For the second half we set

$$Y_i = \{U \in X_{\mathcal{E}}(V) \mid U \in Z \text{ or } \dim U \geq 2n+3-i\}, \kappa_i = \text{Flag } Y_i, \quad n \leq i \leq 2n+2.$$

Then $T_{\mathcal{E}}(V) = \kappa_{2n+2}$. We start by showing the existence of a cone function for κ_0 , where we use the filtration of κ_0 from [Abr96, Proposition 13].

Lemma 6.13. *With the definitions as above, we have that κ_0 has an $(n-1)$ -cone function Cone_{κ_0} such that*

$$\text{Rad}_j(\text{Cone}_{\kappa_0}) \leq 2n+3 \quad \forall -1 \leq j \leq n-1.$$

Proof. Let $T_{\mathcal{E}}(V) \in \mathbf{C}_C$ as above. Thus $\dim \kappa = n$. In particular, any totally isotropic subspace of V has dimension at most $n+1$.

We will apply Theorem 3.1 inductively to two different filtrations of κ_0 . The first one is defined as follows, using X^1, X^2, X^3 as defined above.

$$A_0 = X^1$$

$$A_i = Y_{i-1} \cup \{U \in X^2 \mid \dim U = i\} = X^1 \cup \{U \in X^2 \mid \dim U \leq i\} \text{ for } 1 \leq i \leq n+1.$$

We set $\alpha_i = \text{Flag } A_i$. Note that $A_{n+1} = X^1 \cup X^2$.

The second filtration is given by

$$B_0 = A_{n+1}$$

$$\begin{aligned} B_i &= B_{i-1} \cup \{U \in X^3 \mid \dim U = n+1-(i-1)\} \\ &= B_0 \cup \{U \in X^3 \mid \dim U \geq n+1-(i-1)\} \text{ for } 1 \leq i \leq n+1. \end{aligned}$$

We set $\beta_i = \text{Flag } B_i$. Note that $B_{n+1} = X^1 \cup X^2 \cup X^3 = X_0$ and hence $\beta_{n+1} = \kappa_0$.

Next, we want to show that for each α_i there exists an $(n-1)$ -cone function Cone_{α_i} such that for every $-1 \leq j \leq n-1$ we have $\text{Rad}_j(\text{Cone}_{\alpha_i}) \leq i+1$ for all $0 \leq i \leq n+1$. We do this by induction on i .

For $i=0$ we have $\alpha_0 = \text{st}_{\kappa}(\ell) = \{\ell\} * \text{lk}_{\kappa}(\ell)$. Hence by Theorem 2.12 there exists an $(n-1)$ -cone function Cone_{α_0} with $\text{Rad}_j(\text{Cone}_{\alpha_0}) \leq 1$ for all $-1 \leq j \leq n-1$.

For the following step, fix $1 \leq i \leq n+1$. We want to apply Theorem 3.1 to the following set-up. Here κ_0 corresponds to the complex called X in Theorem 3.1, α_{i-1} corresponds to X' , and $\mathcal{W}_i := A_i \setminus A_{i-1}$ to W i.e. the set of vertices that we want to add. We check that the conditions of Theorem 3.1 are satisfied.

For assumption 1 note that $\mathcal{W}_i \cap \alpha_{i-1} = \emptyset$ by definition. Assumption 2 holds since all subspaces in $\mathcal{W}_i$ have the same dimension, hence they are not connected in κ .

For 3 and 4, let $W \in \mathcal{W}_i$. Then $\text{lk}_{\kappa_0}(W) = \text{Flag } \{U \in \kappa_0(0) \mid U < W \text{ or } W > U\}$. Furthermore, for $U \in A_{i-1}$ we have $U + \ell \in X_0$, since by Fact 6.12 we have $U \cap \mathcal{E} + \ell \iff U + \ell \cap \mathcal{E}$ if $\ell \not\leq U$. If $\ell \leq U$ then we have $U \in X^1 \subseteq X_0$. Note that $\ell \not\leq W$ because $W \in X^2$. Furthermore, since

$\ell \leq W + \ell$, we have $W + \ell \in A_0 \subset A_{i-1}$. Next, we want to show that $\text{lk}_{\kappa_0}(W) \cap \alpha_{i-1}$ is a join of $\{W + \ell\}$ with the flag complex of all other vertices from $\text{lk}_{\kappa_0}(W) \cap \alpha_{i-1}$. Since $\text{lk}_{\kappa_0}(W) \cap \alpha_{i-1}$ is a flag complex, it suffices to check that every vertex different from $W + \ell$ is adjacent to $W + \ell$. Let $U \in (\text{lk}_{\kappa_0}(W) \cap \alpha_{i-1})(0) \setminus \{W + \ell\}$. Then we differentiate two cases:

- Case 1: $\dim U \leq i - 1$: then $U \leq W \leq W + \ell$;
- Case 2: $\dim U > i$ then $U \in A_0$, thus $W \leq U$ and $\ell \leq U$ and hence $W + \ell \leq U$.

Hence, any other vertex is connected to $W + \ell$ by an edge and thus

$$\text{lk}_{\kappa_0}(W) \cap \alpha_{i-1} = \{W + \ell\} * \text{Flag}((\text{lk}_{\kappa_0}(W) \cap \alpha_{i-1})(0) \setminus \{W + \ell\}).$$

By Theorem 2.12 we have $\text{Rad}_j(\text{Cone}_{\text{lk}_{\kappa_0}(W) \cap \alpha_{i-1}}) \leq 1$ for all $-1 \leq j \leq n - 1$.

Thus we can apply Theorem 3.1 and get an $(n-1)$ -cone function Cone_{α_i} satisfying $\text{Rad}_j(\text{Cone}_{\alpha_i}) \leq i + 1$.

We treat the second filtration similarly. We want to show that for every $0 \leq i \leq n + 1$ there is an $(n-1)$ -cone function Cone_{β_i} such that $\text{Rad}_j(\text{Cone}_{\beta_i}) \leq n + 2 + i$. We again proceed by induction on i .

For $i = 0$ we have that $\beta_0 = \alpha_{n+1}$ has a cone function with radius $\leq n + 2$ by the above reasoning.

Now fix $1 \leq i \leq n + 1$. We want to use Theorem 3.1 with κ_0 corresponding to X , β_{i-1} to X' , and $B_i \setminus B_{i-1}$ to W . Then assumptions 1 and 2 of Theorem 3.1 are satisfied by the same argument as above.

For assumptions 3 and 4, fix $W \in B_i \setminus B_{i-1}$ and consider

$$\text{lk}_{\kappa_0}(W) \cap \beta_{i-1} = \text{Flag} \{U \in B_{i-1} \mid U < W \text{ or } U > W\}.$$

First, note that $W \in X^3$, thus $\ell^\perp \not\leq W$ and hence $W \cap \ell^\perp \neq W$. Next, we show that $W \cap \ell^\perp \in B_0 = X^1 \cup X^2$. Clearly $W \cap \ell^\perp \leq \ell^\perp$. Thus it suffices to show that $W \cap \ell^\perp \not\in \mathcal{E} \cup (\mathcal{E} + \ell)$. Since $W \in X^3$ we know $W \not\in \mathcal{E} \cup (\mathcal{E} + \ell)$. Fix $E \in \mathcal{E}$. Then $E \not\leq \ell^\perp$ since otherwise $E^\perp \geq \ell$, but $E^\perp \in \mathcal{E}$ and $\ell \not\in \mathcal{E}$. Similarly, $E + \ell \not\leq \ell^\perp$ since otherwise $(E + \ell)^\perp \geq \ell$, while $(E + \ell)^\perp = E^\perp \cap \ell^\perp \in \mathcal{E} \cap \ell^\perp$ and $\ell \not\in \mathcal{E} \cap \ell^\perp$. Thus, by Fact 6.12, it suffices to show that $W \not\in (\mathcal{E} \cap \ell^\perp) \cup ((\mathcal{E} + \ell) \cap \ell^\perp)$, which holds since $W \in X^3$.

Next, we show that every vertex (different from $W \cap \ell^\perp$) is connected to $W \cap \ell^\perp$ hence

$$\text{lk}_{\kappa_0}(W) \cap \beta_{i-1} = \{W \cap \ell^\perp\} * \text{Flag} \{U \in B_{i-1} \setminus \{W \cap \ell^\perp\} \mid U < W \text{ or } U > W\}.$$

Let $U \in \text{lk}_{\kappa_0}(W) \cap \beta_{i-1}(0) = \{U \in B_{i-1} \mid U < W \text{ or } U > W\}$. If $W \leq U$ then $W \cap \ell^\perp \leq U$ and thus $\{U, W \cap \ell^\perp\} \in \text{lk}_{\kappa_0}(W) \cap \beta_{i-1}$. If $U \leq W$, then $\dim U < \dim W = n + 1 - i + 1$ and hence $U \in B_0$. Therefore either $U \in X^1$ or $U \in X^2$. But if $U \in X^1$, then $\ell \leq U \leq W$, a contradiction to $W \in B_i \setminus B_{i-1}$. Thus, $U \in X^2$, and in particular $U \leq \ell^\perp$. Therefore, $U \leq W \cap \ell^\perp$ and hence $\{U, W \cap \ell^\perp\} \in \text{lk}_{\kappa_0}(W) \cap \beta_{i-1}$.

Therefore, by Theorem 2.12, $\text{lk}_{\kappa_0}(W) \cap \beta_{i-1}$ has a cone radius bounded by 1. Applying Theorem 3.1 we get a cone function Cone_{β_i} satisfying

$$\text{Rad}(\text{Cone}_{\beta_i}) \leq \text{Rad}(\text{Cone}_{\beta_{i-1}}) + 1 \leq n + 2 + i.$$

Since $\kappa_0 = \beta_{n+1}$ we get the desired result. $\square$

From [Abr96, Proposition 13] we get the following.

Fact 6.14. *Let $\kappa_0 \subset \dots \subset \kappa_{2n+2} = \kappa$ be the filtration as above. Then for every $w \in \kappa_i(0) \setminus \kappa_{i-1}(0)$, $1 \leq i \leq 2n+2$ we have $\text{lk}_{\kappa_i}(w) \cap \kappa_{i-1}$ is in $\mathbf{C} = \mathbf{C}_A \cup \mathbf{C}_C$ or is a join of two elements from $\mathbf{C}$.*

Combining the above results, we get:

Corollary 6.15. *The class $\mathbf{C}$ satisfies the requirements of Theorem 6.1.*

Corollary 6.16. *Let $\Delta = \text{Flag } X(V)$ be a classical C_n building as described in Section 5.3.3 (note that $\dim \Delta = n-1$). Assume that K is infinite or $K = \mathbb{F}_q$ and*

- $q \geq 2^{2n-2}$ if f is alternating and $Q = 0$,
- $q \geq 2^{4n-4}$ if $m = 2n$ and $\sigma \neq \text{id}$,
- $q \geq 2^{2n-1}$ if $m = 2n+1$ or $2n+2$.

In particular, if Δ is the building coming from the BN-pair of a Chevalley group of type B_n or C_n over the field $\mathbb{F}_q$ we require $q \geq 2^{2n-1}$.

Then for any $a \in \Delta$ there exists an $(n-2)$ -cone function $\text{Cone}_{\Delta^0(a)}$ of $\Delta^0(a)$ with

$$\text{Rad}_j(\text{Cone}_{\Delta^0(a)}) \leq \mathcal{R}(n-1) \quad \forall -1 \leq j \leq n-2$$

where $\mathcal{R}(n-1)$ is as in Theorem 6.1 with $f(n) = 2n+1$ and $\ell_n = 2n+2$.

6.4 Buildings of type D_n

The proof that opposite complexes of buildings of type D_n are coboundary expanders is the most involved case. We will use the notation introduced in Section 5.3.4. We will later apply an adapted version of the general induction argument to a class of complexes containing flag complexes of the form $T_{\mathcal{E}}(V) = \text{Flag } X_{\mathcal{E}}(V)$. But by Theorem 5.14, the opposite complexes in a building of type D_n are the oriflamme complexes $\tilde{T}_{\mathcal{E}}(V)$. Thus we start by showing that a cone function for $T_{\mathcal{E}}(V)$ also gives rise to a cone function for $\tilde{T}_{\mathcal{E}}(V)$. This can be seen as a quantitative version of [Abr96, Corollary 17 (ii)].

Proposition 6.17. *Let $\tau = \{E_1 < \dots < E_r\} \in \tilde{\Delta}$ and set $\mathcal{E}(\tau) = \{E_i, E_i^\perp \mid 1 \leq i \leq r\}$. If $T_{\mathcal{E}(\tau)}(V)$ has an $(n-1)$ -cone function $\text{Cone}_{T_{\mathcal{E}(\tau)}(V)}$ with $\text{Rad}(\text{Cone}_{T_{\mathcal{E}(\tau)}(V)}) \leq c$ for some $c \geq 0$ then there exists an $(n-1)$ -cone function $\text{Cone}_{\tilde{T}_{\mathcal{E}(\tau)}(V)}$ with $\text{Rad}(\text{Cone}_{\tilde{T}_{\mathcal{E}(\tau)}(V)}) \leq 2c$.*

Proof. For shorter notation, we write $T = T_{\mathcal{E}(\tau)}(V)$, $Y = X_{\mathcal{E}(\tau)}(V)$ and $\tilde{T} = \tilde{T}_{\mathcal{E}(\tau)}(V)$, $\tilde{Y} = \tilde{X}_{\mathcal{E}(\tau)}(V)$.

The idea of the proof is to construct an $(n-1)$ -cone function for $\tilde{T}$ given a cone function

$$\text{Cone}_T : \bigoplus_{j=-1}^{n-1} C_j(T) \rightarrow \bigoplus_{j=0}^n C_j(T)$$

by defining maps $f_j : C_j(\tilde{T}) \rightarrow C_j(T)$ for $-1 \leq j \leq n-1$ and $g_j : C_j(T) \rightarrow C_j(\tilde{T})$ for $0 \leq j \leq n$ such that $g_{j+1} \circ \text{Cone}_T|_{C_j(T)} \circ f_j$ gives rise to a cone function of $\tilde{T}$.

As already observed in [Abr96], T is a subdivision of $\tilde{T}$ in the following way. Note that $\tilde{Y} \subset Y$ and $Y \setminus \tilde{Y} = \{U \in Y \mid \dim U = n-1\}$. If two subspaces $U, W \in \tilde{Y}$ are connected by an edge in T then the same is true in $\tilde{T}$, but we have additional edges in $\tilde{T}$, namely in the case when $\dim U = \dim W = n$ and $\dim(U \cap W) = n-1$. To go from $\tilde{T}$ to T , we subdivide the edge $\{U, W\}$ by adding the vertex $U \cap W$ and connecting U, W and every vertex $U' \in \tilde{Y}$ for which $\{U', U, W\} \in \tilde{T}$ to $U \cap W$.

Recall from the discussion in Section 5.3.4 that there are two different types of n -dimensional spaces in $\tilde{X}$ and that spaces from the same type cannot have an intersection of dimension $n-1$. Furthermore, each $U \in X$ of dimension $n-1$ is contained in exactly two totally isotropic spaces of dimension n . If $U \in Y$ of dimension $n-1$, i.e. $U \tilde{\cap} \mathcal{E}$, with $W_1, W_2 \in X$ of dimension n such that $U = W_1 \cap W_2$ then by [Abr96, Lemma 31 (iii)] we have that $W_i \tilde{\cap} \mathcal{E}$ and hence $W_1, W_2 \in Y$.

In particular, any simplex in $\tilde{T}$ contains at most two vertices which are spaces of dimension n .

For each $U \in Y \setminus \tilde{Y}$ pick one of the two n -dimensional subspaces containing it and denote it by W_U . Note that if $W \in Y$ is connected to U by an edge in T , i.e. $W \subseteq U$ or $U \subseteq W$ then $\{W_U, W\} \in T(1)$ if and only if $W \neq W_U$.

We define the following two subsets of $T(j)$ for $0 \leq j \leq n$:

$$I_j = \{\sigma \in T(j) \mid \exists U \in \sigma(0) : \dim U = n-1\}, \quad J_j = \{\sigma \in T(j) \mid \exists U \in \sigma(0) : \dim U = n-1 \text{ and } W_U \notin \sigma(0)\}.$$

Note that $I_j = T(j) \setminus \tilde{T}(j)$. Let $\sigma \in J_j$ then we define $\sigma \langle U, W_U \rangle \in \overrightarrow{\tilde{T}(j)}$ to be the oriented simplex with vertex W_U at the position of U instead of U . This is well defined by the above observation. Similarly, if $\sigma \in \tilde{T}(j)$ with $W_1, W_2 \in \sigma(0)$, $\dim W_1 = \dim W_2 = n$ and $U = W_1 \cap W_2$ has dimension $n-1$, we write $\sigma \langle W_i, U \rangle$ for the oriented simplex where we replaced W_i with U . This is now a well-defined simplex in $\overrightarrow{\tilde{T}(j)}$.

Next, we define the function $f_j : C_j(\tilde{T}) \rightarrow C_j(T)$, $0 \leq j \leq n-1$ by its action on chains of the form $\mathbb{1}_\sigma$ and then extending $\mathbb{Z}$ -linearly:

$$f_j(\mathbb{1}_\sigma) = \begin{cases} \mathbb{1}_\sigma & \text{if } \sigma \in \overrightarrow{\tilde{T}(j)} \cap \overrightarrow{\tilde{T}(j)} \\ \mathbb{1}_{\sigma \langle W_1, U \rangle} + \mathbb{1}_{\sigma \langle W_2, U \rangle} & \text{if } \sigma \in \overrightarrow{\tilde{T}(j)} \setminus \overrightarrow{\tilde{T}(j)}, W_1, W_2 \in \sigma(0), \dim W_i = n, U = W_1 \cap W_2, \dim U = n-1. \end{cases}$$

This is indeed well-defined since the only simplices in $\tilde{T}$ that are not simplices in T are those that have two n -dimensional subspaces as vertices.

On the other hand, we define

$$g_j : C_j(T) \rightarrow C_j(\tilde{T})$$

$$\mathbb{1}_\sigma \mapsto \begin{cases} \mathbb{1}_\sigma & \text{if } \sigma \in \overrightarrow{\tilde{T}(j)} \cap \overrightarrow{\tilde{T}(j)} \\ \mathbb{1}_{\sigma \langle U, W_U \rangle} & \text{if } \sigma \in J_j \\ 0 & \text{if } \sigma \in I_j \setminus J_j. \end{cases}$$

We define $\text{Cone}_{\tilde{T}}^j = g_{j+1} \circ \text{Cone}_T^j \circ f_j$ for all $-1 \leq j \leq n-1$, where $\text{Cone}_T^j := \text{Cone}_T|_{C_j(T)}$ and $\text{Cone}_{\tilde{T}} = \bigoplus_{j=-1}^{n-1} \text{Cone}_{\tilde{T}}^j$. The majority of the rest of the proof is dedicated to showing that this is indeed a cone function. As concatenation of $\mathbb{Z}$ -linear functions it is $\mathbb{Z}$ -linear. If $[U]$ is the apex of Cone_T then the apex of $\text{Cone}_{\tilde{T}}$ is again $[U]$ if $\dim U \neq n-1$ and otherwise it is $[W_U]$.

Hence, we are left with showing the cone equation. Let $\sigma \in \overrightarrow{T(j)}$. We write $\text{Cone}_T(\mathbb{1}_\sigma) = \sum_{\tau \in \overrightarrow{T(j+1)}} \lambda_\tau \mathbb{1}_\tau$ for $\lambda_\tau \in \mathbb{Z}$. In the following, if $\tau \in I_{j+1}$ is a simplex, then U will always denote the subspace of dimension $n-1$ in $\tau(0)$. Additionally, $[\sigma : \tau]$ will denote the oriented incidence number, which is 0 if $\tau \not\subset \sigma$ and 1 or -1 if $\tau \subset \sigma$ depending on the position of the vertex of σ that was removed to obtain τ . It is chosen in such a way that for $\sigma \in \overrightarrow{T(j+1)}$ we have $\partial_{j+1} \mathbb{1}_\sigma = \sum_{\tau \in \overrightarrow{T(j)}} [\sigma : \tau] \mathbb{1}_\tau$. In the calculations below we will slightly abuse notation as $\text{Cone}_T(\mathbb{1}_\sigma)$ is not a chain in $\tilde{T}$, but this is fine since the problematic terms sum to zero. We have

$$\begin{aligned} \partial_{j+1}(g_{j+1}(\text{Cone}_T(\mathbb{1}_\sigma))) &= \partial_{j+1}(\text{Cone}_T(\mathbb{1}_\sigma)) - \sum_{\tau \in I_{j+1}} \lambda_\tau \partial_{j+1} \mathbb{1}_\tau + \sum_{\tau \in J_{j+1}} \lambda_\tau \partial_{j+1} \mathbb{1}_{\tau\langle U, W_U \rangle} \\ &= \mathbb{1}_\sigma - \text{Cone}_T(\partial_j \mathbb{1}_\sigma) - \sum_{\tau \in I_{j+1}} \lambda_\tau \sum_{\gamma \in \overrightarrow{T(j)}} [\tau : \gamma] \mathbb{1}_\gamma \\ &\quad + \sum_{\tau \in J_{j+1}} \lambda_\tau [\tau : \tau \setminus \{U\}] \mathbb{1}_{\tau \setminus \{U\}} + \sum_{\tau \in J_{j+1}} \lambda_\tau \sum_{\gamma \in \overrightarrow{T(j)} : \gamma \neq \tau \setminus \{U\}} [\tau : \gamma] \mathbb{1}_{\gamma\langle U, W_U \rangle} \end{aligned}$$

where we used that

$$\partial_{j+1} \mathbb{1}_{\tau\langle U, W_U \rangle} = \sum_{\gamma \in \overrightarrow{T(j)} : \gamma \neq \tau \setminus \{U\}} [\tau : \gamma] \mathbb{1}_{\gamma\langle U, W_U \rangle} + [\tau : \tau \setminus \{U\}] \mathbb{1}_{\tau \setminus \{U\}}.$$

Note that $(\tau\langle U, W_U \rangle) \setminus \{W_U\} = \tau \setminus \{U\}$ and $[\tau : \gamma\langle U, W_U \rangle] = [\tau : \gamma]$.

We take a closer look at the second half of that expression:

$$\begin{aligned} & - \sum_{\tau \in I_{j+1}} \lambda_\tau \sum_{\gamma \in \overrightarrow{T(j)}} [\tau : \gamma] \mathbb{1}_\gamma + \sum_{\tau \in J_{j+1}} \lambda_\tau [\tau : \tau \setminus \{U\}] \mathbb{1}_{\tau \setminus \{U\}} + \sum_{\tau \in J_{j+1}} \lambda_\tau \sum_{\gamma \in \overrightarrow{T(j)} : \gamma \neq \tau \setminus \{U\}} [\tau : \gamma] \mathbb{1}_{\gamma\langle U, W_U \rangle} \\ &= - \sum_{\tau \in I_{j+1}} \lambda_\tau \sum_{\gamma \in \overrightarrow{T(j)} : \gamma \neq \tau \setminus \{U\}} [\tau : \gamma] \mathbb{1}_\gamma + \sum_{\tau \in J_{j+1}} \lambda_\tau \sum_{\gamma \in \overrightarrow{T(j)} : \gamma \neq \tau \setminus \{U\}} [\tau : \gamma] \mathbb{1}_{\gamma\langle U, W_U \rangle} \\ &\quad - \sum_{\tau \in I_{j+1}} \lambda_\tau [\tau : \tau \setminus \{U\}] \mathbb{1}_{\tau \setminus \{U\}} + \sum_{\tau \in J_{j+1}} \lambda_\tau [\tau : \tau \setminus \{U\}] \mathbb{1}_{\tau \setminus \{U\}} \\ &= - \sum_{\tau \in I_{j+1}} \lambda_\tau \sum_{\gamma \in \overrightarrow{T(j)} : \gamma \neq \tau \setminus \{U\}} [\tau : \gamma] \mathbb{1}_\gamma + \sum_{\tau \in J_{j+1}} \lambda_\tau \sum_{\gamma \in \overrightarrow{T(j)} : \gamma \neq \tau \setminus \{U\}} [\tau : \gamma] \mathbb{1}_{\gamma\langle U, W_U \rangle} \\ &\quad - \sum_{\tau \in I_{j+1} \setminus J_{j+1}} \lambda_\tau [\tau : \tau \setminus \{U\}] \mathbb{1}_{\tau \setminus \{U\}} \end{aligned}$$

On the other hand, let $\lambda'_\gamma \in \mathbb{Z}$ such that $\text{Cone}_T(\partial \mathbb{1}_\sigma) = \sum_{\gamma \in \overrightarrow{T(j)}} \lambda'_\gamma \mathbb{1}_\gamma$. Then

$$g_j(\text{Cone}_T(\partial \mathbb{1}_\sigma)) = \text{Cone}_T(\partial \mathbb{1}_\sigma) - \sum_{\gamma \in I_j} \lambda'_\gamma \mathbb{1}_\gamma + \sum_{\gamma \in J_j} \lambda'_\gamma \mathbb{1}_{\gamma\langle U, W_U \rangle}.$$

What are the λ'_γ in terms of the λ_τ ?

$$\begin{aligned}
\text{Cone}_T(\partial \mathbb{1}_\sigma) &= \mathbb{1}_\sigma - \partial_{j+1} \text{Cone}_T(\mathbb{1}_\sigma) = \mathbb{1}_\sigma - \partial_{j+1} \sum_{\tau \in \overrightarrow{T(j+1)}} \lambda_\tau \mathbb{1}_\tau \\
&= \mathbb{1}_\sigma - \sum_{\tau \in \overrightarrow{T(j+1)}} \lambda_\tau \sum_{\gamma \in \overrightarrow{T(j)}} [\tau : \gamma] \mathbb{1}_\gamma \\
&= \mathbb{1}_\sigma - \sum_{\gamma \in \overrightarrow{T(j)}} \left(\sum_{\tau \in \overrightarrow{T(j+1)}} \lambda_\tau [\tau : \gamma] \right) \mathbb{1}_\gamma.
\end{aligned}$$

Hence if $\gamma \neq \sigma$ then

$$\lambda'_\gamma = - \sum_{\tau \in \overrightarrow{T(j+1)}} \lambda_\tau [\tau : \gamma]$$

and otherwise

$$\lambda'_\sigma = 1 - \sum_{\tau \in \overrightarrow{T(j+1)}} \lambda_\tau [\tau : \sigma].$$

To compare the expression of $g_j(\text{Cone}_T(\partial_j \mathbb{1}_\sigma))$ with $\partial_{j+1}(g_{j+1}(\text{Cone}_T(\mathbb{1}_\sigma)))$ we furthermore need the following observation comparing the sets I_j with I_{j+1} and J_j with J_{j+1} .

$$\left\{ (\tau, \gamma) : \gamma \in I_j, \tau \in \overrightarrow{T(j+1)}, \gamma \subset \tau \right\} = \left\{ (\tau, \gamma) : \tau \in I_{j+1}, \gamma \in \overrightarrow{T(j)}, \gamma \subset \tau, \gamma \neq \tau \setminus \{U\} \right\}$$

and

$$\begin{aligned}
&\left\{ (\tau, \gamma) : \gamma \in J_j, \tau \in \overrightarrow{T(j+1)}, \gamma \subset \tau \right\} \\
&= \left\{ (\tau, \gamma) : \tau \in J_{j+1}, \gamma \in \overrightarrow{T(j)}, \gamma \subset \tau, \gamma \neq \tau \setminus \{U\} \right\} \cup \left\{ (\tau, \gamma) : \tau \in I_{j+1} \setminus J_{j+1}, \gamma = \tau \setminus \{W_U\} \right\}.
\end{aligned}$$

For $\sigma \in \overrightarrow{T(j)}$ we define

$$h_\sigma = \begin{cases} 0 & \text{if } \sigma \notin I_j \\ \mathbb{1}_\sigma & \text{if } \sigma \in I_j \setminus J_j \\ \mathbb{1}_\sigma - \mathbb{1}_{\sigma \setminus \langle U, W_U \rangle} & \text{if } \sigma \in J_j \end{cases}$$

We use the above result and the new notation to further investigate the second half of the expression

for $g_j(\text{Cone}_T(\partial \mathbb{1}_\sigma))$.

$$\begin{aligned}
& - \sum_{\gamma \in I_j} \lambda'_\gamma \mathbb{1}_\gamma + \sum_{\gamma \in J_j} \lambda'_\gamma \mathbb{1}_{\gamma \langle U, W_U \rangle} = - \sum_{\gamma \in I_j \setminus \{\sigma\}} \left(- \sum_{\tau \in \overrightarrow{T(j+1)}} \lambda_\tau [\tau : \gamma] \right) \mathbb{1}_\gamma + \sum_{\gamma \in J_j \setminus \{\sigma\}} \left(- \sum_{\tau \in \overrightarrow{T(j+1)}} \lambda_\tau [\tau : \gamma] \right) \mathbb{1}_{\gamma \langle U, W_U \rangle} \\
& - \begin{cases} 0 & \text{if } \sigma \notin I_j \\ 1 - \sum_{\tau \in \overrightarrow{T(j+1)}} \lambda_\tau [\tau : \sigma] \mathbb{1}_\sigma & \text{if } \sigma \in I_j \end{cases} + \begin{cases} 0 & \text{if } \sigma \notin J_j \\ 1 - \sum_{\tau \in \overrightarrow{T(j+1)}} \lambda_\tau [\tau : \sigma] \mathbb{1}_{\sigma \langle U, W_U \rangle} & \text{if } \sigma \in J_j \end{cases} \\
& = - \sum_{\gamma \in I_j} \left(- \sum_{\tau \in \overrightarrow{T(j+1)}} \lambda_\tau [\tau : \gamma] \right) \mathbb{1}_\gamma + \sum_{\gamma \in J_j} \left(- \sum_{\tau \in \overrightarrow{T(j+1)}} \lambda_\tau [\tau : \gamma] \right) \mathbb{1}_{\gamma \langle U, W_U \rangle} - h_\sigma \\
& = \sum_{\tau \in I_{j+1}} \sum_{\gamma \in \overrightarrow{T(j)} : \gamma \neq \tau \setminus \{U\}} [\tau : \gamma] \lambda_\tau \mathbb{1}_\gamma - \sum_{\tau \in J_{j+1}} \sum_{\gamma \in \overrightarrow{T(j)} : \gamma \neq \tau \setminus \{U\}} [\tau : \gamma] \lambda_\tau \mathbb{1}_{\gamma \langle U, W_U \rangle} \\
& \quad - \sum_{\tau \in I_{j+1} \setminus J_{j+1}} [\tau : \tau \setminus \{W_U\}] \lambda_\tau \mathbb{1}_{(\tau \setminus \{W_U\}) \langle U, W_U \rangle} - h_\sigma.
\end{aligned}$$

We now look at almost the complete cone equation for $\text{Cone}_{\tilde{T}}$, still not considering f_j . Let $\sigma \in \overrightarrow{T(j)}$, then we get

$$\begin{aligned}
& \partial_{j+1} g_{j+1}(\text{Cone}_T(\mathbb{1}_\sigma)) + g_j(\text{Cone}_T(\partial_j \mathbb{1}_\sigma)) = \underbrace{\mathbb{1}_\sigma - \text{Cone}_T(\partial_j \mathbb{1}_\sigma)}_{(1)} \\
& - \underbrace{\sum_{\tau \in I_{j+1}} \lambda_\tau \sum_{\gamma \in \overrightarrow{T(j)} : \gamma \neq \tau \setminus \{U\}} [\tau : \gamma] \mathbb{1}_\gamma}_{(2)} + \underbrace{\sum_{\tau \in J_{j+1}} \lambda_\tau \sum_{\gamma \in \overrightarrow{T(j)} : \gamma \neq \tau \setminus \{U\}} [\tau : \gamma] \mathbb{1}_{\gamma \langle U, W_U \rangle}}_{(3)} \\
& - \underbrace{\sum_{\tau \in I_{j+1} \setminus J_{j+1}} \lambda_\tau [\tau : \tau \setminus \{U\}] \mathbb{1}_{\tau \setminus \{U\}}}_{(4)} \\
& + \underbrace{\text{Cone}_T(\partial \mathbb{1}_\sigma)}_{(1)} + \underbrace{\sum_{\tau \in I_{j+1}} \sum_{\gamma \in \overrightarrow{T(j)} : \gamma \neq \tau \setminus \{U\}} [\tau : \gamma] \lambda_\tau \mathbb{1}_\gamma}_{(2)} - \underbrace{\sum_{\tau \in J_{j+1}} \sum_{\gamma \in \overrightarrow{T(j)} : \gamma \neq \tau \setminus \{U\}} [\tau : \gamma] \lambda_\tau \mathbb{1}_{\gamma \langle U, W_U \rangle}}_{(3)} \\
& - \underbrace{\sum_{\tau \in I_{j+1} \setminus J_{j+1}} [\tau : \tau \setminus \{W_U\}] \lambda_\tau \mathbb{1}_{(\tau \setminus \{W_U\}) \langle U, W_U \rangle}}_{(4)} - h_\sigma.
\end{aligned}$$

The parts marked (1), (2), and (3) cancel each other out. It remains to show that the same is true for the parts marked (4). Note that the sums in the parts marked (4) run over the same elements. Hence we need to show that for any $\tau \in I_{j+1} \setminus J_{j+1}$ we have

$$[\tau : \tau \setminus \{U\}] \mathbb{1}_{\tau \setminus \{U\}} = -[\tau : \tau \setminus \{W_U\}] \mathbb{1}_{(\tau \setminus \{W_U\}) \langle U, W_U \rangle}$$

Note that as unoriented simplices $\tau \setminus \{U\}$ is the same as $(\tau \setminus \{W_U\})\langle U, W_U \rangle$ (in both cases we have all the vertices of τ except U). Let $\tau = [U_0, \dots, U_{j+1}]$ and assume that $U = U_\ell$ and $W_U = U_k$ for some $0 \leq \ell, k \leq j+1$. Then $[\tau : \tau \setminus \{U\}] = (-1)^\ell$ and $[\tau : \tau \setminus \{W_U\}] = (-1)^k$. Further note that we can obtain $(\tau \setminus \{W_U\})\langle U, W_U \rangle$ from $\tau \setminus \{U\}$ by cyclically permuting the vertices of index between $\min\{\ell, k\}$ and $\max\{\ell, k\} - 1$. The cycle has length $|k - \ell| - 1$ and thus signature $(-1)^{k-\ell-1}$. Hence $(-1)^k \mathbb{1}_{(\tau \setminus \{W_U\})\langle U, W_U \rangle} = (-1)^k \cdot (-1)^{k-\ell-1} \mathbb{1}_{\tau \setminus \{U\}} = -(-1)^\ell \mathbb{1}_{\tau \setminus \{U\}}$ which is what we wanted to show.

To summarize, we get

$$\partial_{j+1}g_{j+1}(\text{Cone}_T(\mathbb{1}_\sigma)) + g_j(\text{Cone}_T(\partial_j \mathbb{1}_\sigma)) = \mathbb{1}_\sigma - h_\sigma = \begin{cases} \mathbb{1}_\sigma & \text{if } \sigma \notin I_j \\ 0 & \text{if } \sigma \in I_j \setminus J_j \\ \mathbb{1}_{\sigma\langle U, W_U \rangle} & \text{if } \sigma \in J_j \end{cases}.$$

Note that if $\sigma \in \overrightarrow{T(j)} \cap \overrightarrow{\widetilde{T}(j)}$, then $f_j(\mathbb{1}_\sigma) = \mathbb{1}_\sigma$ which shows that the cone equation holds in this case. To conclude the proof, we need to check the cone equation also for the case where $\sigma \in \overrightarrow{\widetilde{T}(j)} \setminus \overrightarrow{T(j)}$. In this case $\sigma(0)$ contains two subspaces W_1, W_2 with $\dim W_i = n$ and $U := W_1 \cap W_2$ has dimension $n - 1$. Without loss of generality, assume that $W_U = W_1$. We get $f_j(\mathbb{1}_\sigma) = \mathbb{1}_{\sigma\langle W_1, U \rangle} + \mathbb{1}_{\sigma\langle W_2, U \rangle}$. We have that $\sigma\langle W_1, U \rangle \in J_j$ and $\sigma\langle W_2, U \rangle \in I_j \setminus J_j$ thus we can apply the above reasoning to both of them separately and get, using linearity:

$$\begin{aligned} & \partial_{j+1}g_{j+1}(\text{Cone}_T(f_j(\mathbb{1}_\sigma))) + g_j(\text{Cone}_T(f_{j-1}(\partial_j \mathbb{1}_\sigma))) \\ &= \partial_{j+1}g_{j+1}(\text{Cone}_T(\mathbb{1}_{\sigma\langle W_1, U \rangle})) + g_j(\text{Cone}_T(\partial_j \mathbb{1}_{\sigma\langle W_1, U \rangle})) \\ & \quad + \partial_{j+1}g_{j+1}(\text{Cone}_T(\mathbb{1}_{\sigma\langle W_2, U \rangle})) + g_j(\text{Cone}_T(\partial_j \mathbb{1}_{\sigma\langle W_2, U \rangle})) \\ &= \mathbb{1}_{(\sigma\langle W_1, U \rangle)\langle U, W_U \rangle} + 0 = \mathbb{1}_\sigma. \end{aligned}$$

For the last equality, we used that $(\sigma\langle W_1, U \rangle)\langle U, W_U \rangle = \sigma$ since we first replace W_1 in σ by U and by applying $\langle U, W_U \rangle$ we replace U by $W_U = W_1$ and end up where we started. We furthermore use that $f_{j-1}\partial_j \mathbb{1}_\sigma = \partial_j f_j \mathbb{1}_\sigma$. This is indeed true as the following argument shows. We have

$$\partial_j \mathbb{1}_\sigma = \sum_{\{W_1, W_2\} \subset \gamma \subset \sigma} [\sigma : \gamma] \mathbb{1}_\gamma + \sum_{i \in \{1, 2\}} [\sigma : \sigma \setminus \{W_i\}] \mathbb{1}_{\sigma \setminus \{W_i\}}$$

and thus

$$f_{j-1}\partial_j \mathbb{1}_\sigma = \sum_{i \in \{1, 2\}} \left([\sigma : \sigma \setminus \{W_i\}] \mathbb{1}_{\sigma \setminus \{W_i\}} + \sum_{\{W_1, W_2\} \subset \gamma \subset \sigma} [\sigma : \gamma] \mathbb{1}_{\gamma\langle W_i, U \rangle} \right).$$

On the other hand,

$$\partial_j f_j \mathbb{1}_\sigma = \sum_{i \in \{1, 2\}} \partial_j \mathbb{1}_{\sigma\langle W_i, U \rangle} = \sum_{i \in \{1, 2\}} \sum_{\gamma \subset \sigma\langle W_i, U \rangle} [\sigma\langle W_i, U \rangle : \gamma] \mathbb{1}_\gamma.$$

The set $\{\gamma : \{W_1, W_2\} \subset \gamma \subset \sigma\}$ is in bijection with the set $\{\gamma : \{U, W_{3-i}\} \subset \gamma \subset \sigma\langle W_i, U \rangle\}$ via the map $\gamma \mapsto \gamma\langle W_i, U \rangle$, which preserves the oriented incidence number. Thus,

$$\begin{aligned} & \sum_{\{W_1, W_2\} \subset \gamma \subset \sigma} [\sigma : \gamma] \mathbb{1}_{\gamma\langle W_i, U \rangle} = \sum_{\{U, W_{3-i}\} \subset \gamma \subset \sigma\langle W_i, U \rangle} [\sigma\langle W_i, U \rangle : \gamma] \mathbb{1}_\gamma \\ &= \partial_j \mathbb{1}_{\sigma\langle W_i, U \rangle} - [\sigma\langle W_i, U \rangle : \sigma\langle W_i, U \rangle \setminus \{W_{3-i}\}] \mathbb{1}_{\sigma\langle W_i, U \rangle \setminus \{W_{3-i}\}} - [\sigma\langle W_i, U \rangle : \sigma \setminus \{U\}] \mathbb{1}_{\sigma\langle W_i, U \rangle \setminus \{U\}}. \end{aligned}$$

The last term is equal to $[\sigma : \sigma \setminus \{W_i\}] \mathbb{1}_{\sigma \setminus \{W_i\}}$ (since deleting W_i or swapping W_i to U and then deleting U does not change orientation). Adding everything up, it remains to check that

$$[\sigma \langle W_1, U \rangle : \sigma \langle W_1, U \rangle \setminus \{W_2\}] \mathbb{1}_{\sigma \langle W_1, U \rangle \setminus \{W_2\}} + [\sigma \langle W_2, U \rangle : \sigma \langle W_2, U \rangle \setminus \{W_1\}] \mathbb{1}_{\sigma \langle W_2, U \rangle \setminus \{W_1\}} = 0.$$

To see that this holds, we assume without loss of generality (by applying cyclic shifts if necessary) that $\sigma = [U_0, \dots, U_j]$ with $U_0 = W_1$ and $U_k = W_2$. Then we have

$$\begin{aligned} [\sigma \langle W_1, U \rangle : \sigma \langle W_1, U \rangle \setminus \{W_2\}] &= (-1)^k, \\ [\sigma \langle W_2, U \rangle : \sigma \langle W_2, U \rangle \setminus \{W_1\}] &= 1. \end{aligned}$$

Since $\sigma \langle W_2, U \rangle \setminus \{W_1\} = [U_2, \dots, U_{k-1}, U, U_{k+1}, \dots, U_j]$ may be obtained from $\sigma \langle W_1, U \rangle \setminus \{W_2\} = [U, U_2, \dots, U_{k-1}, U_{k+1}, \dots, U_j]$ via cyclically shifting the coordinates between 0 and $k-1$, which is a cycle of length k and has signature $(-1)^{k-1}$, we get

$$\mathbb{1}_{\sigma \langle W_2, U \rangle \setminus \{W_1\}} = (-1)^{k-1} \mathbb{1}_{\sigma \langle W_2, U \rangle \setminus \{W_2\}}$$

which concludes the argument.

For bounding the cone radius of $\text{Cone}_{\tilde{T}}$ note that f_j at most doubles the size of support of a chain, while applying g_j will not increase the size of the support of a chain. Hence in the worst case, i.e. if $\sigma \in I_j$ then

$$|\text{supp}(\text{Cone}_{\tilde{T}}(\mathbb{1}_\sigma))| \leq |\text{supp}(\text{Cone}_T(\mathbb{1}_{\sigma \langle W_1, U \rangle}))| + |\text{supp}(\text{Cone}_T(\mathbb{1}_{\sigma \langle W_2, U \rangle}))| \leq 2 \text{Rad}_j \text{Cone}_T \leq 2c.$$

Hence we get the desired bound on the cone radius. $\square$

To show that the flag complexes $T_{\mathcal{E}}(V)$ as above admit cone functions, we first need a weaker and more specialized version of Theorem 6.1 which allows for arguments via a filtration also for the links.

Theorem 6.18. *Let $\mathbf{C}$ be a class of pairs (κ, Y) where $Y \neq \emptyset$ is a poset and $\kappa = \text{Flag } Y$ (with inclusion as incidence relation). Let $n = \dim \kappa$, assume there exists $\ell_n \in \mathbb{N}$ and a filtration $\kappa_0 \subset \kappa_1 \subset \dots \subset \kappa_\ell = \kappa$ with $\ell \leq \ell_n$ satisfying:*

1. *There exists an $(n-1)$ -cone function Cone_{κ_0} such that*

$$\text{Rad}_j(\text{Cone}_{\kappa_0}) \leq f(n), \forall -1 \leq j \leq n-1$$

where $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function depending only on $\mathbf{C}$.

2. *Every κ_i is a full subcomplex of κ , i.e. $\kappa_i = \text{Flag } Y_i$ for some $Y_i \subseteq Y$ and all elements in $Y_i \setminus Y_{i-1}$ are not comparable (i.e. not incident).*
3. *For every $U \in Y_i \setminus Y_{i-1}$ we have that $\text{lk}_{\kappa_i}(U) \cap \kappa_{i-1} = \kappa_{i-1}^{<U} * \kappa_{i-1}^{>U}$ is such that $\kappa_{i-1}^{<U}, \kappa_{i-1}^{>U} \in \mathbf{C}$ or one (or both) have again a filtration $R_0 \subset \dots \subset R_h, h \leq \ell_{\dim R_0}$ which starts with an element of $\mathbf{C}$ (i.e. $R_0 \in \mathbf{C}$), such that condition 2. is satisfied and such that for every $W \in R_j(0) \setminus R_{j-1}(0)$ we have $R_{j-1}^{<W}, R_{j-1}^{>W} \in \mathbf{C}$.*

Then there exist constants $\mathcal{R}(n)$ depending only on $\mathbf{C}$ and n , and an $(n-1)$ -cone function Cone_κ such that

$$\text{Rad}_j(\text{Cone}_\kappa) \leq \mathcal{R}(n), \forall -1 \leq j \leq n-1.$$

Proof. The proof is analogous to the one of Theorem 6.1. The main difference is that to get a cone function for $\text{lk}_{\kappa_i}(U) \cap \kappa_{i-1}$ it might happen that we cannot just use the induction hypothesis, but that we need to do another induction on the filtration of $\kappa_{i-1}^{<U}$ or $\kappa_{i-1}^{>U}$, using Theorem 3.1 in the step of the induction. This gives an even weaker and more complicated bound on the cone radius of the final cone function, thus we decided to not keep track of it. But it will still only depend on n and the class $\mathbf{C}$, not on the field K . $\square$

We now describe the set up for the rest of this section.

1. We define a class $\mathbf{C}_1$.
2. We define a filtration $\kappa_0 \subset \dots \subset \kappa_\ell = \kappa$ for each $\kappa \in \mathbf{C}_1$.
3. We show that κ_0 admits a cone function with bounded radius.
4. We conclude that $\mathbf{C}_1$ satisfies the assumptions of the adapted general induction argument.
5. We define a class $\mathbf{C}_2$.
6. We proceed as for $\mathbf{C}_1$ (for this class the original induction argument suffices).
7. We define a class $\mathbf{C}_3$.
8. We define a filtration $\kappa_0 \subset \dots \subset \kappa_\ell = \kappa$ for each $\kappa \in \mathbf{C}_3$.
9. We show that κ_0 admits a cone function with bounded radius.
10. We show that $\mathbf{C}_1 \cup \mathbf{C}_2 \cup \mathbf{C}_3$ satisfies the assumptions of the adapted general induction argument.
11. We conclude that all opposite complexes of type D_n admit a cone function.

We fix n to be the Witt index of (V, Q, f) hence $\dim V = 2n$ and we are dealing with a D_n building, which is $n-1$ -dimensional. Recall $X = \{0 < U < V \mid U \text{ totally isotropic}\}$.

Definition 6.19 ([Abr96, Definition 12]). *Let $U, E < V$, $\dim U < n$, $\dim E = n$. Assume that E is not totally isotropic. Then*

$$U @ E : \iff U^\perp \cap E \text{ is not totally isotropic.}$$

Set

$$\begin{aligned} \mathcal{U}_n &= \mathcal{U}_n(V) = \{A < V \mid \dim A = n, A \notin X\} \\ \mathcal{M}(A) &= \{M < A \mid \dim M = n-1, M \in X\} \text{ for any } A \in \mathcal{U}_n. \end{aligned}$$

Definition 6.20. *We define the class $\mathbf{C}_1$ to consist of complexes $\text{Flag } X_{\mathcal{E};\mathcal{F}}(U; V)$ where*

- (V, Q, f) is a thick pseudo-quadratic space of Witt index n , in particular $\dim V = 2n$;

- $U \in X$ with $\dim U = k \geq 2$;
- $\mathcal{E}$ a finite set of subspaces of U , set $e_j = |\mathcal{E}_j|$, $1 \leq j \leq k-1$ like before;
- Let $\mathcal{F} = \mathcal{F}_1 \supset \mathcal{F}_2 \supset \dots \supset \mathcal{F}_{k-1}$ be finite subsets of $\mathcal{U}_n(V)$ satisfying
 1. $U \cap F \in \mathcal{E} \cup \{0\}$ for all $F \in \mathcal{F}$;
 2. $\mathcal{F}_j^\perp = \mathcal{F}_j$;
 3. $\dim(U \cap F) \leq k-1-j$ for all $F \in \mathcal{F}_j$, $1 \leq j \leq k-1$.
- Assume that $|K| \geq \sum_{j=1}^{k-1} \binom{k-2}{j-1} e_j + 2^{k-1}s$, where $s = |\mathcal{F}|$.

Set

$$X_{\mathcal{E};\mathcal{F}}(U;V) := \{0 < W < U \mid W \pitchfork_U \mathcal{E} \text{ and } W @_V \mathcal{F}_j \text{ for } \dim W = j\}.$$

Fact 6.21 ([Abr96, Lemma 33, Step 1]). *Let $\kappa = \text{Flag } X_{\mathcal{E};\mathcal{F}}(U;V) \in \mathbf{C}_1$ with $k = \dim U \geq 2$, in particular κ has dimension $k-2$. Then there exists a 1-dimensional subspace $\ell \in X_{\mathcal{E};\mathcal{F}}(U;V)$.*

For $\kappa = \text{Flag } X_{\mathcal{E};\mathcal{F}}(U;V) \in \mathbf{C}_1$ with $k = \dim U \geq 3$, we set $Y = X_{\mathcal{E};\mathcal{F}}(U;V)$ and define a filtration as follows: We fix a 1-dimensional subspace $\ell \in X_{\mathcal{E};\mathcal{F}}(U;V)$ and set

$$Y_0 := \{A \in Y \mid A + \ell \in Y\}.$$

The filtration will then consist of two dimension decreasing filtrations. First we define

$$\begin{aligned} Z &:= \{W \in Y \mid \ell \leq W \text{ or } W \pitchfork_U (\mathcal{F} \cap U) + \ell\} \\ Y_i &:= \{W \in Z \mid W \in Y_0 \text{ or } \dim W \geq k-i\}, \\ \kappa_i &:= \text{Flag } Y_i, \quad 1 \leq i \leq k-1. \end{aligned}$$

The second part of the filtration of κ is defined as follows:

$$\begin{aligned} Z_i &:= \{W \in Y \mid W \in Z \text{ or } \dim W \geq k-1\}, \\ \kappa_{k+i} &:= \text{Flag } Z_i, \quad 0 \leq i \leq k-1. \end{aligned}$$

Fact 6.22 ([Abr96, Step 2, Lemma 33]). *We have*

$$Y_0 = \left\{ W \in Y \mid \ell \leq W \vee (\dim W < k-1 \wedge W \pitchfork_U \mathcal{E} + \ell \wedge W @_V (\mathcal{F}_{\dim W} \cup ((\mathcal{F}_{\dim W+1} \cap \ell^\perp) + \ell))) \right\} \subseteq Z$$

using that $W \pitchfork_U \mathcal{E} + \ell$ implies $W \pitchfork_U (\mathcal{F} \cap U) + \ell$ since by assumption $\mathcal{F} \cap U \subseteq \mathcal{E}$.

Lemma 6.23. *With the definitions as above, we have that κ_0 has a $(k-3)$ -cone function Cone_{κ_0} with*

$$\text{Rad}_j(\text{Cone}_{\kappa_0}) \leq k, \quad -1 \leq j \leq k-3.$$

Proof. We proceed similar to the proof of Theorem 6.7. Set $B_0 := \{A \in Y \mid \ell \leq A\}$, $\beta_0 = \text{Flag } B_0$. Then $\beta_0 = \{\ell\} * \text{lk}_Y(\{\ell\})$ and hence it has a $(k-3)$ -cone function with cone radius bounded by 1. Next, set $B_i = \{A \in Y_0 \mid \ell \leq A \text{ or } \dim A \leq i\}$, $\beta_i = \text{Flag } B_i$ for $1 \leq i \leq k-1$. Then all subspaces in $B_i \setminus B_{i-1}$ have dimension i . Let $A \in B_i \setminus B_{i-1}$. Then $\text{lk}_{\beta_i}(A) \cap \beta_{i-1} = \{A + \ell\} * \text{Flag}\{W \in B_{i-1} \mid W \neq A + \ell, A < W \text{ or } W < A\}$ by the same argument as in Theorem 6.7. Applying Theorem 3.1 inductively, we get a $(k-3)$ -cone function $\text{Cone}_{\beta_{k-1}} = \text{Cone}_{\kappa_0}$ with

$$\text{Rad}_j(\text{Cone}_{\kappa_0}) \leq k, \quad -1 \leq j \leq k-3.$$

□

We collect some facts about the filtration of κ from [Abr96, Lemma 33].

Fact 6.24 ([Abr96, Step 3-5, Lemma 33]). *Let $W \in Y_i \setminus Y_{i-1}$. Then $Y_{i-1}^{<W} = Y_0^{<W}$ (this can be seen directly by looking at the dimension). Furthermore, $Y_0^{<W} = X_{\mathcal{E}', \mathcal{F}'}(W; V)$ for some $\mathcal{E}', \mathcal{F}'$. Thus $\kappa_{i-1}^{<W} \in \mathbf{C}_1$. On the other hand, $Y_{i-1}^{>W} = Z^{>W}$. While $\text{Flag } Z^{>W}$ is not necessarily in $\mathbf{C}_1$ it admits itself a filtration of length i satisfying the necessary properties.*

Fact 6.25 ([Abr96, Step 4-5, Lemma 33]). *Let $W \in Z_i \setminus Z_{i-1}$. Then $Z_{i-1}^{>W} = Y^{>W}$ and $\text{Flag } Y^{>W} \in \mathbf{C}_1$. On the other hand, $Z_{i-1}^{<W} = Z^{<W}$ for which we have $\text{Flag } Z^{<W} \in \mathbf{C}_1$.*

Lemma 6.26. *The class $\mathbf{C}_1$ satisfies the assumptions of Theorem 6.18. In particular, any complex in $\mathbf{C}_1$ of dimension $k - 2$ has a $(k - 3)$ -cone function with cone radius bounded by a constant depending only on k .*

Proof. Let $\kappa = \text{Flag } X_{\mathcal{E}, \mathcal{F}}(U; V) \in \mathbf{C}_1$ and $k = \dim U = \dim \kappa + 2$. By Fact 6.21 we can fix a 1-dimensional subspace $\ell \in X_{\mathcal{E}, \mathcal{F}}(U; V) =: Y$. In particular, if κ is 0-dimensional (i.e. $k = 2$), it is not empty. Thus we can define a (-1) -cone function for κ with cone radius 1. Hence, we get a filtration of κ with $\ell_0 = 0$. For $k \geq 3$, we use the filtration defined above. By Theorem 6.23, there exists a $(k - 3)$ -cone function Cone_{κ_0} such that

$$\text{Rad}_j(\text{Cone}_{\kappa_0}) \leq k =: f(k), \forall -1 \leq j \leq k - 3.$$

Combining Facts 6.24, 6.25 we see that the filtration defined above indeed satisfies the requirements of Theorem 6.18. $\square$

Definition 6.27. *We define the class $\mathbf{C}_2$ to consist of complexes $\text{Flag } Z_{\mathcal{E}, \mathcal{F}}(U; V)$ where*

- (V, Q, f) is a thick pseudo-quadratic space of Witt index n , in particular $\dim V = 2n$;
- $U \in X$ with $\dim U = k \geq 2$;
- $\mathcal{E}$ a finite set of subspaces of U , set $e_j = |\mathcal{E}_j|, 1 \leq j \leq k - 1$ like before;
- $\mathcal{F} \subset \mathcal{U}_n(V)$ with $|\mathcal{F}| = s < \infty$ such that
 1. $U \cap \mathcal{F} \subseteq \mathcal{E} \cup \{0\}$,
 2. $U \cap F^\perp = 0$ and $\dim(U \cap F) \leq 1$ for all $F \in \mathcal{F}$.
- Assume $|K| \geq \sum_{j=1}^{k-1} \binom{k-2}{j-1} e_j + 2s$.
- Set

$$Z_{\mathcal{E}, \mathcal{F}}(U; V) := \{0 < W < U \mid W \pitchfork_U \mathcal{E} \text{ and } W @_V \mathcal{F}\}.$$

Let $\kappa = \text{Flag } Z_{\mathcal{E}, \mathcal{F}}(U; V) \in \mathbf{C}_2$, $\dim U = k \geq 2$, in particular $\dim \kappa = k - 2$. Set $Y = Z_{\mathcal{E}, \mathcal{F}}(U; V)$.

Fact 6.28 ([Abr96, Lemma 34, Step 3]). *There exists $H < U$ with $\dim H = k - 1$ and $H \in Y$.*

Using such an H , we define a filtration for κ as follows. We set

$$Y_0 := \{B \in Y \mid B \cap H \in Y\}$$

and

$$\begin{aligned} Y_i &:= \{W \in Y \mid W \in Y_0 \text{ or } \dim W \leq i\}, \\ \kappa_i &:= \text{Flag } Y_i, \quad 1 \leq i \leq k - 1. \end{aligned}$$

Lemma 6.29. *The complex $\kappa_0 = \text{Flag } Y_0$ has a $(k-3)$ -cone function Cone_{κ_0} with*

$$\text{Rad}_j(\text{Cone}_{\kappa_0}) \leq k, \quad -1 \leq j \leq k-3.$$

Proof. The proof is similar to the proofs of Theorem 6.7 and Theorem 6.23, the difference being that we now look at the dual version, where we fix a hyperplane instead of a line.

Set $B_0 = \{A \in Y \mid A \leq H\}$, $\beta_0 = \text{Flag } B_0$. Then $\beta_0 = \{H\} * \text{lk}_{Y_0}(\{H\})$ hence it has a $(k-3)$ -cone function with cone radius bounded by 1.

Next, set $B_i = \{A \in Y_0 \mid A \leq H \text{ or } \dim A \geq k-i\}$, $\beta_i = \text{Flag } B_i$ for $1 \leq i \leq k-1$. Then the subspaces in $B_i \setminus B_{i-1}$ are all of dimension $k-i$. Let $A \in B_i \setminus B_{i-1}$, then $\text{lk}_{\beta_i}(A) \cap \beta_{i-1} = \{A \cap H\} * \text{Flag}\{W \in B_{i-1} \mid W \neq A \cap H, W < A \text{ or } A < W\}$. Hence it has a $(k-4)$ -cone function with cone radius bounded by 1. Thus we can apply Theorem 3.1 inductively and obtain a $(k-3)$ -cone function $\text{Cone}_{B_{k-1}} = \text{Cone}_{\kappa_0}$ with

$$\text{Rad}_j(\text{Cone}_{\kappa_0}) \leq k, \quad -1 \leq j \leq k-3.$$

□

We recall the following fact about the filtration of κ .

Fact 6.30 ([Abr96, Step 4, Step 5, Lemma 34]). *Let $W \in Y_i \setminus Y_{i-1}$. Then $Y_{i-1}^{<W} = Y^{<W} = \{0 < A < W \mid A \cap W \in \mathcal{E} \cap W\} = Z_{\mathcal{E} \cap W; \emptyset}(W; V)$ and hence $\text{Flag } Y^{<W} \in \mathbf{C}_2$. On the other hand, $Y_{i-1}^{>W} = Y_0^{>W}$ can be described in such a way that $\text{Flag } Y_0^{>W} \in \mathbf{C}_2$.*

Lemma 6.31. *The class $\mathbf{C}_2$ satisfies the assumptions of Theorem 6.1. In particular, any complex in $\mathbf{C}_2$ of dimension $k-2$ has a $(k-3)$ -cone function with cone radius bounded depending only on k .*

Proof. Let $\kappa = \text{Flag } Z_{\mathcal{E}; \mathcal{F}}(U; V) \in \mathbf{C}_2$. Then using Fact 6.28 we fix $H < U$ such that $\dim H = \dim U - 1 = k-1$ and $H \in Z_{\mathcal{E}; \mathcal{F}}(U; V) =: Y$. In particular, $\kappa \neq \emptyset$. We prove the existence of a desired filtration by induction on k . For $k=2$ we have that $\dim \kappa = 0$ hence the desired filtration and cone function exist trivially. Fix $k \geq 3$. We consider the filtration for κ as defined above. This clearly satisfies assumption 2. of Theorem 6.1. Assumption 3. is satisfied by Fact 6.30. Since $\kappa_{k-1} = \kappa$, the filtration is indeed as desired. □

Next, we will define a third class, which will be the one containing the relevant opposite complexes, see also Theorem 6.40.

Definition 6.32. *We define the class $\mathbf{C}_3$ to consist of complexes of the form $\text{Flag } Y_{\mathcal{E}}(V)$ where*

- (V, Q, f) is a thick pseudo-quadratic space of Witt index n , in particular $\dim V = 2n$;
- $\mathcal{E} = \mathcal{E}^\perp$ is a finite set of subspaces of V , set $\mathcal{E}_j = \{E \in \mathcal{E} : \dim E = j\}$, $e_j = |\mathcal{E}_j|$;
- $|K| \geq 2 \sum_{j=1}^{2n-1} \binom{2n-2}{j-1} e_j$;
- $\hat{\mathcal{E}}_n := \mathcal{E}_n \cap \mathcal{U}_n(V)$;
- $X = \{0 < U < V \mid U \text{ is totally isotropic}\}$.

Set

$$Y_{\mathcal{E}}(V) := \{U \in X \mid U \tilde{\cap} \mathcal{E} \text{ and if } \dim U < n \text{ then } U @ \hat{\mathcal{E}}_n\}.$$

Let $\kappa = \text{Flag } Y_{\mathcal{E}}(V) \in \mathbf{C}_3$ with $\dim V = 2n$.

Fact 6.33 ([Abr96, Proposition 14, Step 1]). *There exists a 1-dimensional subspace $\ell \in Y = Y_{\mathcal{E}}(V)$.*

Assume $n \geq 3$ and fix an ℓ as above. Set

$$\begin{aligned} \mathcal{F} &:= \mathcal{E} \cup (\mathcal{E} \cap \ell^\perp) \cup (\mathcal{E} + \ell) \cup ((\mathcal{E} \cap \ell^\perp) + \ell) \text{ and} \\ \hat{\mathcal{F}}_n &:= \mathcal{F}_n \cap \mathcal{U}_n(V). \end{aligned}$$

We further define

$$\begin{aligned} X^1 &= \{U \in X \mid \ell \leq U, U \tilde{\cap} \mathcal{E} \text{ and if } \dim U < n \text{ then } U @ \hat{\mathcal{E}}_n\} \\ X^2 &= \left\{ U \in X \mid \ell \not\leq U \leq \ell^\perp \text{ and } \begin{cases} U \cap \mathcal{E}, U @ \hat{\mathcal{E}}_n & \text{if } \dim U = n-1 \\ U \cap \mathcal{E} \cup (\mathcal{E} + \ell), U @ \hat{\mathcal{F}}_n & \text{if } \dim U \leq n-2 \end{cases} \right\} \\ X^3 &= \left\{ U \in X \mid U \not\leq \ell^\perp, \dim U > 1, U \tilde{\cap} \mathcal{F} \text{ and if } \dim U < n \text{ then } U @ \hat{\mathcal{F}}_n \right\} \\ X^4 &= \left\{ U \in X \mid U \not\leq \ell^\perp \text{ and } \begin{cases} U \tilde{\cap} \mathcal{F} & \text{if } \dim U = n \\ U \cap \mathcal{F} \text{ and } U @ \hat{\mathcal{F}}_n & \text{if } \dim U = n-1 \\ U \cap \mathcal{E} \cup (\mathcal{E} + \ell) \text{ and } U @ \hat{\mathcal{F}}_n & \text{if } \dim U = n-2. \end{cases} \right\} \end{aligned}$$

We can now define the filtration for κ . It will contain two parts, one dimension increasing and one dimension decreasing filtration. We start with

$$Y_0 = \{U \in X \mid U \in X^1 \cup X^2 \cup X^3\},$$

and set

$$\begin{aligned} Z &:= X^1 \cup X^2 \cup X^4 \\ Y_i &:= \{U \in Z \mid U \in Y_0 \text{ or } \dim U \leq i\}, 1 \leq i \leq n-2 \\ Y_{n-1} &:= Z \cup \{U \in Y \mid \dim U = n-1 \text{ and } Y_0^{>U} \neq \emptyset\}. \end{aligned}$$

Note that $Y_0 \subseteq Z$ since if $\dim U < n$ then $U \tilde{\cap} \mathcal{E} \iff U \cap \mathcal{E}$. The dimension increasing filtration is defined by setting

$$Y_i := \{U \in Y \mid U \in Y_{n-1} \text{ or } \dim U \geq 2n-i\}, n \leq i \leq 2n-1.$$

As usual, we define $\kappa_i = \text{Flag } Y_i$ and have $\kappa_{2n-1} = \kappa$.

Before constructing a cone function for κ_0 we recall the following fact.

Fact 6.34 ([Abr96, Step 3, Proposition 14]). *For every $U \in Y_0$, we have that $U \cap \ell^\perp \in Y_0$ and $(U \cap \ell^\perp) + \ell \in Y_0$.*

Lemma 6.35. *The complex $\kappa_0 = \text{Flag } Y_0$ has an $(n-2)$ -cone function Cone_{κ_0} with*

$$\text{Rad}_j(\text{Cone}_{\kappa_0}) \leq 2n+1, \quad -1 \leq j \leq n-2.$$

Proof. Let $\kappa = \text{Flag } Y_{\mathcal{E}}(V) \in \mathbf{C}_3$ with $\dim V = 2n$ and let Y_0 be defined as above. Recall that $X = \{0 < U < V \mid U \text{ is totally isotropic subspace}\}$. We define a filtration for Y_0 and we want to apply Theorem 3.1 inductively, using Fact 6.34.

Set

$$\begin{aligned} A_0 &:= X^1 \\ A_i &:= X^1 \cup \{U \in X^2 \mid \dim U \leq i\}, \quad 1 \leq i \leq n; \\ \alpha_i &:= \text{Flag } A_i, \quad 0 \leq i \leq n. \end{aligned}$$

Then $A_0 = \{\ell\} * \text{Flag}\{U \in A_0 \mid U \neq \ell\}$ and hence has a cone function with cone radius bounded by 1. Clearly elements in $A_i \setminus A_{i-1}$ have the same dimension and are hence not adjacent. Let $U \in A_i \setminus A_{i-1}$. Then

$$\text{lk}_{\alpha_i}(U) \cap \alpha_{i-1} = \text{Flag}\{W \in X \mid (W \in A_0 \text{ and } U < W) \text{ or } (W \in A_{i-1} \text{ and } W < U)\}.$$

We want to show that each subspace in $\{W \in X \mid (W \in A_0 \text{ and } U < W) \text{ or } (W \in A_{i-1} \text{ and } W < U)\}$ is either contained in or contains $U + \ell$. Note that we have $U \cap \ell^\perp + \ell = U + \ell \in A_0 \subseteq A_{i-1}$, since we know that $U \leq \ell^\perp$ because $U \in X^2$, $U + \ell \in Y_0$ and $\ell \leq \ell^\perp$ (because ℓ is totally isotropic) and thus $U + \ell$ cannot lie in $X^1 \cup X^3$ and hence has to lie in X^2 . If $W \in A_0$ and $U < W$ then $\ell \leq W$ and hence $U + \ell \leq W$. If $W \in A_{i-1}$ and $W < U$ then $W < U + \ell$ as needed. The second part of the filtration is defined as follows. We set

$$\begin{aligned} B_i &:= A_n \cup \{U \in X^3 \mid \dim U \geq n - i + 1\}, \\ \beta_i &:= \text{Flag } B_i, \quad 0 \leq i \leq n. \end{aligned}$$

Then $A_n = B_0$. Let $U \in B_i \setminus B_{i-1}$. We want to show that $\text{lk}_{\beta_i}(U) \cap \beta_{i-1}$ can be written as the join of $\{U \cap \ell^\perp\}$ and $\text{Flag}\{W \in B_{i-1} \mid W \neq U \cap \ell^\perp, W < U \text{ or } U < W\}$. First, note that $U \cap \ell^\perp \in Y_0$ and hence $U \cap \ell^\perp \in A_n \subseteq B_{i-1}$ (since $U \cap \ell^\perp \leq \ell^\perp$ and thus $U \cap \ell^\perp \notin X^3$). Let $W \in B_{i-1} \setminus \{U \cap \ell^\perp\}$. If $W < U$ then $\dim W < n - i$, hence $W \in A_n \setminus A_0$, in particular $W \leq \ell^\perp$. Hence $W = W \cap \ell^\perp \leq U \cap \ell^\perp$. If $U < W$ then $U \cap \ell^\perp \leq U \leq W$. Thus we get a filtration of κ_0 of length $2n$. By applying Theorem 3.1 at each step, we get an $(n-2)$ -cone function Cone_{κ_0} with

$$\text{Rad}_j(\text{Cone}_{\kappa_0}) \leq 2n + 1, \quad -1 \leq j \leq n - 2.$$

□

We collect some of the relevant facts from [Abr96] to show that the filtration above has the desired properties.

Fact 6.36 ([Abr96, Step 4, Step 5, Proposition 14]). *Let $U \in Y_i \setminus Y_{i-1}$. We have that $U \not\leq \ell^\perp$ and $\dim U = i \leq n - 2$. We get $Y_{i-1}^{<U} = Z^{<U}$. Since $\ell \not\leq U$ (otherwise $U \leq U^\perp \leq \ell^\perp$), we get that $\text{Flag } Z^{<U} \in \mathbf{C}_2$.*

On the other hand, $Y_{i-1}^{>U} = Y_0^{>U}$ and we have $\text{Flag } Y_0^{>U} \in \mathbf{C}_3$.

Fact 6.37 ([Abr96, Step 20 (b), Proposition 14]). *Let $n \leq i \leq 2n - 1$, $U \in Y_i \setminus Y_{i-1}$ with $\dim U < n$. Then $Y^{>U} \in \mathbf{C}_3$.*

Fact 6.38 ([Abr96, Step 2, Proposition 14]). *Let $\kappa = \text{Flag } Y_{\mathcal{E}}(V) \in \mathbf{C}_3$ in the case where $n = 2$, i.e. $\dim \kappa = 1$. Fix a 1-dimensional subspace $\ell \in Y_{\mathcal{E}}(V) =: Y$ (see Fact 6.33). Then each vertex $U \in Y$ can be connected to ℓ by a path of length at most 5.*

We now have all the necessary ingredients to show that the relevant class of simplicial complexes satisfies the conditions of Theorem 6.18. The proof will be similar to the previous proofs of similar results.

Theorem 6.39. *The class $\mathbf{C}_D = \mathbf{C}_1 \cup \mathbf{C}_2 \cup \mathbf{C}_3$ satisfies the conditions of Theorem 6.18.*

Proof. Let $\kappa = \text{Flag } Y \in \mathbf{C}_D$. If $\kappa \in \mathbf{C}_1 \cup \mathbf{C}_2$ then the result follows from Theorem 6.26 and Theorem 6.31.

Now assume $\kappa = \text{Flag } Y_{\mathcal{E}}(V) \in \mathbf{C}_3$. In this case the proof follows closely the proof of [Abr96, Proposition 14]. First of all, we fix a 1-dimensional subspace $\ell \in Y_{\mathcal{E}}(V) =: Y$ (see Fact 6.33). Let $n = \frac{1}{2} \dim V = \dim \kappa + 1$. We will again prove the existence of a filtration by induction on n , but this time starting with $n = 2$.

Let $n = 2$, hence $\dim \kappa = 1$. By Fact 6.38 each vertex $U \in Y$ can be connected to ℓ by a path of length at most 5. Thus we define a 0-cone function of κ as follows:

$$\begin{aligned} \text{Cone}_{\kappa}(\emptyset) &= \mathbb{1}_{\ell} \\ \text{Cone}_{\kappa}(\mathbb{1}_{[U]}) &= \sum_{i=0}^{k-1} \mathbb{1}_{[U_i, U_{i+1}]}, \end{aligned}$$

where $U = U_0, \dots, U_k = \ell$ is the path from U to ℓ for $U \in Y \setminus \{\ell\}$. In particular, $k \leq 5$. By Theorem 2.6, this is indeed a cone function. We can read off the cone radius from the explicit description and get

$$\text{Rad}_{-1}(\text{Cone}_{\kappa}) = 1, \text{Rad}_0(\text{Cone}_{\kappa}) \leq 5.$$

Now assume $n \geq 3$. Consider the filtration of κ described above. By Theorem 6.35 we know that there exists an $(n-2)$ -cone function Cone_{κ_0} with $\text{Rad}_j(\text{Cone}_{\kappa_0}) \leq 2n+1 = f(n)$ for all $-1 \leq j \leq n-2$.

We now check that the rest of the filtration has the required form. Start with the first half, i.e. $1 \leq i \leq n-2$. Let $U \in Y_i \setminus Y_{i-1}$. Fact 6.36 implies that the link of U has the desired form.

Next we consider $i = n-1$. Let $U \in Y_{n-1} \setminus Y_{n-2}$. Then $Y_{n-2}^{<U} = Z^{<U}$ and $\text{Flag } Z^{<U} \in \mathbf{C}_2$. Furthermore, we have $Y_{n-2}^{>U} = Y_0^{>U} \neq \emptyset$. Note that $\dim \text{Flag } Y_0^{>U} = 0$ since $Y_0^{>U}$ only contains subspaces of dimension n . Hence it trivially has a (-1) -cone function with cone radius 1.

We now consider the dimension increasing filtration, i.e. $n \leq i \leq 2n-1$. Let $U \in Y_i \setminus Y_{i-1}$ and set $k = 2n - i = \dim U$. We distinguish the cases $k = n$ and $k < n$.

If $k = n$, then $Y_{n-1}^{<U} = \emptyset$ since there is no totally isotropic subspace of dimension $> n$. This is fine for our sake, since $\emptyset * A = A$ for an arbitrary simplicial complex A . On the other hand, $Y_{n-1}^{<U}$ has again a filtration starting from an element in $\mathbf{C}_1$. To describe the filtration, we set $\mathcal{E}' := (\mathcal{E} \cap U) \cup ((\mathcal{E} + \ell) \cap U)$, $\mathcal{F}' := \mathcal{F} \cap U$, $\mathcal{H} := \mathcal{H}_1 := \dots := \mathcal{H}_{n-2} = \widehat{\mathcal{F}}_n$ and $\mathcal{H}_{n-1} := \{F \in \widehat{\mathcal{F}}_n \mid U \cap F = 0\}$. Furthermore, we set

$$S := X_{\mathcal{E}', \mathcal{H}}(U; V) \cap \{0 < W < U \mid F' \not\leq W \text{ for all } 0 \neq F' \in \mathcal{F}'\}.$$

By Step (20) in [Abr96, Proposition 14], we know that $S \leq Y_{n-1}^{<U}$ and given a cone function of $\text{Flag } S$ we get a cone function of $\text{Flag } Y_{n-1}^{<U}$ by one application of Theorem 3.1. To see that $\text{Flag } S$ has a cone function, we note that if $W \pitchfork_U F'$ then this implies that $F' \not\leq W$. Thus we set $\mathcal{E}'' = \mathcal{E}' \cup \mathcal{F}'$ and get $X_{\mathcal{E}'', \mathcal{H}}(U; V) \leq S$. We set

$$A_0 := X_{\mathcal{E}'', \mathcal{H}}(U; V), A_i = \{W \in S \mid W \in A_0 \text{ or } \dim W \leq i\}, 1 \leq i \leq n-1.$$

We follow the by now standard procedure to check that this is a valid filtration. Let $W \in A_i \setminus A_{i-1}$. Then $A_{i-1}^{>W} = A_0^{>W} = X_{\mathcal{E}'', \mathcal{H}}(U; V)^{>W}$ and by Step 4. in [Abr96, Lemma 33] (see also Fact 6.24) we have $\text{Flag } X_{\mathcal{E}'', \mathcal{H}}(U; V)^{>W} \in \mathbf{C}_1$. On the other hand, $A_{i-1}^{<W} = S^{<W}$. Note that if $W \in S$ and $W' \in X_{\mathcal{E}', \mathcal{H}}(U; V)$ such that $W' \leq W$ then $W' \in S$ since if $F' \not\leq W$ then $F' \not\leq W' \leq W$. Thus

$$S^{<W} = X_{\mathcal{E}', \mathcal{H}}(U; V)^{<W} = \{0 < W' < W \mid W' \cap \mathcal{E}', W' @ \mathcal{H}_j \text{ for } \dim W' = j\} = X_{\mathcal{E}' \cap W, \mathcal{H}}(W; V)$$

and $\text{Flag } X_{\mathcal{E}' \cap W, \mathcal{H}}(W; V) \in \mathbf{C}_1$. Hence we got a filtration

$$\text{Flag } A_0 \subseteq \dots \text{Flag } A_{n-1} = \text{Flag } S \subseteq \text{Flag } Y_{n-1}^{<U}$$

of length $n = \dim U$ which satisfies the assumptions.

If $\dim U < n$, then $Y_{i-1}^{<U} = Z^{<U}$ and we again have $\text{Flag } Z^{<U} \in \mathbf{C}_3$. Additionally, we have $Y_{i-1}^{>U} = Y^{>U}$ for which by Fact 6.37 we have $\text{Flag } Y^{>U} \in \mathbf{C}_3$.

Hence all the conditions of Theorem 6.18 are satisfied. $\square$

The following remark from [Abr96, page 105] gives the connection between $X_{\mathcal{E}}(V)$ and $Y_{\mathcal{E}}(V)$.

Remark 6.40. Let $\tau = \{E_1, \dots, E_r\}$ be a simplex of $\tilde{\Delta} = \text{Orifl } \tilde{X}$ and set $\mathcal{E} = \mathcal{E}(\tau) = \{E_i, E_i^\perp \mid 1 \leq i \leq r\}$. Then $\mathcal{E} \cap \mathcal{U}_n = \emptyset$, $Y_{\mathcal{E}}(V) = X_{\mathcal{E}}(V)$, $e_n \leq 2, e_{n-1} = e_{n+1} = 0$ and $e_i \leq 1$ for all other $1 \leq i \leq 2n - 1$.

Corollary 6.41. Let $\tilde{\Delta}$ be a building of type D_n over the field K , and $a \in \tilde{\Delta}$ be a simplex. If $|K| \geq 2^{2n-1}$, then $\tilde{\Delta}^0(a)$ has an $(n-2)$ -cone function with $\text{Rad}_j(\text{Cone}_{\tilde{\Delta}^0(a)}) \leq 2\mathcal{R}(n-1)$, $-1 \leq j \leq n-2$, where $\mathcal{R}(n)$ does not depend on K .

Proof. By Theorem 6.39, we have that $Y_{\mathcal{E}(a)}(V)$ has a cone function with radius $\leq \mathcal{R}(n)$, where $n = \dim Y_{\mathcal{E}(a)}(V)$. By Theorem 6.40, we have $X_{\mathcal{E}(a)}(V) = Y_{\mathcal{E}(a)}(V)$. Thus Theorem 6.17 yields a cone function for $\tilde{\Delta}^0(a) = \tilde{T}_{\mathcal{E}(a)}(V)$ with cone radius bounded by $2\mathcal{R}(n)$. $\square$

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